Swine viral diseases (II)

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Outline

- Porcine circovirus-associated disease (PCVAD)
- Porcine reproductive and respiratory syndrome (PRRS)
- Porcine parvovirus infection
- Porcine rotavirus infection
- Vaccination of pigs

Porcine circovirus-associated disease (PCVAD)



PCV1, PCV2, PCV3, PCV4

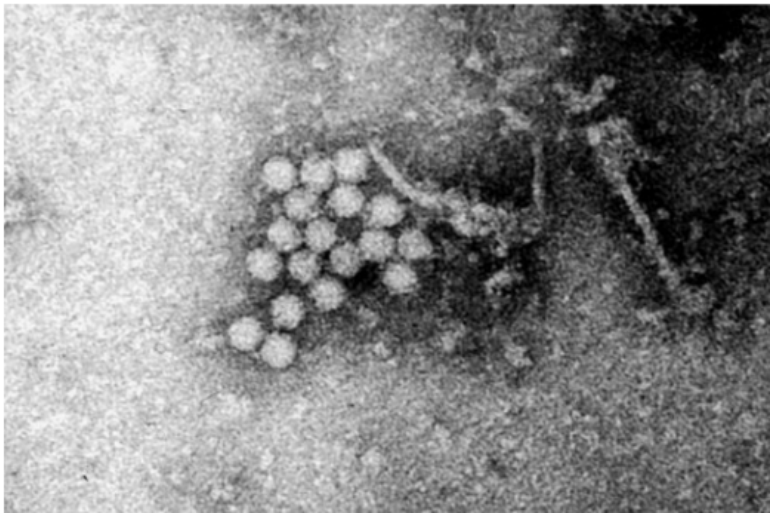
- Four PCVs have been recognized: PCV1, PCV2, PCV3 and PCV4.
  - **PCV1** was first described as a PK-15 cell contaminant in 1974 and was not pathogenic for swine.
  - **PCV2** was first identified in 1991 and has been the most commonly studied species because of its link to PCVAD.
  - **PCV3** was first identified in 2016, showing varied clinical signs, and the virus has been detected in many countries worldwide.
  - **PCV4** was first identified in 2019 and has been reported in Asia, Europe, and the US.

Porcine circovirus disease (PCVD, PCVAD)

- PCV2 is considered to be among the most important pathogens in the global swine industry.
- The term **“porcine circovirus diseases” (PCVDs)** was proposed to group diseases or conditions linked to PCV2.
- In North America, the term **“porcine circovirus-associated diseases” (PCVAD)** is used.
- PCVAD can have different clinical manifestations:
  - PCV2 systemic disease **(PCV2-SD)**: post weaning multisystemic wasting syndrome **(PMWS)**
  - PCV2 reproductive disease **(PCV2-RD)**
  - Porcine dermatitis and nephropathy syndrome **(PDNS)**
  - PCV2 subclinical infection (PCV2-SI)

Porcine circovirus

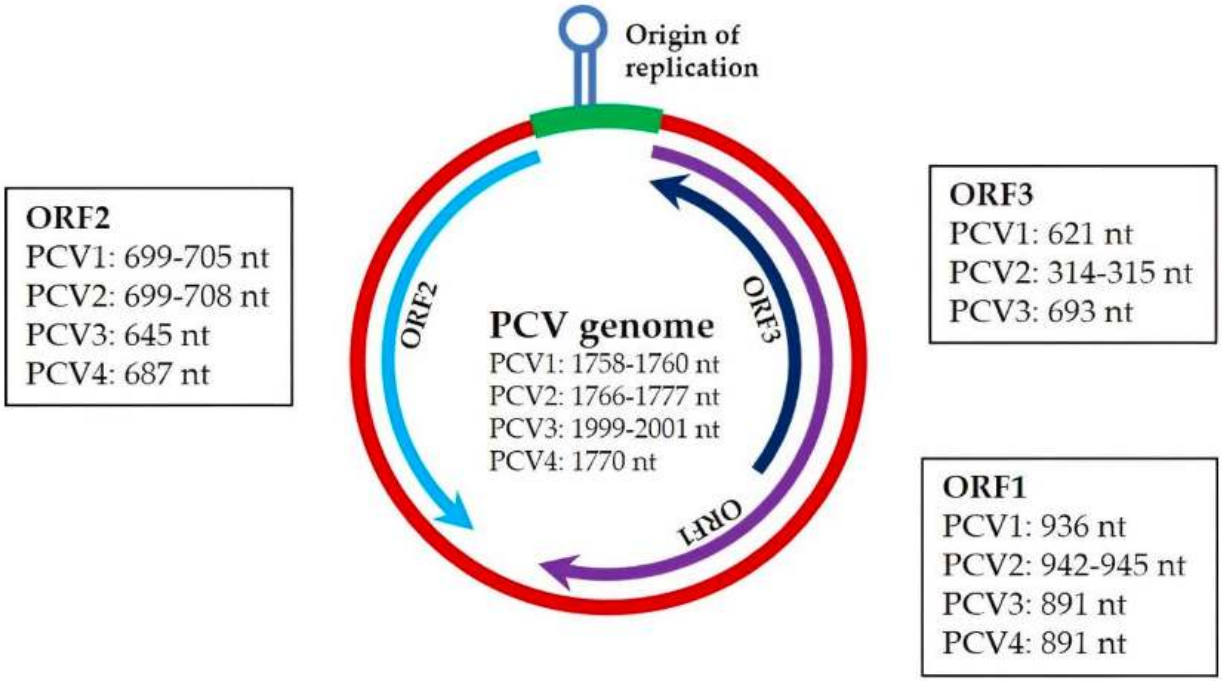
- PCVs belong to the genus *Circovirus* in the family *Circoviridae*.
- PCVs are non-enveloped and 12–23 nm in diameter.
- The nucleocapsids exhibit icosahedral symmetry.
- PCV2 contains a **single-stranded** 1.7 kb **circular DNA**.



Virus inactivation

- Stable at low temperatures
  - Readily isolated from -70°C tissue samples
- Stable at **high** temperatures
  - Resists inactivation at 56°C for 1 hour and 75°C for 15 minutes
  - More effectively inactivated by wet-heat treatment
- Stable at **low** pH
  - Remains viable even at pH < 2
- Resistant to most disinfectants
  - Oxidizing agents, aldehyde-quaternary ammonium combination products show the highest efficacy

Genome of PCV1 to PCV4

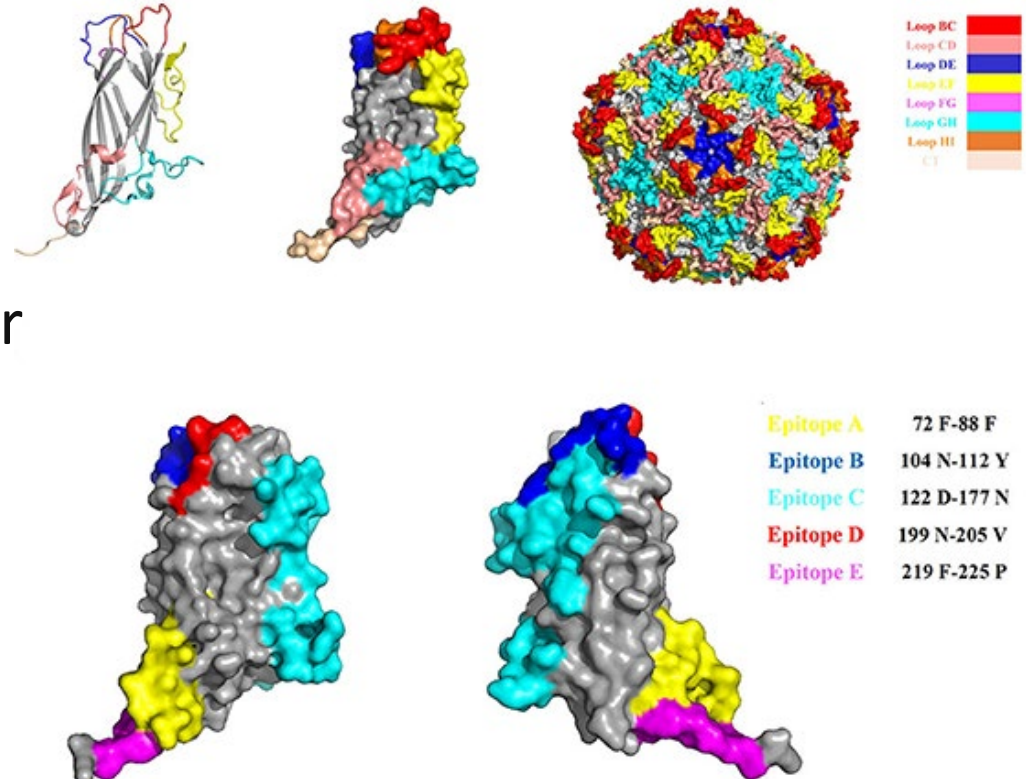


	PCV1–PCV2	PCV1–PCV3	PCV1–PCV4	PCV2–PCV3	PCV2–PCV4	PCV3–PCV4
Complete genome (nt)	68.0–76.0	43.5–44.0	50.3–51.6	42.7–48.0	51.5	42.9–45.0
Replicase (aa)	86.0	45.5–45.9	48.1–50.7	46.3–48.0	16.2–47.2	48.4–49.7
Capsid (aa)	65.0	24.0–25.2	43.1–44.4	25.9–37.0	12.7–45.0	23.2–24.8

Maity et al., 2023

PCV capsid protein

- **ORF2-encoded capsid** protein is the major phylogenetic and epidemiological marker
- Target antigen for subunit vaccines



Wang et al., 2021

PCV1

- Discovered in 1974
- Initially identified as a contaminant of PK-15 cells
- Widely distributed
- Generally considered non-pathogenic in pigs
- Limited pathogenic effects reported under experimental conditions
- Very low genetic diversity among

PCV2

- PMWS was first recognized in Canada in 1991
- Retrospective detection of PCV2 DNA in samples dating back to 1962
- Postweaning Multisystemic Wasting Syndrome (PMWS)
- Now grouped under PCVD / PCVAD
- Worldwide distribution
- Major genotypes: PCV2a, PCV2b, and PCV2d

PCV2 genotype shift

- Globally, at least nine PCV2 genotypes (**2a to 2i**) have been identified.
- **PCV2a**, **PCV2b**, and **PCV2d** display a worldwide distribution, while the other genotypes have been detected sporadically.
- **PCV2a to PCV2b**: PCV2a was the most prevalent genotype in clinically affected pigs; after 2003, PCV2b predominated ("**genotype shift**") and was associated with the appearance of more severe clinical disease outbreaks.
- **PCV2b to PCV2d**: After 2012, a second "**genotype shift**" from PCV2b to PCV2d occurred globally and has been detected in some apparent vaccine-failure cases.

PCV2 genotype shift

PCV2a → 2b → 2d

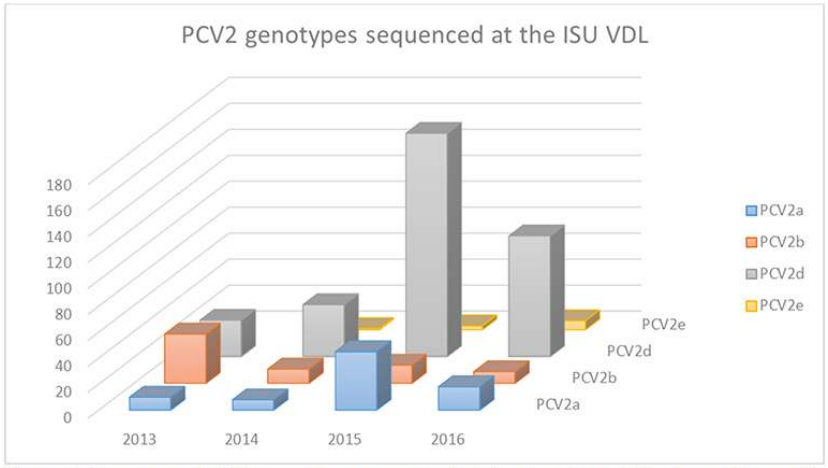


Figure 2: Summary of PCV2 genotypes sequenced at the Iowa State Veterinary Diagnostic Laboratory for the past four years.

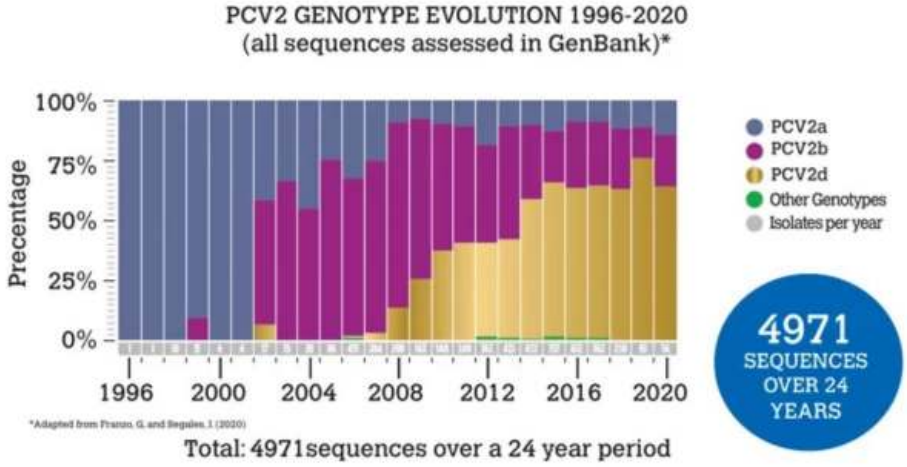


Figure 1: Global PCV2 genotypes

https://www.nationalhogfarmer.com/animal-health/pcv2-genotypes-are-expanding-furthering-challenges

https://www.veterinaryirelandjournal.com/large-animal/271-could-rapidly-mutating-pcv2-be-causing-health-and-production-issues-even-in-the-face-of-vaccination

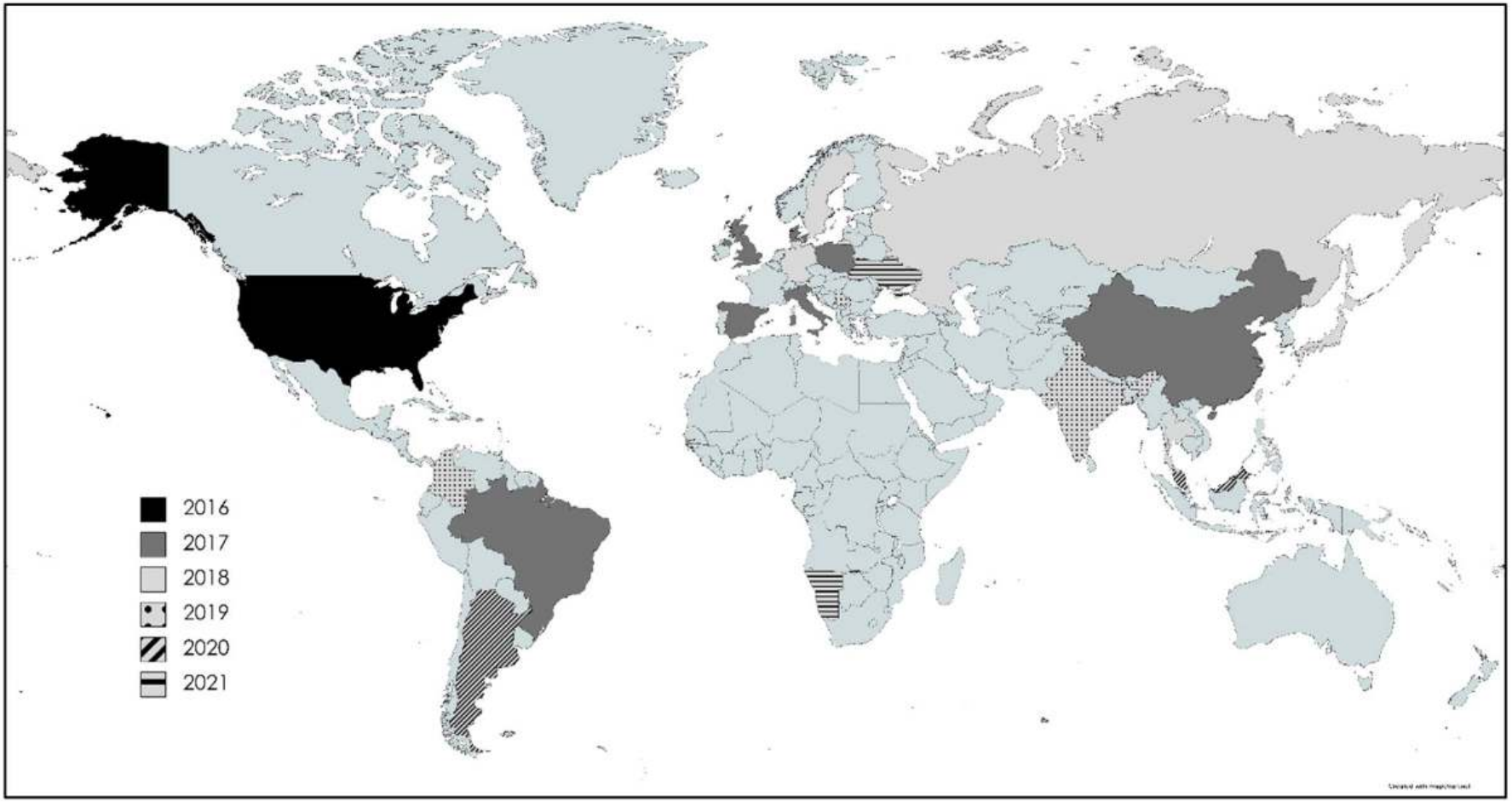
PCV2 a, b, c, d, …..

- PCV2 strains of the same or different genotypes may coinfect the same pig.
- No major antigenic differences have been demonstrated among PCV2 strains.
- Despite genetic and minor antigenic variation, PCV2 is generally considered to comprise a single serotype.

PCV3

- Discovered in US in 2016
- Retrospective evidence of circulation since at least the early 1990s
- Worldwide distribution
- Evolutionarily related to bat-associated circoviruses

The worldwide occurrence of PCV3

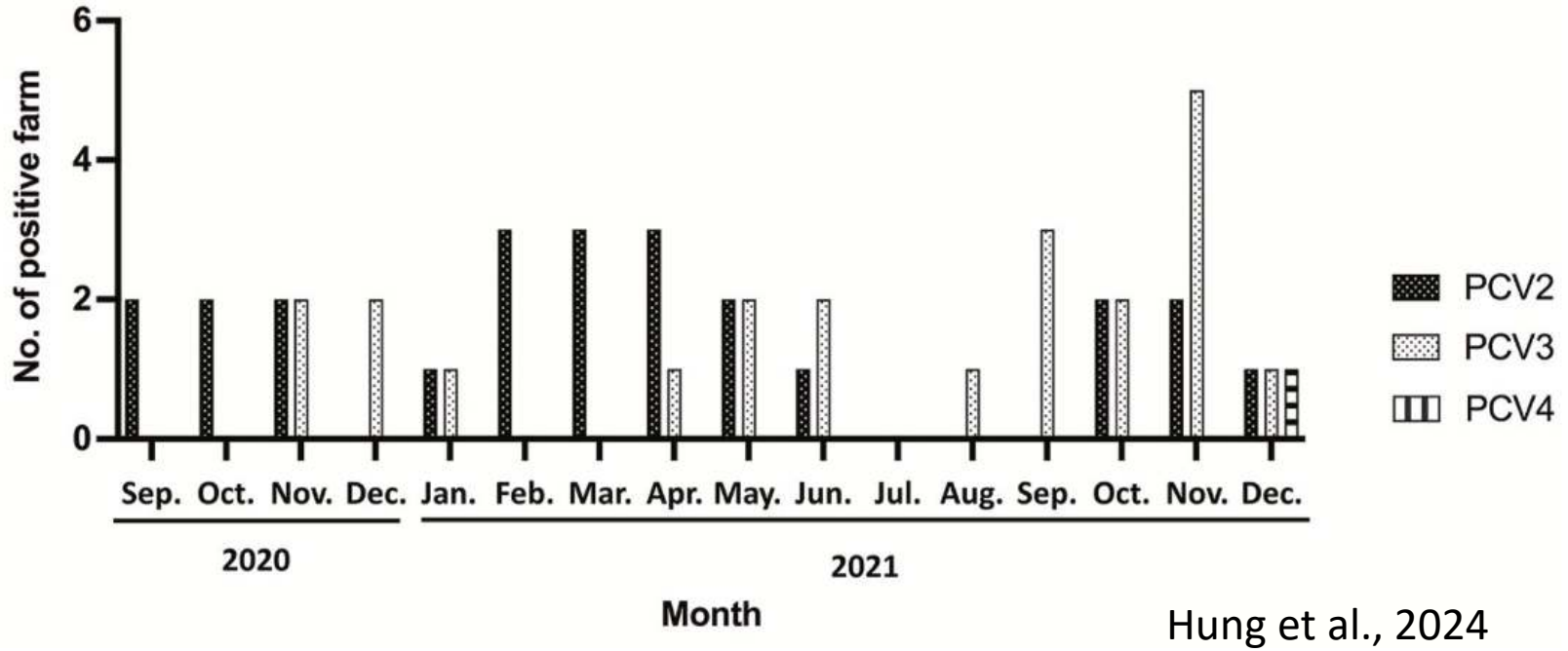


Turlewicz-Podbielska et al., 2022

PCV4

- Discovered in China in 2019, with antibodies detected as early as 2008
- Detected in Asia, Europe, and US
- More closely related to mink circoviruses compared to other PCVs

The occurrence of PCVs in Taiwan



Hung et al., 2024

- The case detection rates were 54.3 % for PCV2, 43.5 % for PCV3, and 2.2 % for PCV4.

The co-infection of PCVs in Taiwan

The cases with co-infection of PCV2, PCV3 or PCV4.

Sample No.	Farm No.	Sampling pig growth stage	Sampling Month	PCV2	PCV3	PCV4
IP2110098	NT01	Growing	Oct.	+	+	N*
IP2106103	ML01	Nursery	Jun.	+	+	N
IP2111121	YL02	Nursery	Nov.	+	+	N
IP2111122	YL02	Nursery	Nov.	+	+	N
IP2110112	CH07	Nursery	Oct.	+	+	N
IP2112138	CH01	Nursery	Dec.	+	N	+

*N: not detected.

Hung et al., 2024

Holgado-Martín *et al. Virology Journal* (2023) 20:230
https://doi.org/10.1186/s12985-023-02181-1

Virology Journal

COMMENT Open Access

First detection of porcine circovirus 4 (PCV-4) in Europe

Rocío Holgado-Martín[1†], José Luís Arnal[2†], Marina Sibila[3,4,5], Giovanni Franzo[6], Desireé Martín-Jurado[2], David Risco[1], Joaquim Segalés[3,4,7*†] and Luís Gómez[1*†]

Abstract

Porcine circovirus 4 (PCV-4) is a novel virus recently discovered (2019) in domestic pigs from China, although several studies have proven its circulation since 2008. Later, PCV-4 was also detected in wild boar populations from China and domestic pigs from South Korea and Thailand. Currently, Asia is so far the only continent where this novel virus has been reported; few studies carried out in South America and Europe failed in the attempt to detect it. The objective of this Comment is to communicate the first detection of PCV-4 in Europe, specifically in wild boar and domestic pigs from Mid-South-Western Spain. A retrospective study was carried out on wild boar and domestic pigs, both extensively (Iberian breed) and intensively raised, from Spain and Italy, sampled between 1998 and 2022. PCV-4 genome detection was attempted using different conventional or quantitative real time PCR (qPCR) protocols and some positive results were confirmed through Sanger sequencing. A total of 57 out of 166 (34.3%) Spanish wild boar and 9 out of 223 (4%) Iberian pigs (both geographically located in the Mid-South-Western Spain) were qPCR positive, while the rest of tested animals from North-Eastern Spain and Italy were negative. Partial sequences of Rep or Cap genes of selected samples confirmed the presence of PCV-4. The relatively high prevalence in wild boar and the low one in Iberian pigs from the same areas suggests intra- and interspecific transmission, being the wild boar a potential viral reservoir. The epidemiological and clinical importance of these findings are currently unknown, but guarantees further research on this novel virus.

Keywords Porcine circovirus 4 (PCV-4), Wild boar, Iberian pig, Spain, Italy

OPEN

First detection of PCV4 in swine in the United States: codetection with PCV2 and PCV3 and direct detection within tissues

Molly Kroeger[1], Diana S. Vargas-Bermudez[2], Jairo Jaime[2], Julian Parada[3], Jennifer Groeltz[1], Philip Gauger[1] & Pablo Piñeyro[1]✉

Since PCV4 was first described in 2019, the virus has been identified in several countries in Southeast Asia and Europe. Most studies have been limited to detecting PCV4 by PCR. Thus, PCV4 has an unclear association with clinical disease. This study utilized 512 porcine clinical lung, feces, spleen, serum, lymphoid tissue, and fetus samples submitted to the ISU-VDL from June–September 2023. PCV4 was detected in 8.6% of samples with an average Ct value of 33. While detection rates among sample types were variable, lymphoid tissue had the highest detection rate (18.7%). Two ORF2 sequences were obtained from lymphoid tissue samples and had 96.36–98.98% nucleotide identity with reference sequences. Direct detection of PCV4 by RNAscope revealed viral replication in B lymphocytes and macrophages in lymph node germinal centers and histiocytic and T lymphocyte infiltration in the lamina propria of the small intestine. PCV4 detection was most commonly observed in nursery to finishing aged pigs displaying respiratory and enteric disease. Coinfection with PCV2, PCV3, and other endemic pathogens was frequently observed, highlighting the complex interplay between different PCVs and their potential roles in disease pathogenesis. This study provides insights into the frequency of detection, tissue distribution, and genetic characteristics of PCV4 in the US.

Keywords PCV4, In situ hybridization, Lymphoid tissue, Coinfection, PCV2, PCV3

PCVAD

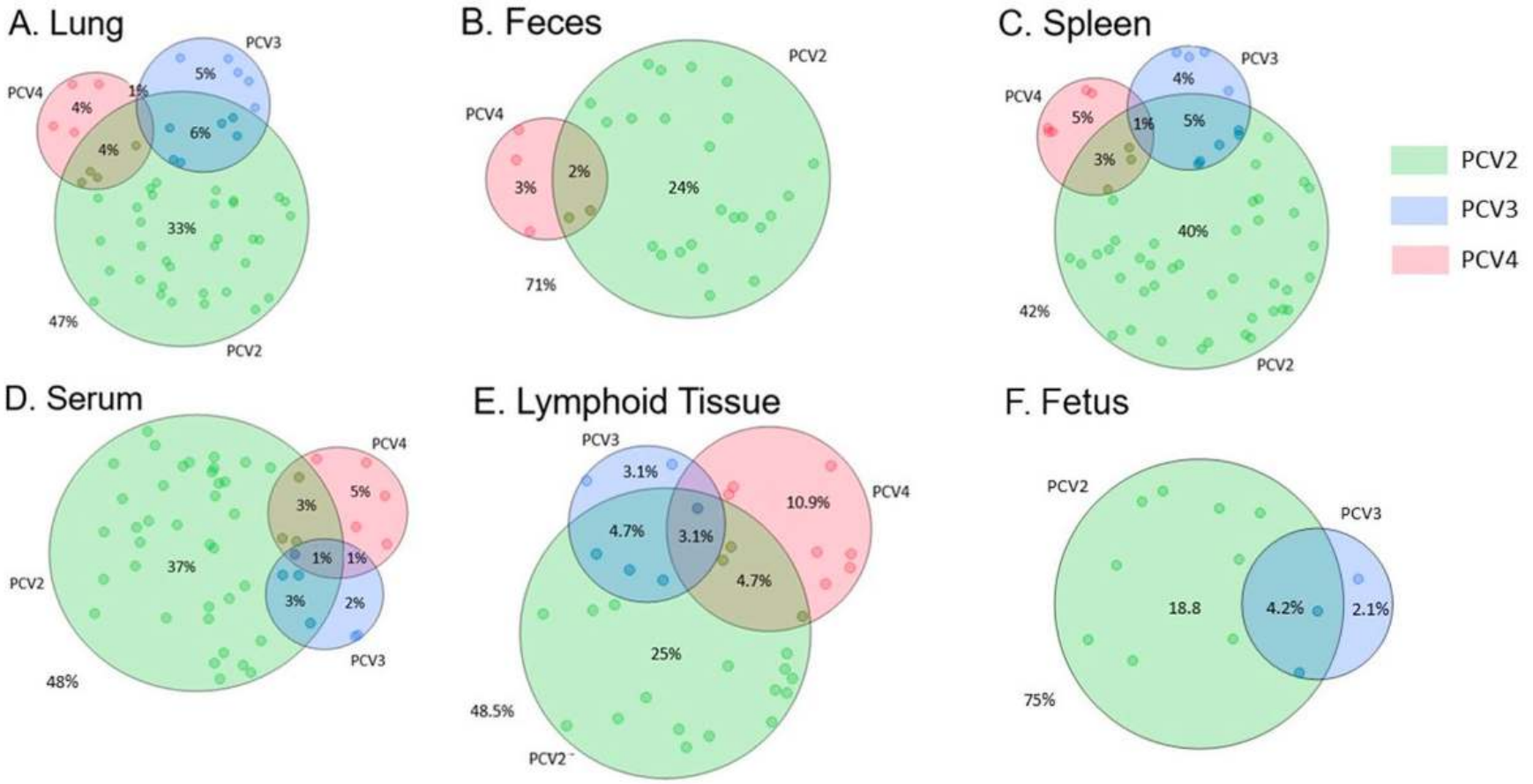


Figure 4. Codetection rate of PCV4 with PCV2 and PCV3 in lung (**A**), feces (**B**), spleen (**C**), serum (**D**), lymphoid tissue (**E**), and fetus (**F**). A total of 512 samples from clinical cases were tested by PCV4 singleplex qPCR and PCV2/3 multiplex qPCR. The Venn diagrams represent proportional areas with the differential proportion of PCV4/3/2 codetection (JMP® Pro 17.1.0; SAS Institute, Cary, NC).

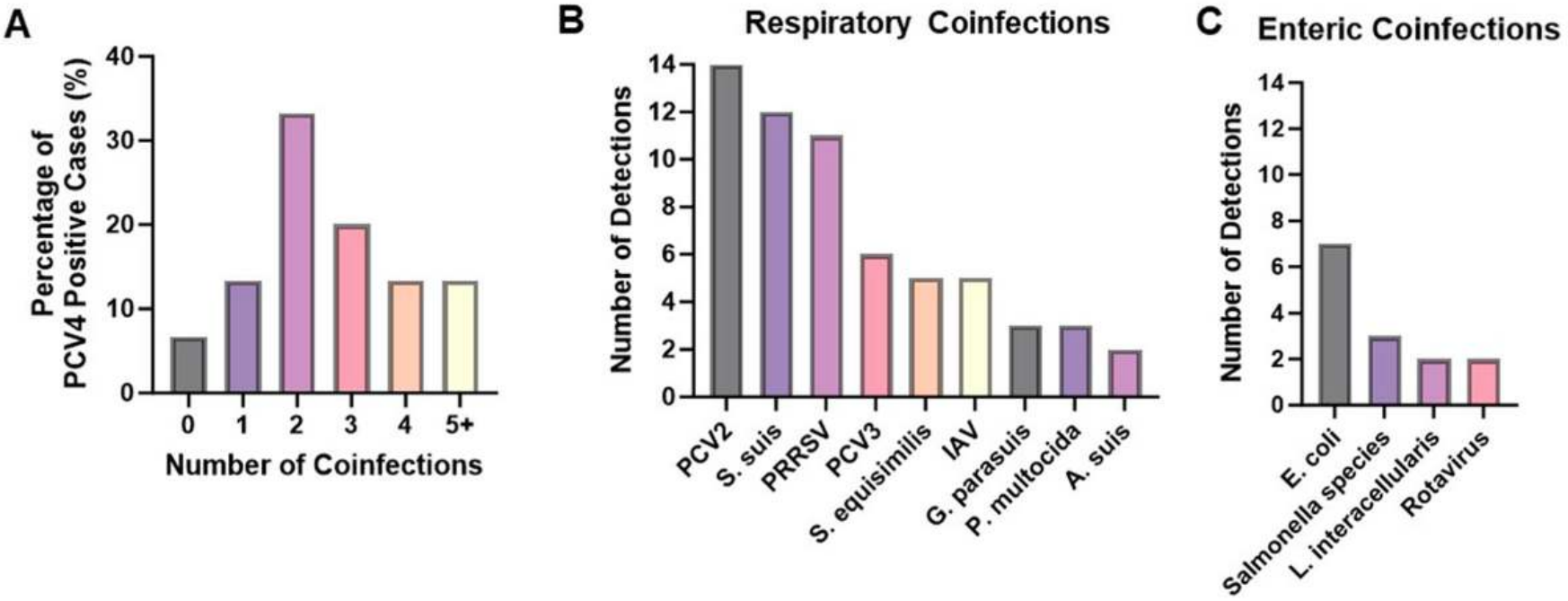
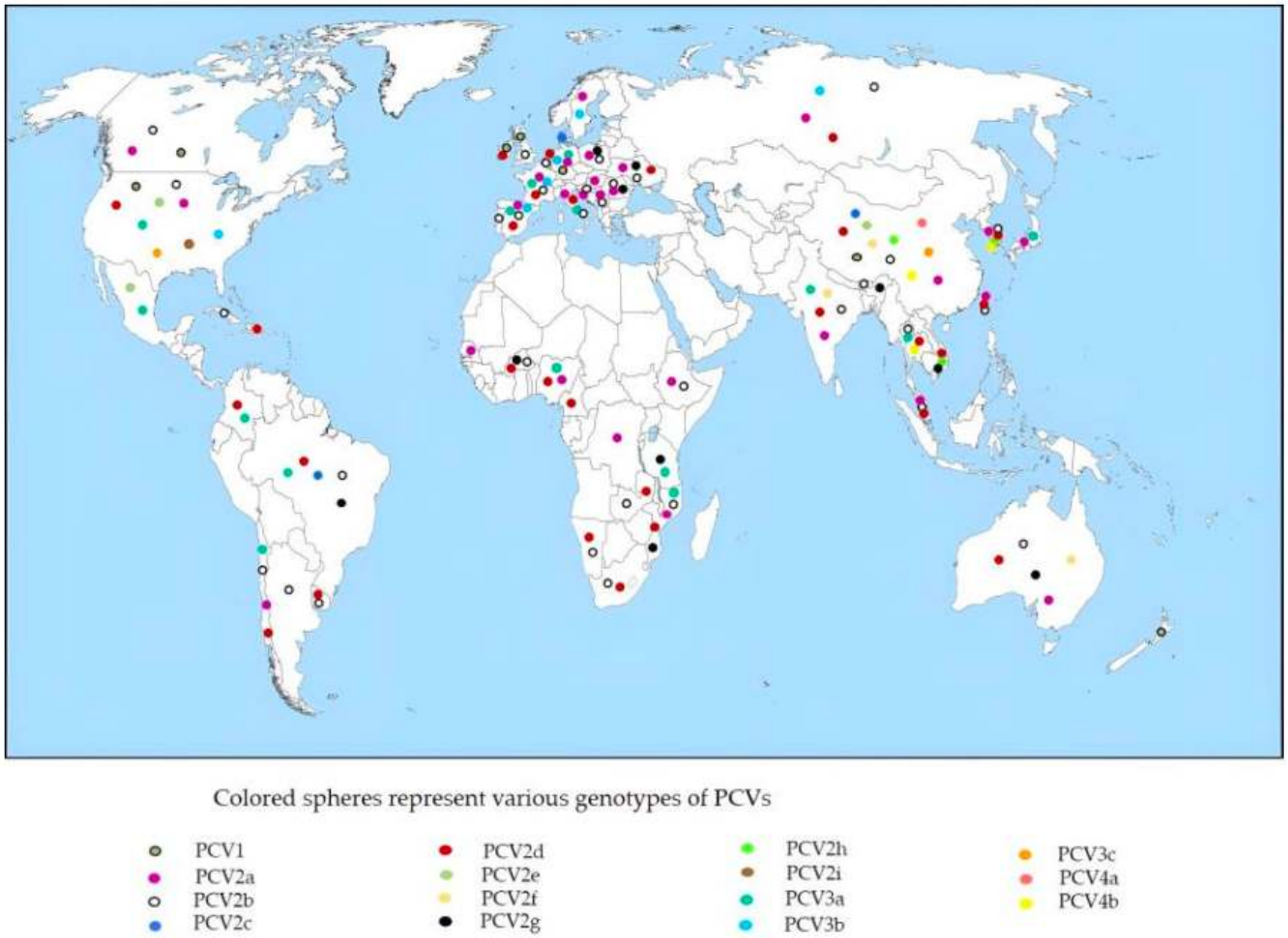


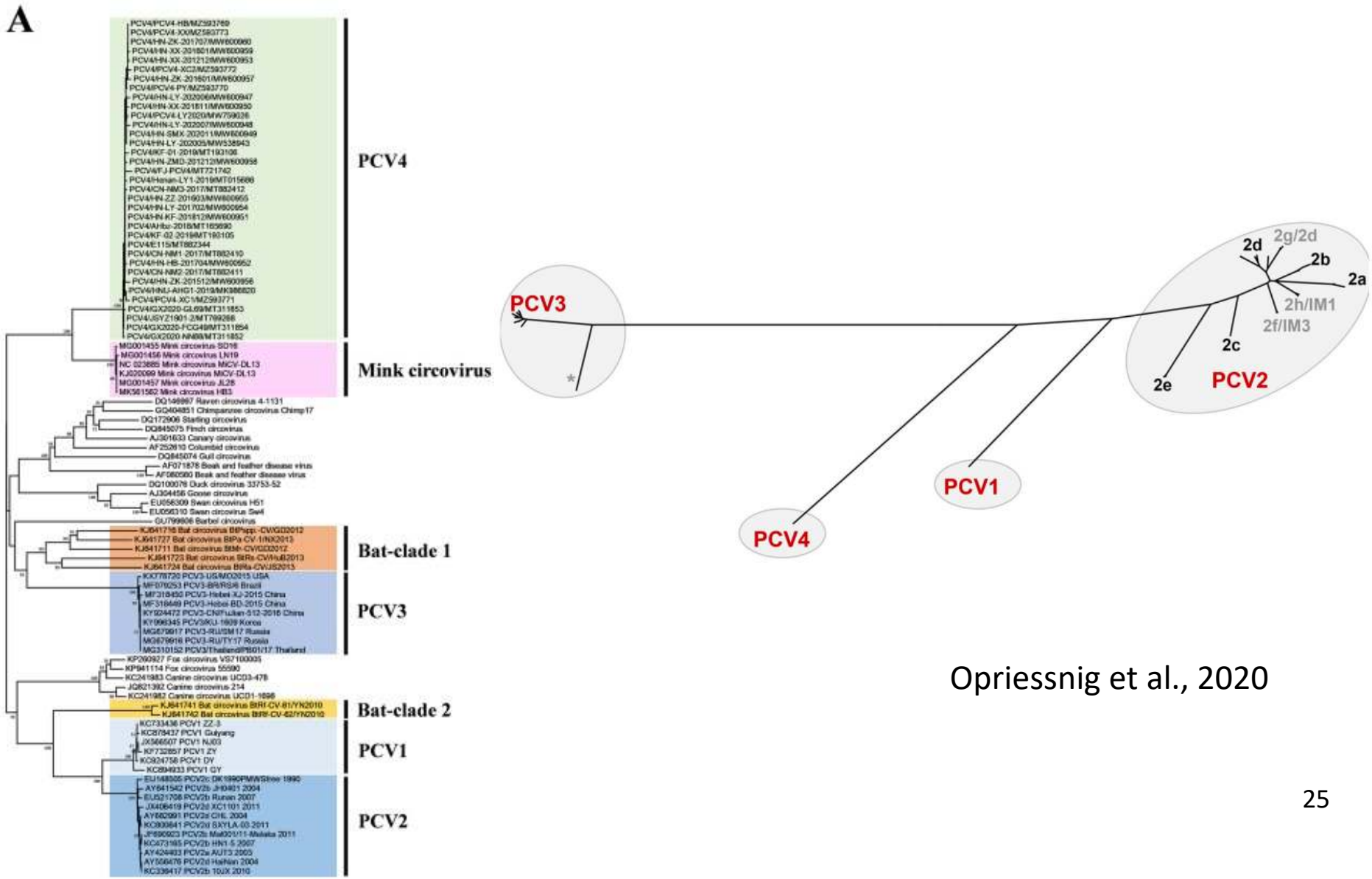
Figure 5. Summary of coinfections from ISU-VDL diagnostic case reports on cases with PCV4 detection in lymphoid tissue, lung, and spleen samples ($n = 30$). Percentage of PCV4 positive cases with 0–5 + coinfections identified in the diagnostic case report (**A**). Respiratory (**B**) or enteric (**C**) pathogens most commonly identified in PCV4 positive cases. Pathogens listed were identified in more than one PCV4 positive case.

Worldwide occurrence of different species of PCVs



Maity et al., 2023

PCV1, PCV2, PCV3, PCV4



Opriessnig et al., 2020

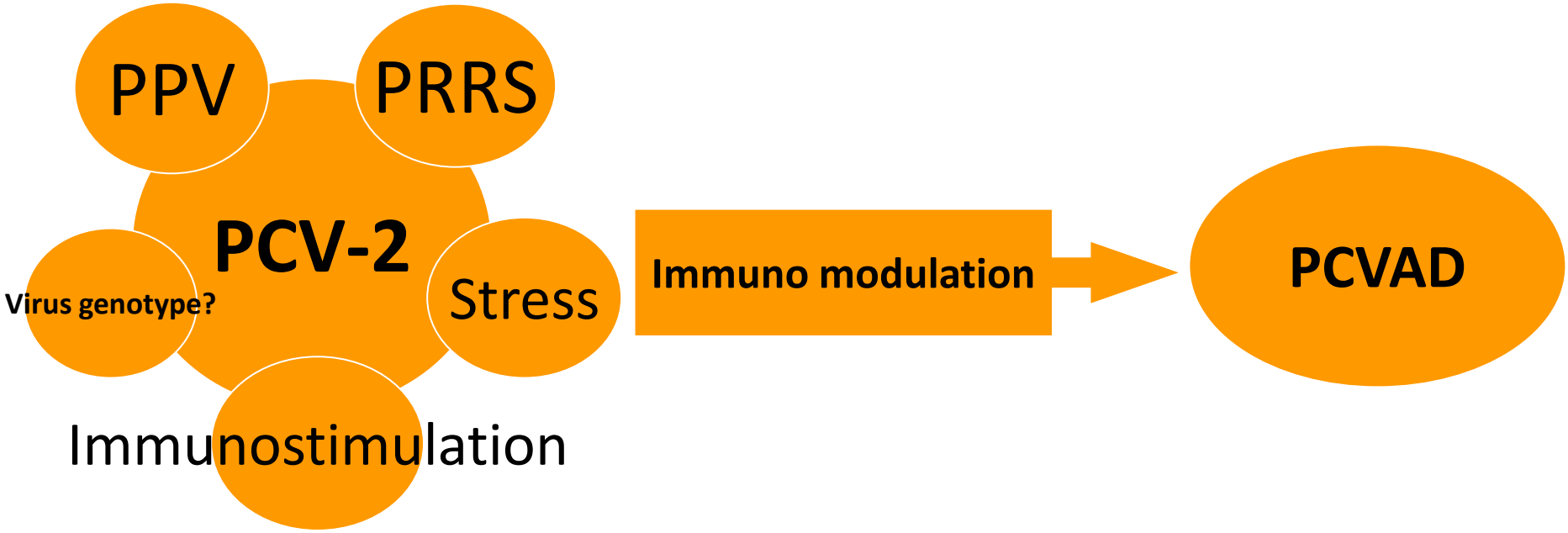
Transmission

- PCV2 is mainly transmitted via the oronasal route through infected secretions and excretions.
- Pigs can carry the virus for weeks after infection.
- Piglets can get infected by the sow in utero since PCV2 can cross the placenta.
- Piglets can also ingest viruses through milk and colostrum.
- Boars tend to shed the virus in their semen.

Pathogenesis

- **PCV2-SD** is characterized by severe impairment of the immune system, and clinical signs and lesions are often the result of the additive effect of co-infectious pathogens in different organs.
- **PCV2-SD** is a multifactorial disease, and PCV2 infection is the necessary cause.
- Genotypes PCV2a, PCV2b, and PCV2d are all considered pathogenic and can cause **PCV2-SD**.

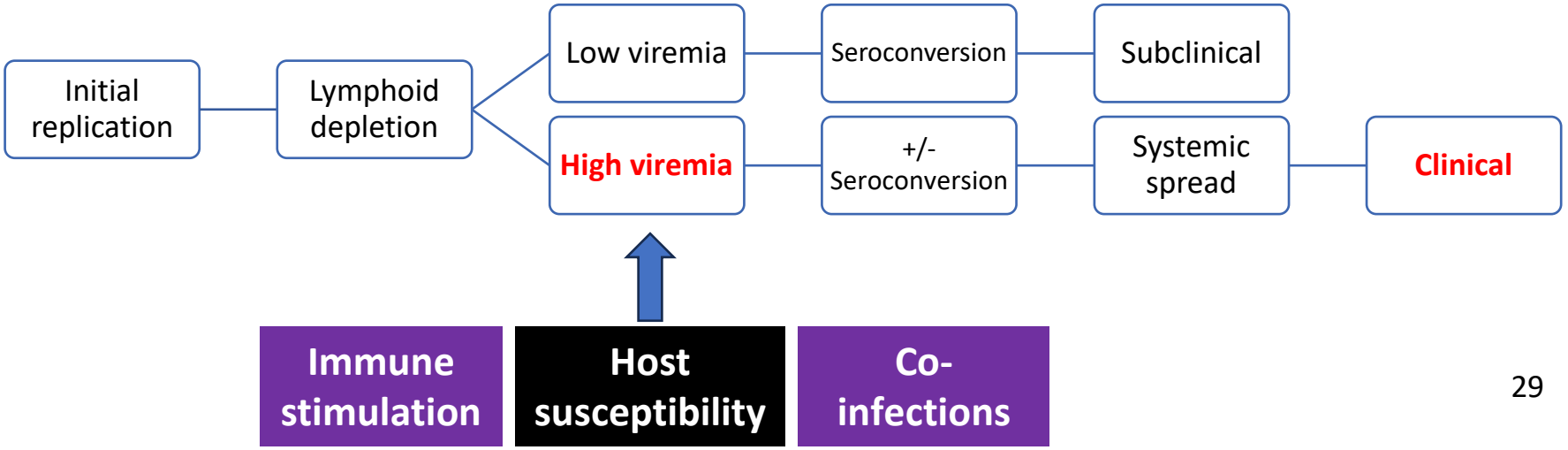
Pathogenesis



Mechanism of upregulation



- Many PCV2 infections are subclinical and may only reduce growth performance.
- Several cofactors can drive subclinical PCV2 infection toward clinical disease.



Pathogenesis

- Pigs infected with PCV2 develop viremia of variable duration, with replication and persistence most extensive in macrophages and monocytes of lymphoid organs and lungs.
- Lymphoid depletion at multiple sites, chronic lymphohistiocytic to granulomatous inflammation.
- **PDNS**: Pigs may develop blotchy purple skin lesions and nephropathy.

Clinical signs

- **PCVAD**
  - Wasting, unthriftiness, rough hair coat, polypnea, dyspnea, pallor, diarrhea, and occasionally icterus.
  - Affected pigs usually die.



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Diseases of Swine, 12th Edition

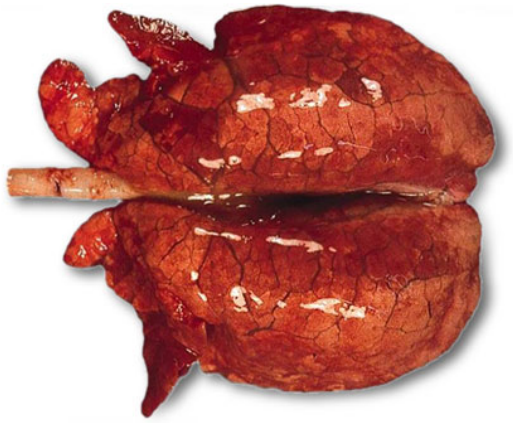
Clinical signs

- PCV2 systemic disease (PCV2-SD)
  - Wasting
  - Weight loss
  - Pallor of the skin
  - Respiratory distress
  - Diarrhea
  - Icterus



Clinical signs

- PCV2 lung disease
  - Respiratory distress
  - Dyspnea

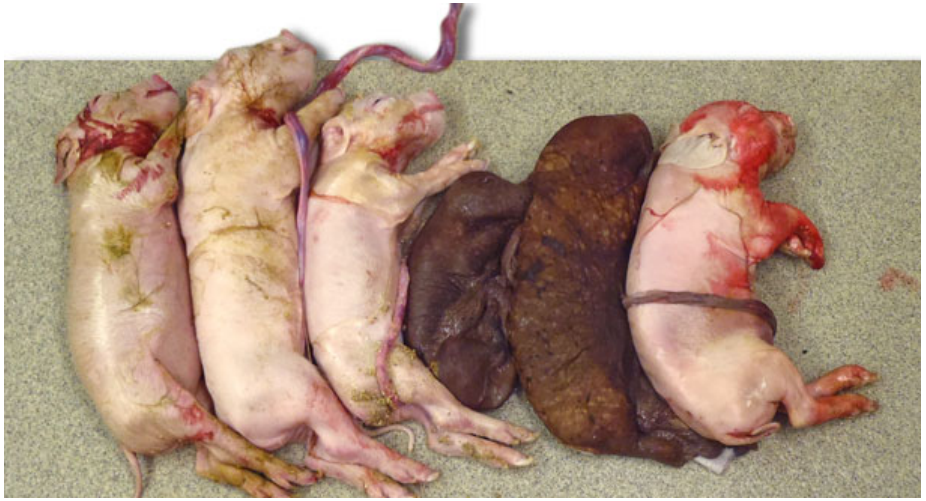


- PCV2 enteric disease
  - Diarrhea



Clinical signs

- **PCV2 reproductive disease (PCV2-RD)**
  - **Abortions or mummifications (SMEDI)**
    - Stillbirth
    - Mummification
    - Embryonic death
    - Infertility
  - **Regular return-to-estrus**
    - Seroconversion



Clinical signs

- **PDNS**
  - Is mainly a condition of pigs from 8-18 weeks of age.
  - There are red-purple blotches on the skin, sometimes slightly raised, most obvious on the hind legs and perineum, but can extend over the abdomen, eventually covering the whole body.
  - Most pigs with PDNS eventually die.



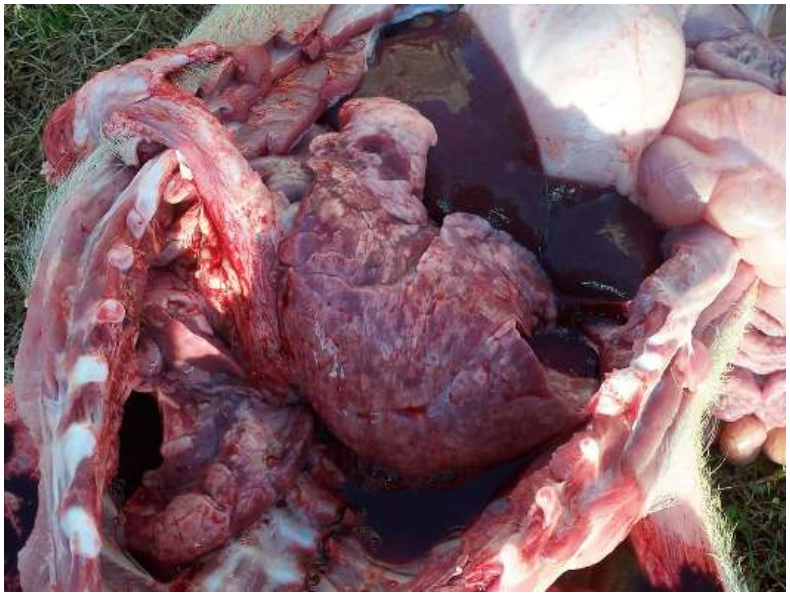


Gross lesions

- Lungs
  - Diffusely red-tan, noncollapsed, rubbery, with variable amounts of cranioventral consolidation
- Lymph nodes
  - Diffusely enlarged
- Other less common lesions
  - Icterus, gastric ulcers, skin lesions (PDNS), enlarged kidneys, swollen or atrophied liver
- Gross lesions are rarely diagnostic - very similar to PRRS

Gross lesions of PCVAD







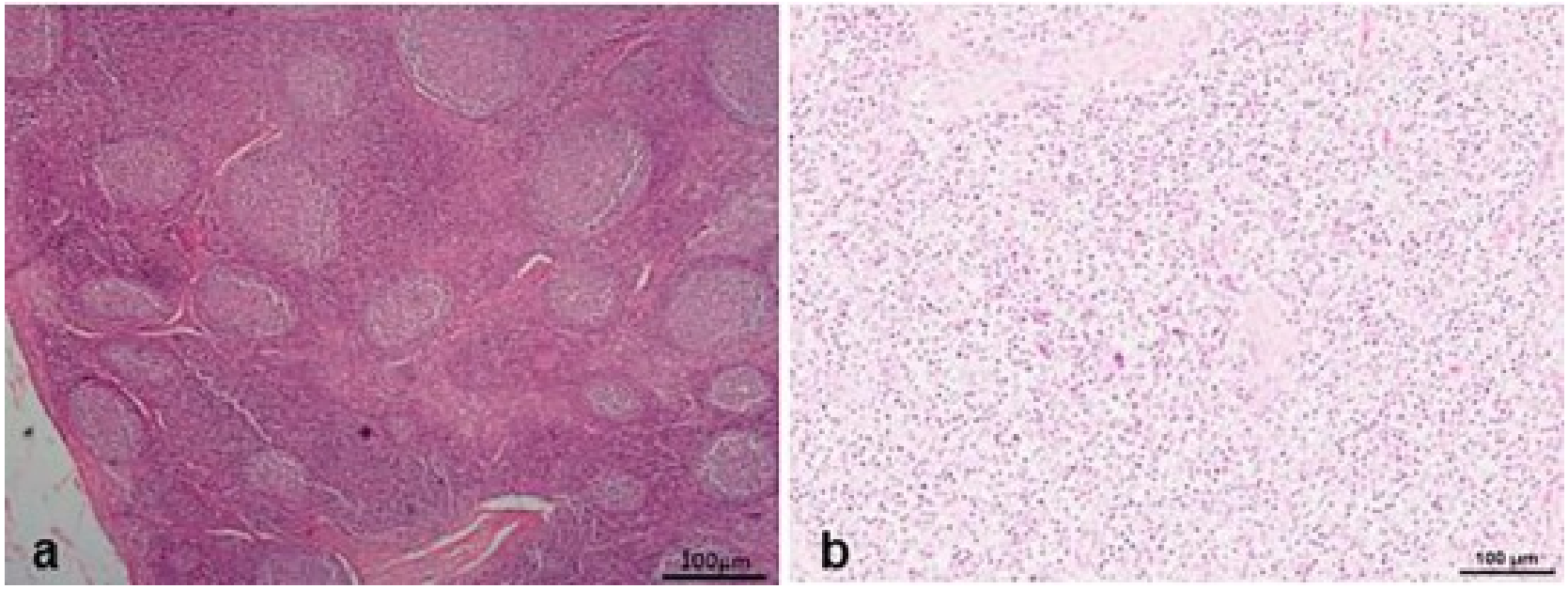


Modified from R. B. Baker, Veterinary Diagostics Laboratories, Iowa State University

Microscopic lesions

- Lung lesions
  - Lymphohistiocytic, proliferative interstitial pneumonia
  - Peribronchiolar and septal fibroplasia
- Lymphoid lesions: Lymphoid depletion and replacement of follicles with macrophages, intracytoplasmic inclusions
- Liver: Hepatocellular necrosis and lymphoplasmacytic inflammation
- Kidney: Lymphoplasmacytic interstitial nephritis

Microscopic lesions



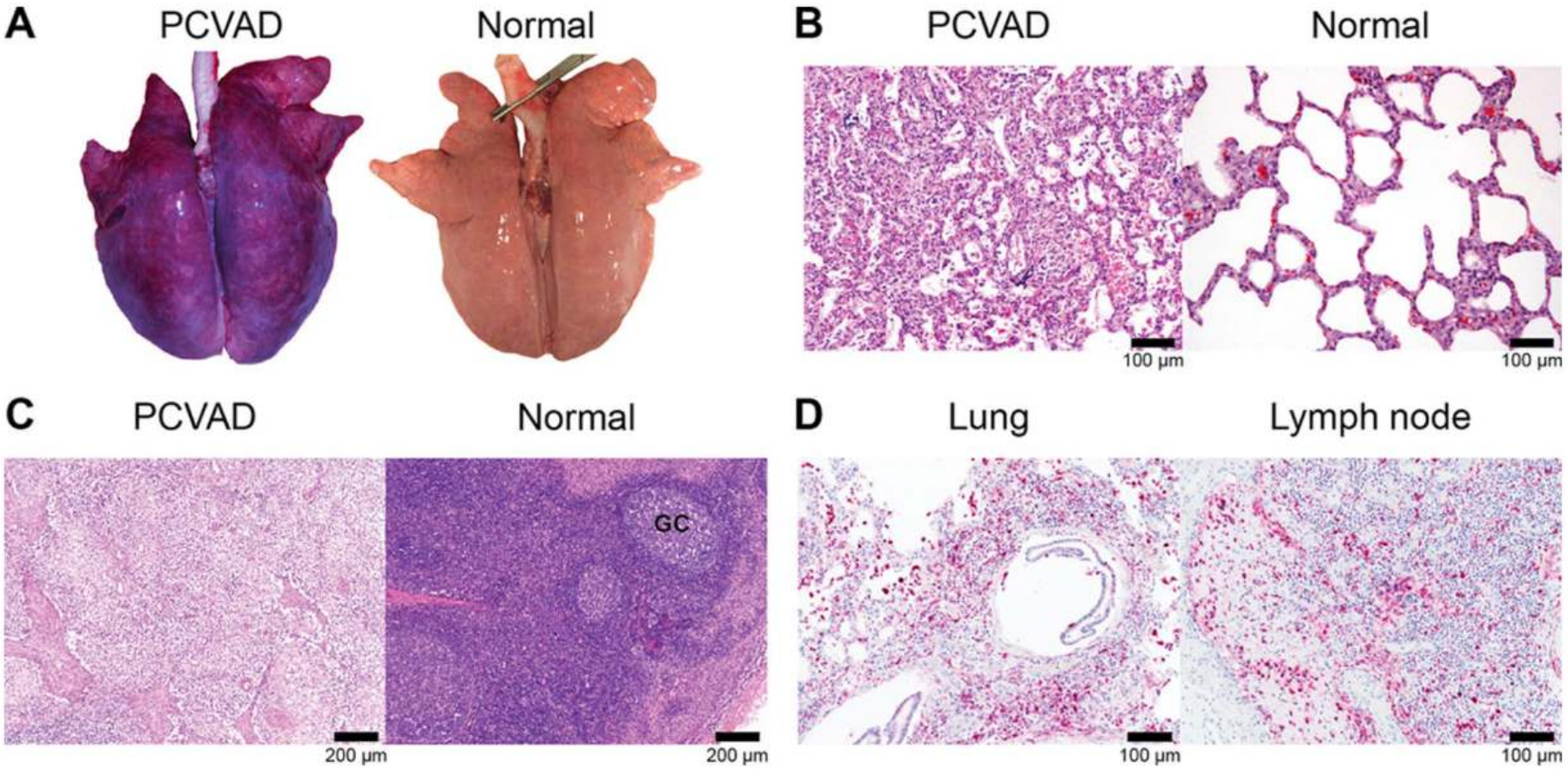
Comparison of a lymph node in a normal pig and in a PCV2-SD affected pig. Lymph node (a) is normal and has the normal follicular architecture while Lymph node (b) has lymphoid depletion without detectable follicles.

Wilfred et al., 2018

Gross and microscopic lesions

Pneumonia, mottling, and consolidation

Multifocal interstitial pneumonia



Lymphoid depletion in lymph node

PCV2 antigen

Niederwerder et al., 2015

Gross lesions of PDNS





Diseases of Swine, 12th Edition

Co-infection of PCV2 and other pathogens





PCV2 + PRRS, Parvovirus, Bacteria, etc

Co-infection with other pathogens

- Porcine parvovirus
  - PCVAD in 6-7% of PCV2 and PPV1 co-infected pigs
- *Mycoplasma hyopneumoniae*
  - PCVAD in 10% of PCV2 and *M. hyopneumoniae* co-infected pigs

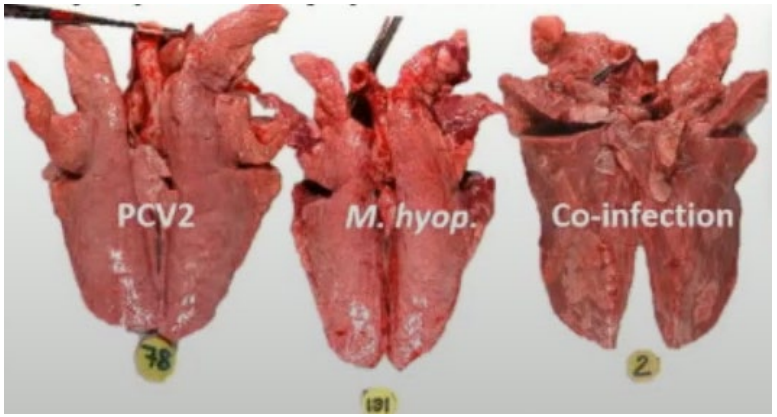
Veterinary Pathology
Volume 41, Issue 6, November 2004, Pages 624-640
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https://doi.org/10.1354/vp.41-6-624

SAGE journals

Article

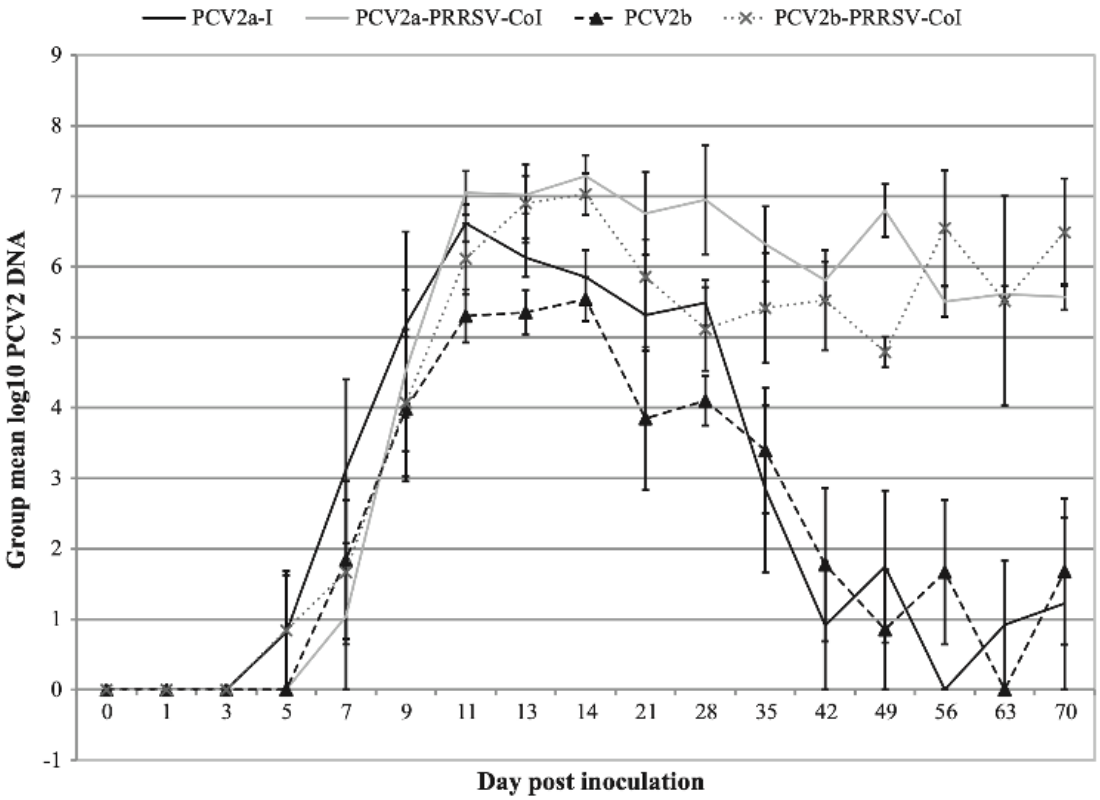
Experimental Reproduction of Postweaning Multisystemic Wasting Syndrome in Pigs by Dual Infection with *Mycoplasma hyopneumoniae* and Porcine Circovirus Type 2

T. Opriessnig[1], E. L. Thacker[2], S. Yu[2], M. Fenaux[3], X.-J. Meng[3], and P. G. Halbur[1]



Co-infection with other pathogens

- Porcine reproductive and respiratory syndrome virus
  - PCV2 and PRRSV co-infection can increase PCV2 viremia and disease expression.



Contents lists available at ScienceDirect
Veterinary Microbiology
journal homepage: www.elsevier.com/locate/vetmic
ELSEVIER

Porcine reproductive and respiratory syndrome virus (PRRSV) influences infection dynamics of porcine circovirus type 2 (PCV2) subtypes PCV2a and PCV2b by prolonging PCV2 viremia and shedding

A. Sinha [a], H.G. Shen [a], S. Schalk [a], N.M. Beach [b], Y.W. Huang [b], X.J. Meng [b], P.G. Halbur [a], T. Opriessnig [a,*]

Sinha et al., 2011

PCV2 infection decreases the efficacy of attenuated CSFV vaccine

Vaccine 37 (2019) 6535–6542

Contents lists available at ScienceDirect

Vaccine

journal homepage: www.elsevier.com/locate/vaccine

The impact of porcine circovirus associated diseases on live attenuated classical swine fever vaccine in field farm applications

Jing-Yuan Chen [a], Chi-Ming Wu [c], Chih-Ming Liao [c], Kun-Chao Chen [c], Chia-Chin You [c], Yu-Wen Wang [c], Chienjin Huang [b,*], Maw-Sheng Chien [a,c,*]

Research in Veterinary Science 96 (2014) 187–195

Contents lists available at ScienceDirect

Research in Veterinary Science

journal homepage: www.elsevier.com/locate/rvsc

Porcine circovirus type 2 decreases the infection and replication of attenuated classical swine fever virus in porcine alveolar macrophages

Yu-Liang Huang [a], Victor Fei Pang [b,c], Ming-Chung Deng [a], Chia-Yi Chang [a], Chian-Ren Jeng [b,c,*]

Huang et al. Veterinary Research 2011, 42:115
http://www.veterinaryresearch.org/content/42/1/115

VETERINARY RESEARCH

RESEARCH Open Access

Porcine circovirus type 2 (PCV2) infection decreases the efficacy of an attenuated classical swine fever virus (CSFV) vaccine

Yu-Liang Huang[1,2], Victor Fei Pang[1,3], Chun-Ming Lin[1], Yi-Chieh Tsai[1], Mi-Yuan Chia[1], Ming-Chung Deng[2], Chia-Yi Chang[2] and Chian-Ren Jeng[1,3*]

Diagnosis of PCVAD

- Clinical symptoms
  - Wasting syndrome, high mortality
- Histopathology
  - Nonsuppurative interstitial pneumonia
  - Multifocal, diffuse lymphoid depletion
- Virology
  - PCV2 positive
- Co-infections
  - PRRSV, SIV, *Strep. suis*, *M. hyopneumoniae*

Diagnosis of PCV2-SD

- The diagnosis of **PCV2-SD** in a pig or group of pigs is warranted if the following criteria are fulfilled:
  - Growth retardation and wasting, frequently with dyspnea and enlargement of inguinal lymph nodes, and occasionally with jaundice.
  - Moderate to severe characteristic histopathological lesions in lymphoid tissues.
  - Moderate to high amounts of PCV2 within the lesions in lymphoid and other tissues of affected pigs.

Diagnosis of PCV2-RD

- The diagnosis of **PCV2-RD** at **late** gestation includes three criteria:
  - Late-term abortions and stillbirths, sometimes with evident hypertrophy of the fetal heart.
  - Heart lesions are characterized by extensive fibrosing and/or necrotizing myocarditis.
  - High levels of PCV2 are present in myocardial lesions and other fetal tissues.

Diagnosis of PCV2-RD

- The putative diagnosis of **PCV2-RD** at **early** gestation has been suggested based on two criteria:
  - Regular return to estrus.
  - PCV2 seroconversion following the return to estrus and/or PCV2 PCR positivity around return-to-estrus occurrence.

Diagnosis of PDNS

- Detection of PCV2 is **not** a diagnostic requirement. Instead, the diagnosis of **PDNS** is based on two criteria:
  - The presence of hemorrhagic and necrotizing skin lesions, primarily on the hind limbs and perineal area, and/or swollen and pale kidneys with generalized cortical petechiae.
  - The presence of systemic necrotizing vasculitis and necrotizing fibrinous glomerulonephritis.

Diagnostic samples

- **Serum** – qPCR and serology
- **Lymph nodes, tonsil, spleen, and lung** – histopathology, IHC/ISH, and qPCR
- Other affected tissues as indicated by clinical signs
- **Oral fluid** – herd-level monitoring
- PCR positivity alone does not confirm PCV2-associated disease.

Differential diagnosis

- PRRS
  - Reproductive failure, wasting, dyspnea
- Swine influenza
  - Respiratory disease
- CSF / ASF
  - Renal hemorrhages and systemic lesions
- Swine erysipelas
  - Skin lesions and reproductive failure

Diagnosis gaps and challenges

- Well-established diagnostic assays are available for PCV2
- More validated commercial tests are needed for PCV3 and PCV4

Assays	PCV1	PCV2 (a-i)	PCV3	PCV4
Serology in-house	✔	✔	✔	✔
Serology commercial	N/A	✔	✔	N/A
PCRs in-house	✔	✔	✔	✔
PCRs commercial	N/A	✔	N/A	N/A
IHC	N/A	✔	N/A	N/A
ISH	✔	✔	✔	✔
Sequencing Illumina/ Sanger	✔	✔	✔	✔

PCV2 immunity

- PCV2 infection can cause marked immune dysregulation, particularly in pigs with PCVAD.

- Pigs with PCVAD are highly correlated with a decrease in the number of circulating B and T cells and in lymphocytes in lymphoid organs.

PCV2 immunity

- 7–14 days: Antibodies detectable by IFA
- 21–35 days: Low titers of **neutralizing** antibodies become detectable

Control of PCVAD

- Vaccines
- Control of other pathogens/diseases
- Management and nutritional changes
- Genetic changes
- Depopulation and Repopulation

PCV2 vaccination

- Commercial PCV2 vaccines have been available since 2006, and economic losses attributed to PCVAD, including subclinical PCV2 infections, have been markedly reduced.
- Vaccination of both sows and piglets has been shown to be beneficial in the continuous control of PCVAD.
- PCV2 is also an important coinfecting agent, and vaccination against PCV2 can assist in the control of clinical diseases associated with polymicrobial infections.



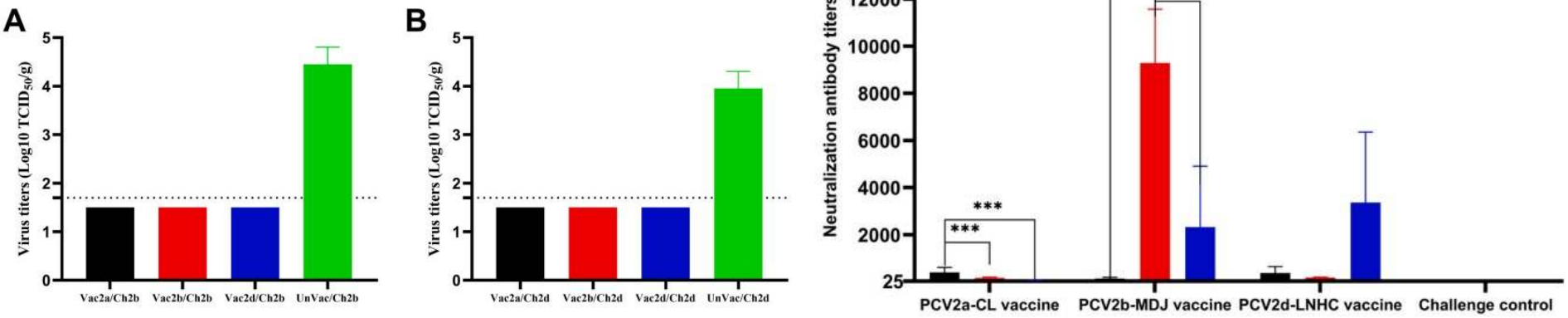
PCV2 vaccination

- Most commercial PCV2 vaccines are based on **PCV2a** strains.
- **PCV2a** and **PCV2b** vaccinations both reduced **PCV2d** viremia and PCV2-associated pathological lesions.
- Due to the great diversity of the PCV2 genotypes, a homologous strain is recommended as the vaccine-based strain.
- PCV2 vaccines are not expected to protect against PCV3 or PCV4.



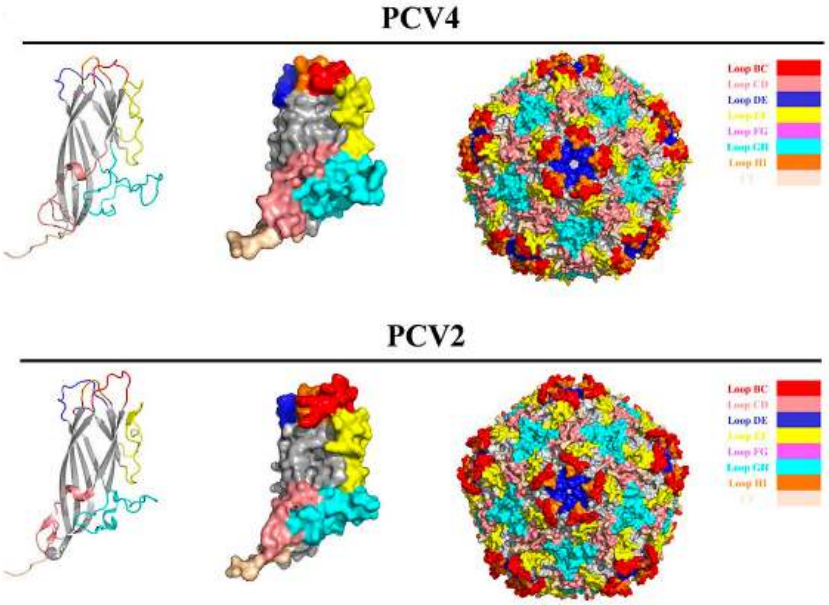
Vaccination—Cross protection

- Limited impact on cross-protection among PCV2 genotypes
  - PCV2a-CL, PCV2b-MDJ and PCV2d-LNHC
- Antigenic differences can significantly affect neutralizing antibody titers

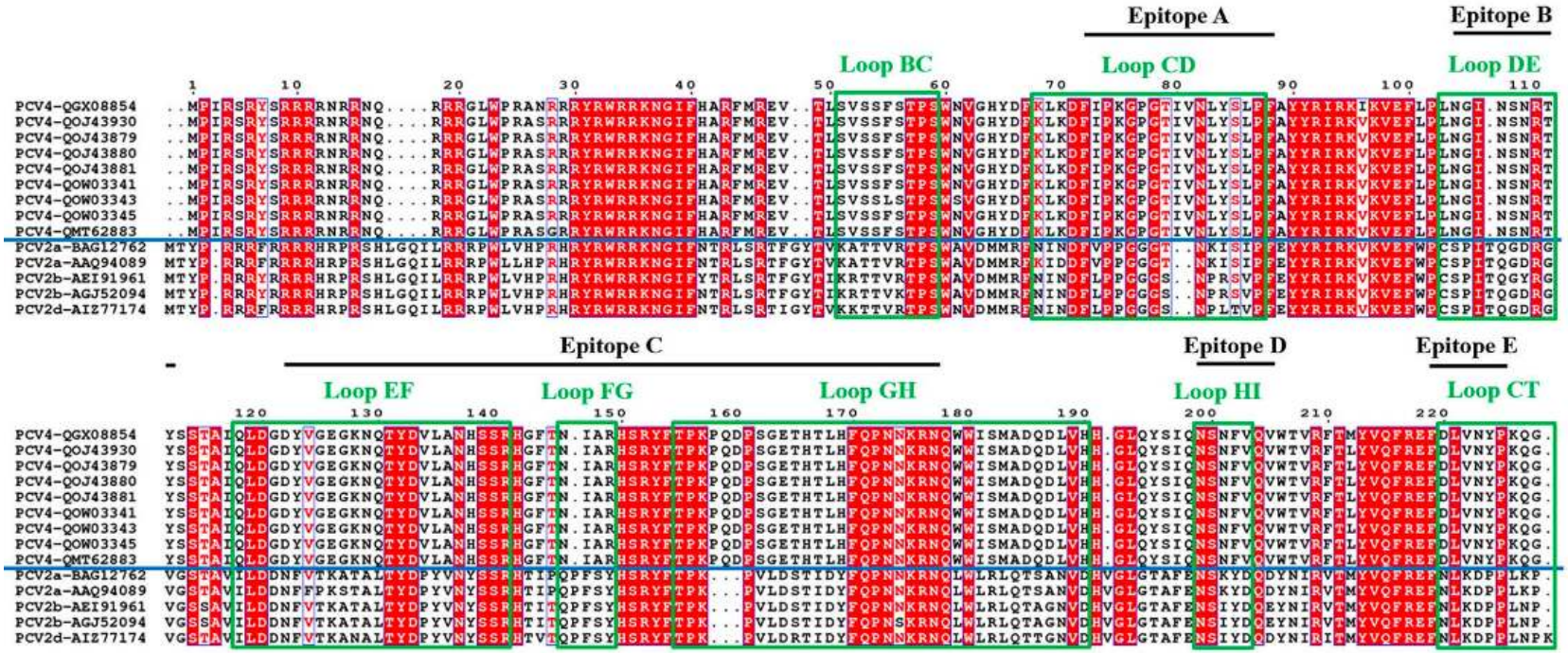


Yu et al., 2023

Antigenicity of PCV 1~4

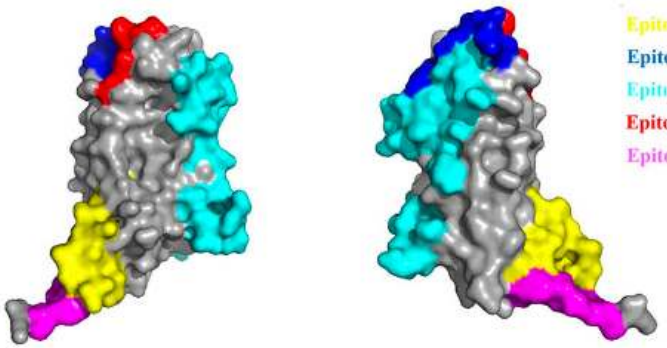


PCV4 vs PCV2 Cap amino acid sequences

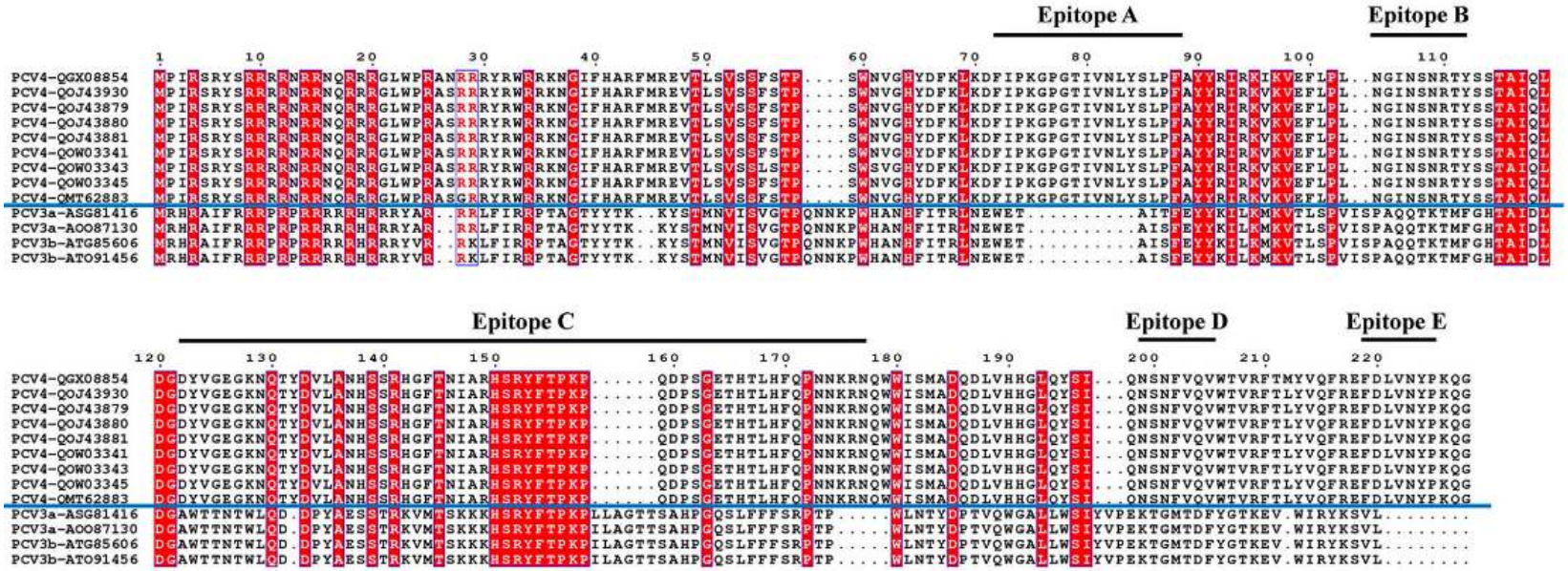


Wang et al., 2021

Antigenicity of PCV 1~4



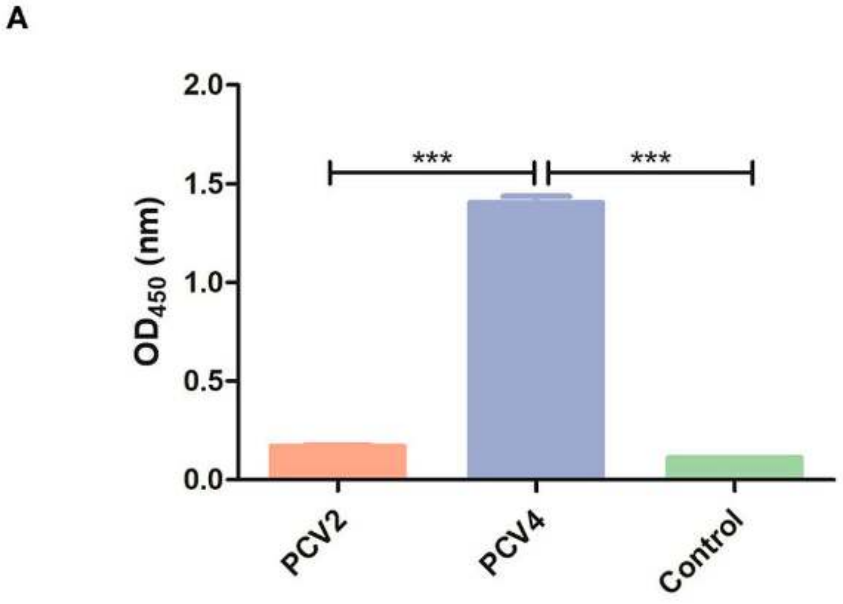
PCV4 vs PCV3 Cap amino acid sequences



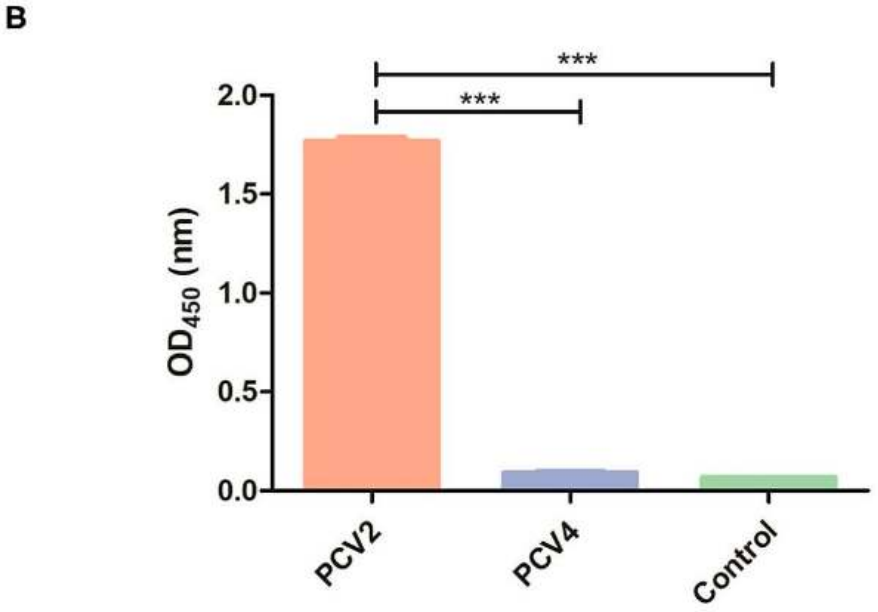
Wang et al., 2021

Antigenicity of PCV 1~4

PCV4-immunized mice sera



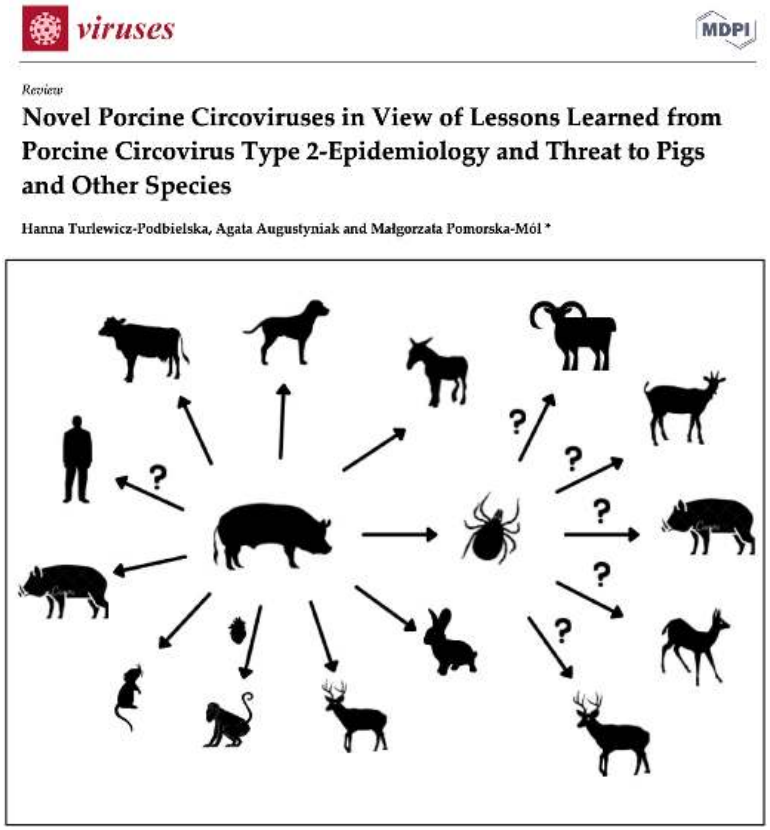
PCV2-immunized mice sera



Wang et al., 2021

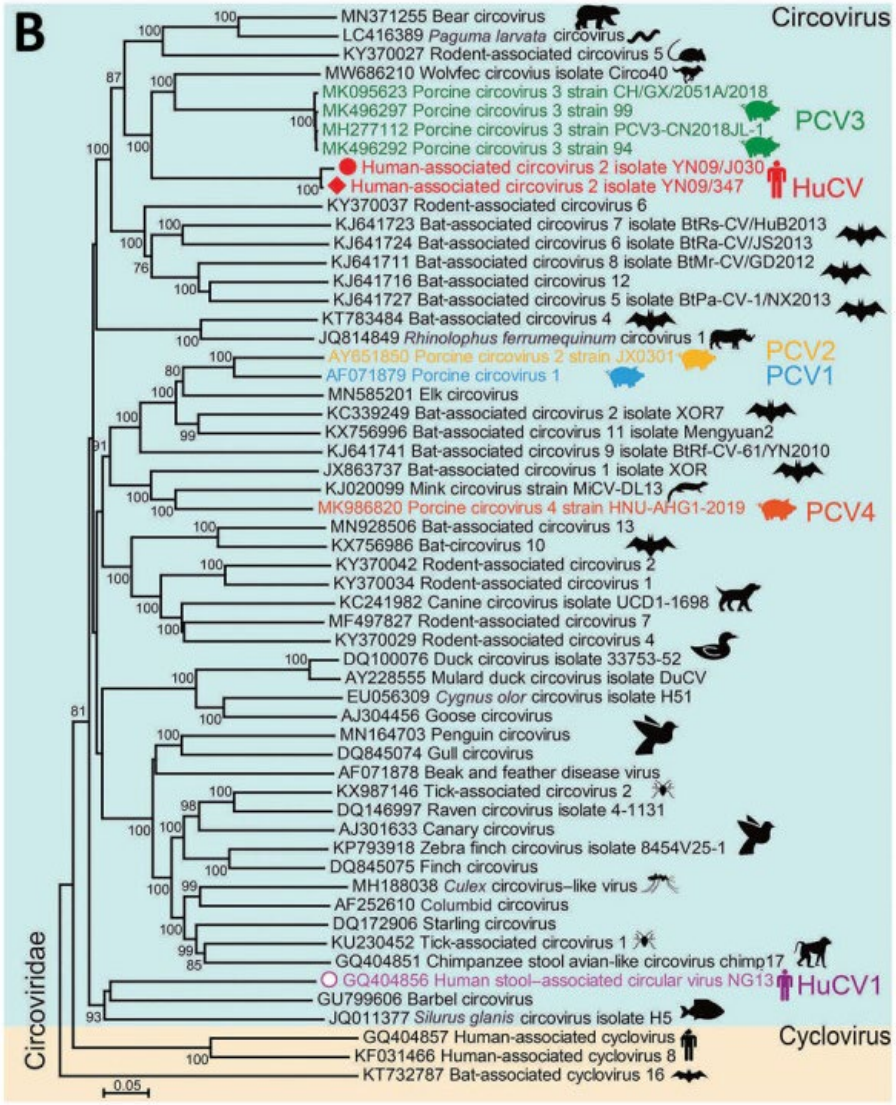
Zoonotic potential

- **PCV2**
  - Detected in several non-porcine species
  - Detected in insects
  - No confirmed zoonotic infection in humans
- **PCV3**
  - Detected in multiple animal species
  - Possible interspecies transmission
  - Human infection not confirmed



Turlewicz-Podbielska et al., 2022

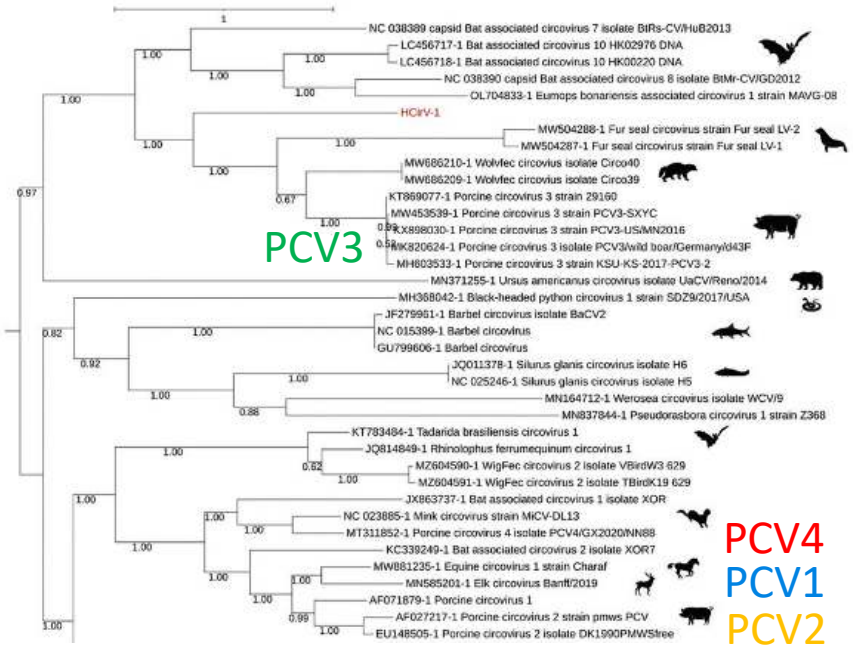
Zoonotic potential



Li et al., 2023

Circovirus Hepatitis Infection in Heart-Lung Transplant Patient, France

Philippe Pérot,[1] Jacques Fourgeaud,[1] Claire Rouzaud,[1] Béatrice Regnault, Nicolas Da Rocha, Hélène Fontaine, Jérôme Le Pavec, Samuel Dolidon, Margaux Garzaro, Delphine Chrétien, Guillaume Morcrette, Thierry Jo Molina, Agnès Ferroni, Marianne Leruez-Ville, Olivier Lortholary, [2] Anne Jamet,[2] Marc Eloit[2]

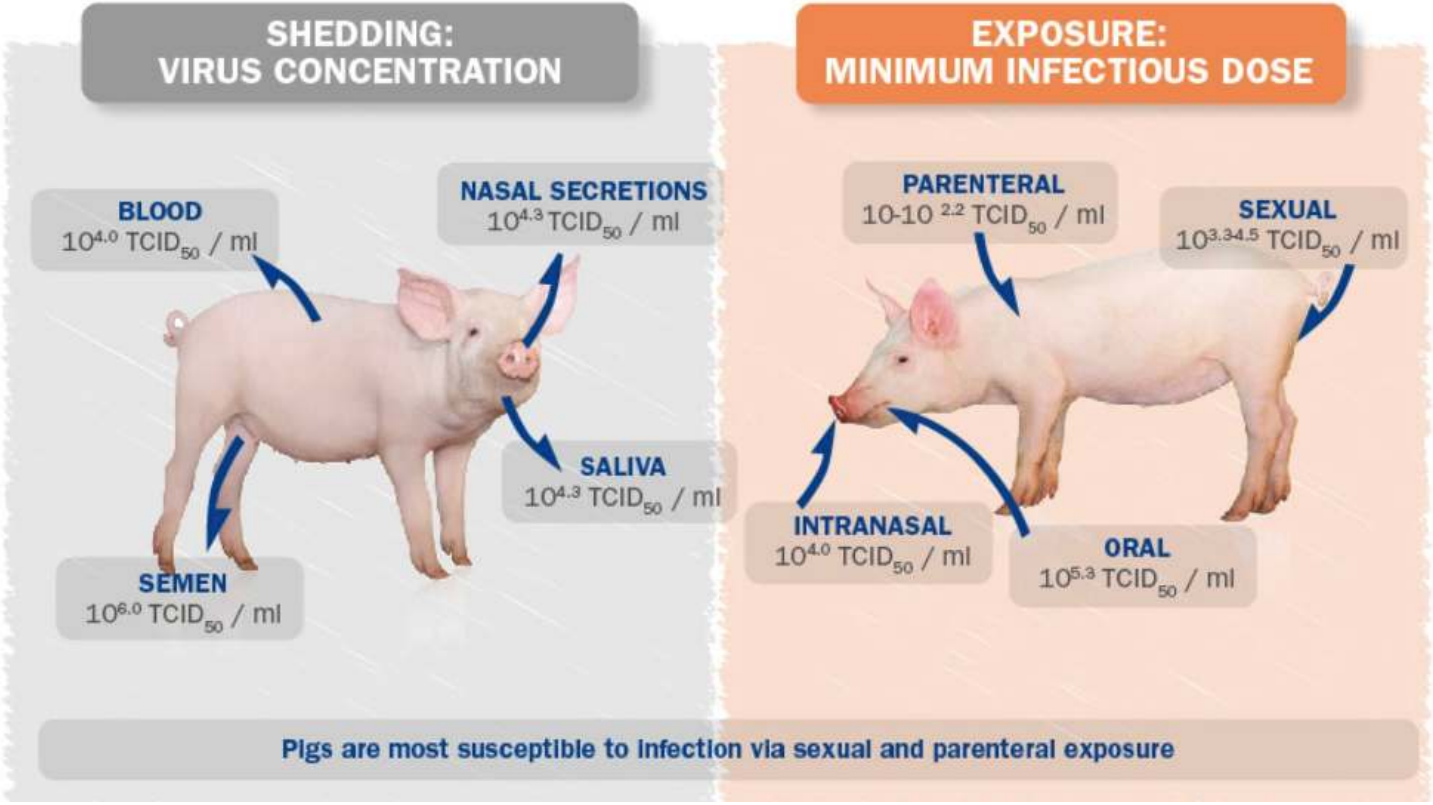


Pérot et al., 2023

Take-home messages

- PCV2 infection is common, but clinical disease develops only in a proportion of infected pigs.
- PCV2-SD is a multifactorial disease, so diagnosis requires compatible clinical signs, characteristic lesions, and PCV2 within the lesions.
- Vaccination greatly reduces PCV2-associated disease and production losses, although infection may still occur.
- PCV2 is well established as a major swine pathogen, whereas PCV3 and PCV4 are still being defined clinically.

Porcine Reproductive and Respiratory syndrome (PRRS)



PRRS

- PRRS, also named “blue ear disease” or “mystery swine disease”, is a widespread viral disease of pigs.
- PRRS was first recognized in US in 1987, and the causative virus was identified in the Netherlands in 1991.
- Symptoms include reproductive failure, pneumonia, and increased susceptibility to secondary bacterial infection.

PRRS

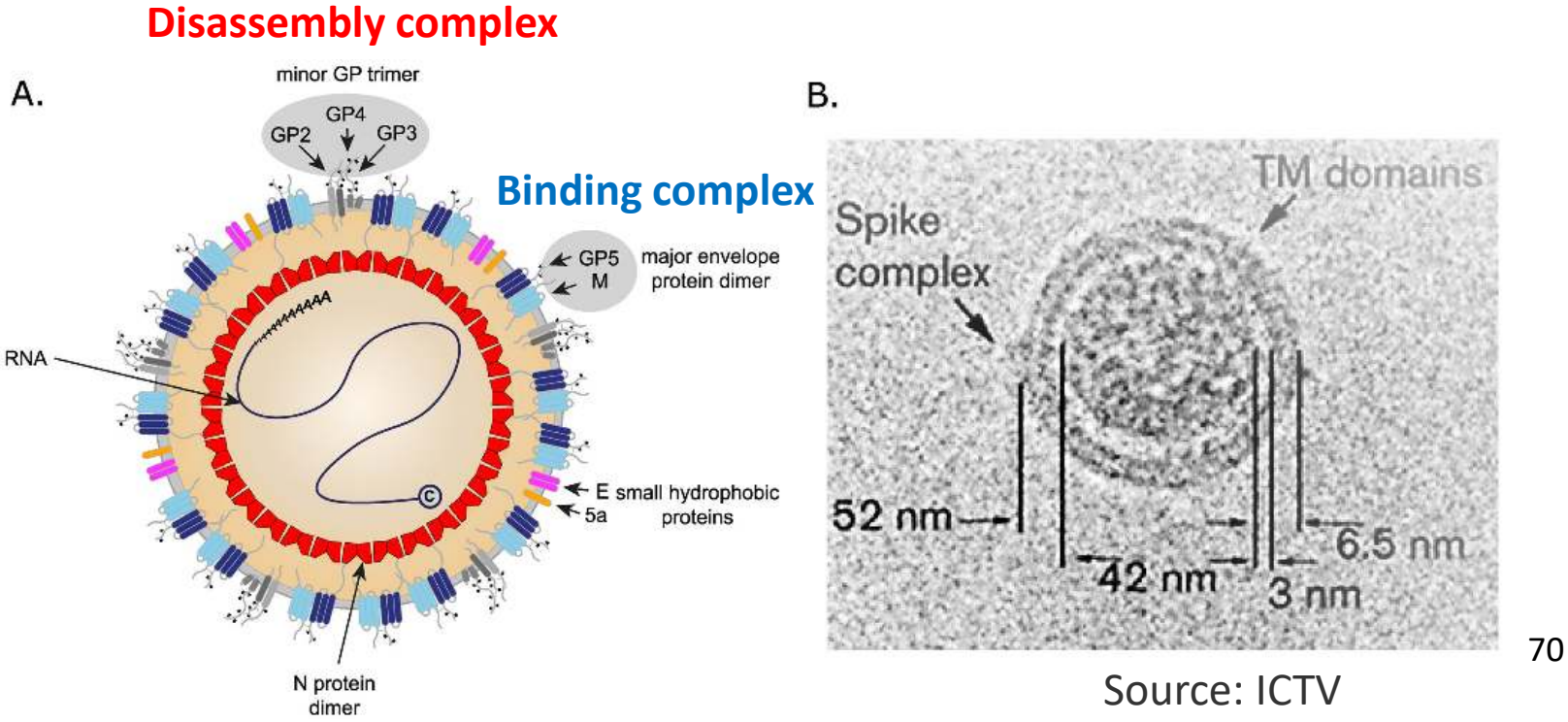
- Multiple strains of the virus with considerable variation in virulence explain the broad spectrum of clinical manifestations of the disease.
- PRRS is one of the most important swine diseases in North America, Europe, and Asia.
- In US, PRRS-associated productivity losses are estimated at approximately US$1.2 billion annually.

Global distribution

- PRRS is widespread in most major pig-producing regions worldwide.
- In Asia, PRRSV has been reported in China, Japan, Korea, Vietnam, the Philippines, Malaysia, and other countries.
- Australia, New Zealand, and India remain free of PRRS.

PRRSV

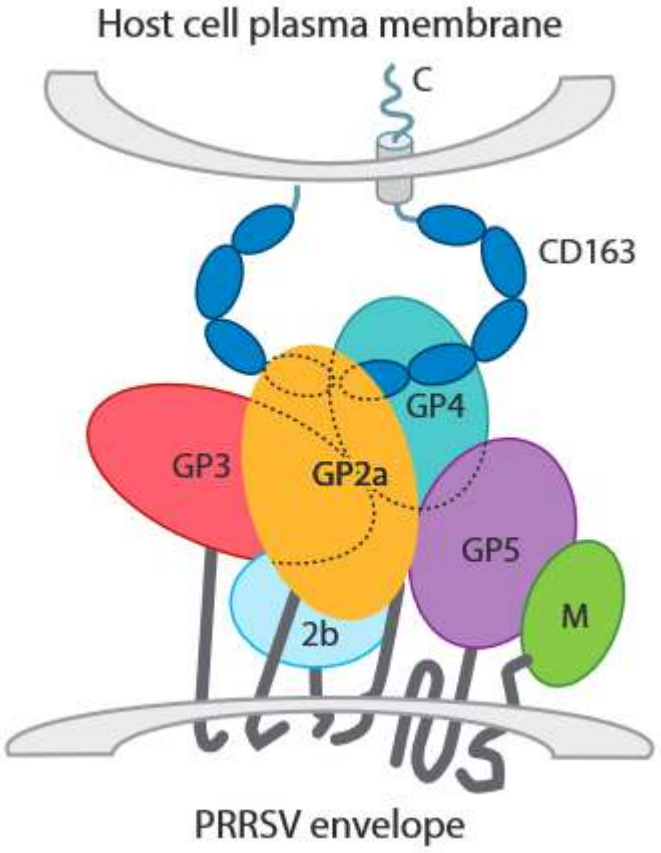
- PRRSV belongs to the family *Arteriviridae* and the genus *Betaarterivirus*.
- An enveloped, single-stranded, positive-sense RNA virus.



Source: ICTV

Structural proteins

- **ORF2, 3, 4, 5, 6, and 7**
  - ORFs 2–5 encode GP2, GP3, GP4, and **GP5**, while ORF6 and ORF7 encode the **M** and **N** proteins, respectively.
  - **GP5**, **M,** and **N** proteins are three major proteins consistently identified in virions and lysates of virus or virus-infected cells.
  - **GP5** is one of the most important structural proteins of PRRSV.



Lunney et al., 2016

PRRSV-1 vs PRRSV-2

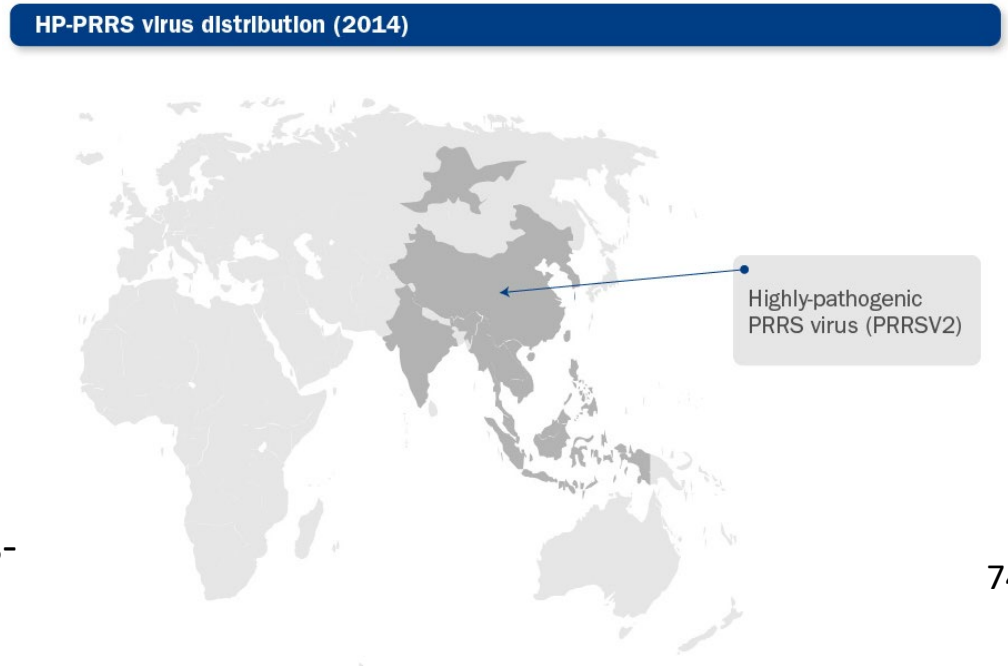
- Historically
  - PRRSV-1: formerly genotype 1 / Type 1 / European type
  - PRRSV-2: formerly genotype 2 / Type 2 / North American type
- Current ICTV classification
  - PRRSV-1 → *Betaarterivirus suid 1*
  - PRRSV-2 → *Betaarterivirus suid 2*
- The two prototype genomes (**PRRSV-1** strain Lelystad and **PRRSV-2** strain VR-2332) were both discovered around 1991 in Europe (**PRRSV-1**) and North America (**PRRSV-2**).
- Today, both species share worldwide distribution, with **PRRSV-1** being predominant in **Europe** and **PRRSV-2** predominant in the **Americas** and **Asia**.

Diversity of PRRSV

- High genetic variability influences virulence, pathogenesis, and vaccine protection
- Approximately 50–70% whole-genome nucleotide identity between the two species
- Numerous lineages and sublineages within each species
- Heterologous protection is variable and often incomplete

HP-PRRS

- In 2006, a **highly pathogenic PRRSV (HP-PRRS)** strain emerged in China
- High fever (40–42°C) in all age groups, abortions in sows, and high mortality in suckling piglets, weaners, and growers
- Morbidity close to 100% and high mortality
- Outbreaks in the East, South, and North of Asia



https://prrscontrol.com/the-prrs-virus/virulent-strains/

PRRS in Taiwan

- **PRRSV-2** was first recognized in Taiwan in 1991, and the virus was isolated in 1992.
- PRRSV-2 has historically predominated in Taiwan.
- The first **PRRSV-1** outbreak in Taiwan was reported in late 2018.

Year	Number	Positive (%)	PRRSV 1 Only (%)	PRRSV 2 Only (%)	PRRSV 1 + PRRSV 2 (%)
2019	8852	2864 (32.4)	108 (3.8)	2747 (95.9)	9 (0.3)
2020	6182	2600 (42.1)	50 (1.9)	2528 (97.2)	22 (0.9)
Total	15,034	5464 (36.3)	158 (2.9)	5275 (96.5)	31 (0.6)

Lin et al., 2020

Physical and chemical characteristics of PRRS virus



THERMAL STABILITY

- One week at 4 °C
- 1 to 6 days at 21 °C
- 3 hrs to 2 days at 37 °C
- 30 minutes at 56 °C



HUMIDITY

Drying conditions » Virus loses 90% of its infectivity



pH

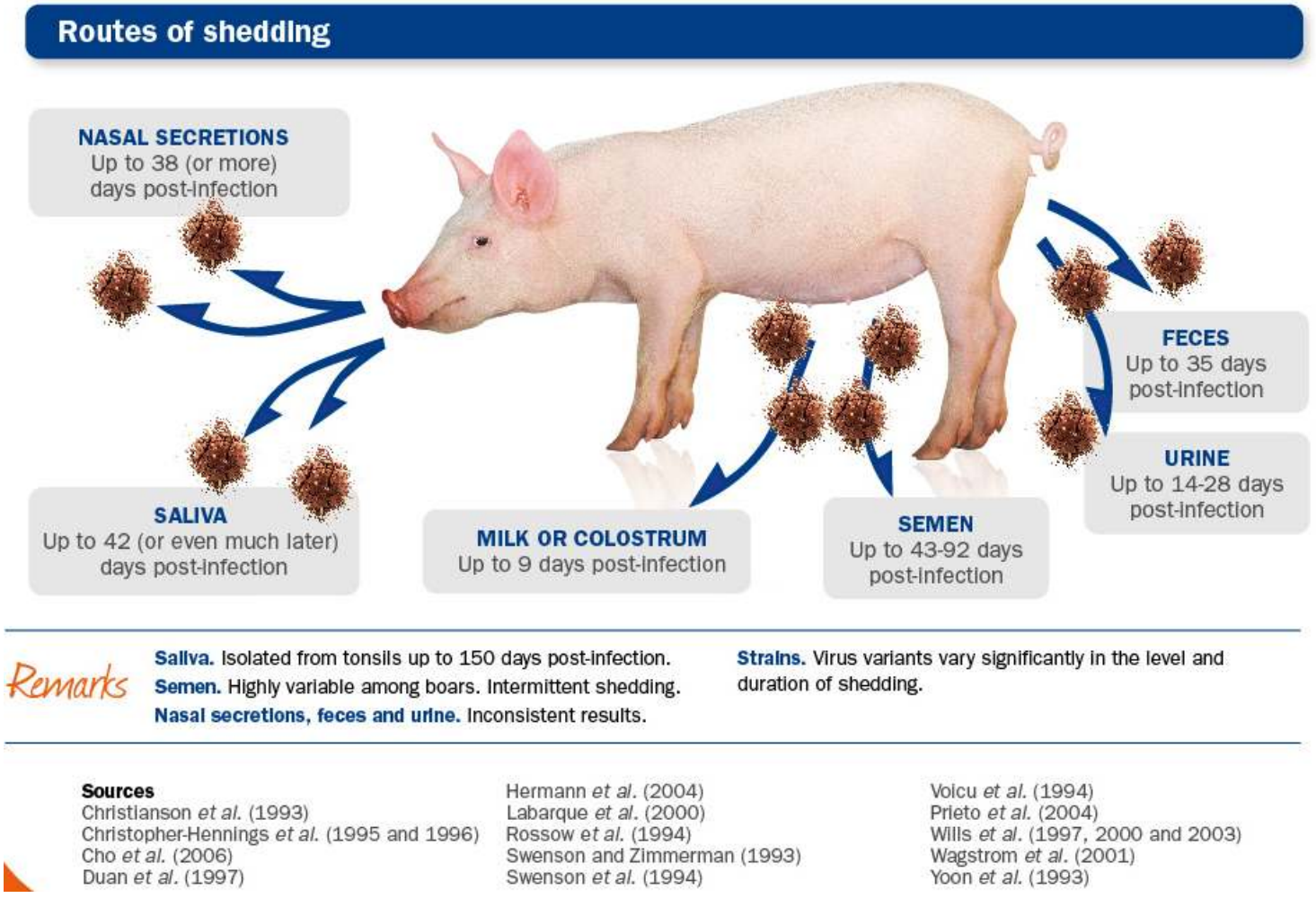
pH below 6 or higher than 7.5 » Virus loses 90% of its infectivity

https://prrscontrol.com/the-prrs-virus/physical-and-chemical-characteristics/

Transmission

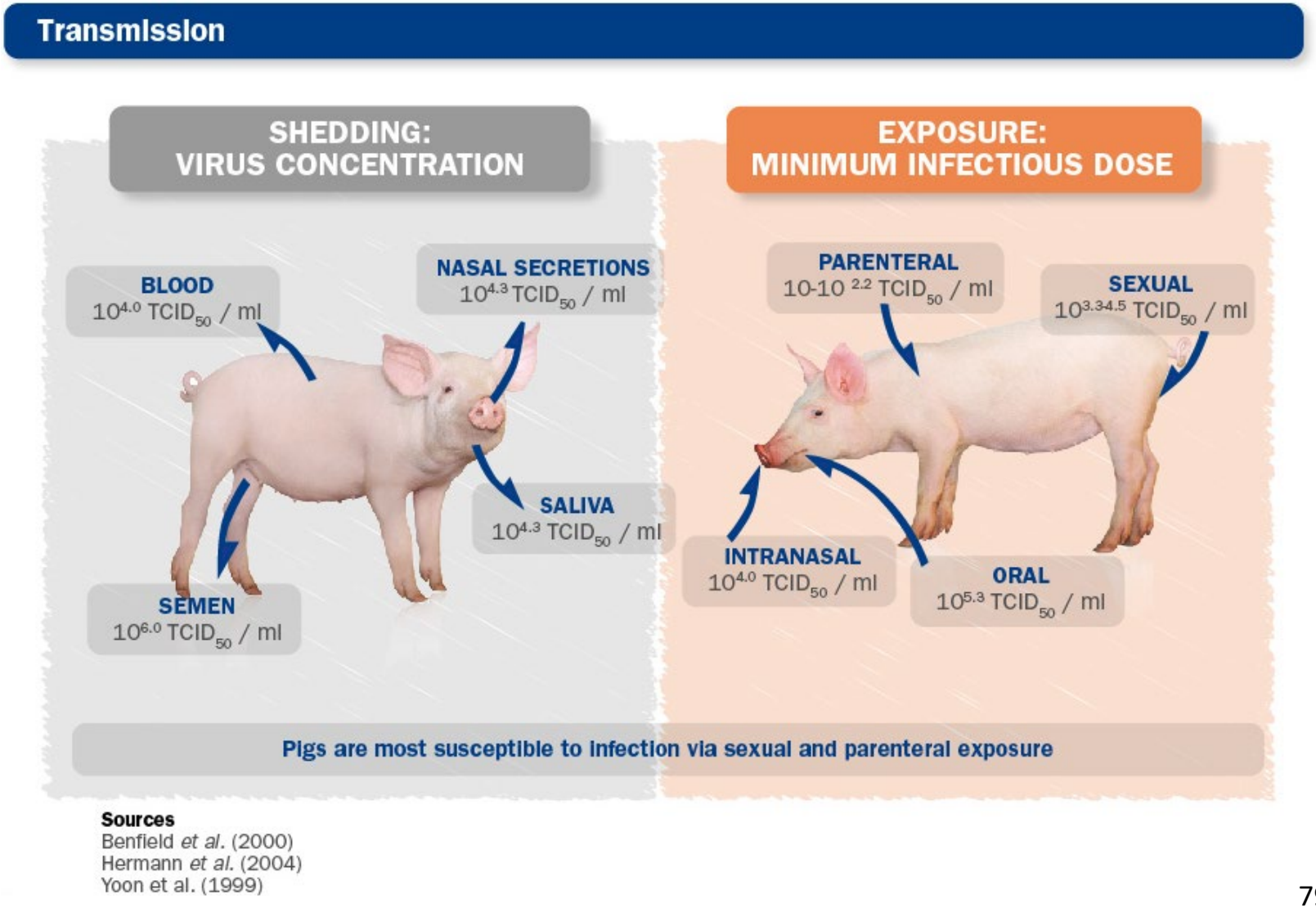
- Mainly transmitted by close contact between infected and susceptible pigs
- Virus shed in saliva, nasal secretions, urine, feces, and semen.
- Nose-to-nose contact and contaminated excretions are important.
- Transmission through contaminated semen and artificial insemination can occur.

Transmission



http://prrscontrol.com/epidemiology/transmission-within-herds/

Transmission



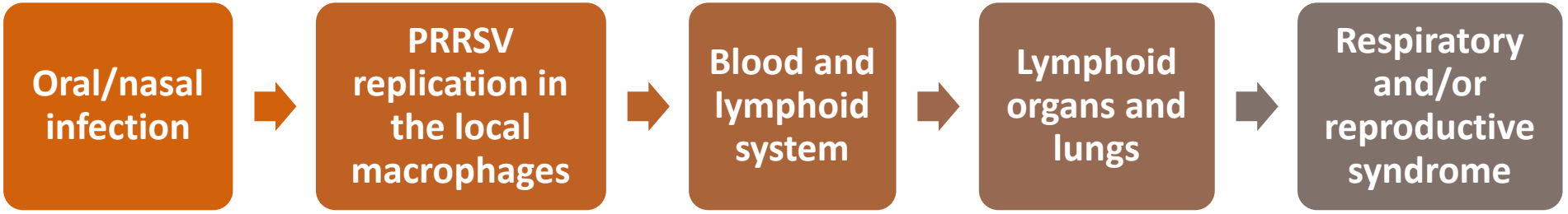
Sources
Benfield *et al.* (2000)
Hermann *et al.* (2004)
Yoon et al. (1999)

http://prrscontrol.com/epidemiology/transmission-within-herds/

Transmission

- Movement of infected pigs — major route of between-herd transmission
- Clinically normal pigs — possible carriers and shedders
- Introduction of weaners and replacement breeding stock — important risk factor

PRRS pathogenesis



- Target cell: Macrophages
  - Pulmonary alveolar macrophages (PAMs)
  - Pulmonary intravascular macrophages (PIMs)
  - Macrophages in lymphoid tissues and respiratory mucosa
  - Macrophages at the maternal–fetal interface during pregnancy

Viremia of PRRSV infection

- **Viremia**
  - 12 hpi: detectable in some pigs.
  - 24 hpi: detectable in most or all experimentally infected pigs.
  - Highest viral loads generally occur during the first 1–2 weeks.
  - Infectious viremia usually declines markedly by about 4 weeks.
- **Persistent infection**
  - Viral RNA may remain detectable for prolonged periods.
  - PRRSV can persist in tonsils and other lymphoid tissues for several months.
  - Persistent PCR positivity does not necessarily indicate ongoing infectious viremia.

Distribution of PRRSV

- Lungs and lymph nodes contain the highest amounts of PRRSV antigen and/or nucleic acid.
- PRRSV antigen and/or nucleic acid can also be detected in perivascular and intravascular macrophages in the heart, brain, kidneys, and other tissues.

Clinical signs

- Acute PRRSV infection in **nursery pigs** is characterized by anorexia, lethargy, cyanosis, rough hair coats, hyperpnea and/or dyspnea without coughing.
- PRRSV infection can cause variable reductions in average daily gain, resulting in uneven growth within a group.



Clinical signs

- **Reproductive failures in sows**
  - Infertility, abortions, stillbirths
  - Litters with normal, weak, stillborn, and mummies
  - The birth of weak piglets that often die soon after birth from respiratory disease and secondary infections

- **Respiratory disease in young pigs**
  - Younger pigs are more affected than older animals
  - Dyspnea
  - Poor growth
  - Red discoloration of the body
  - Blue ears

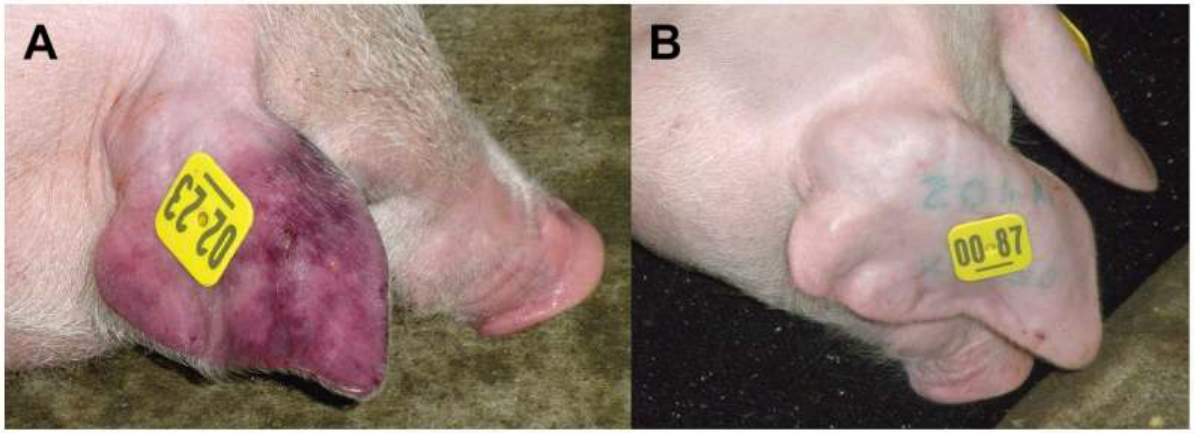
Reproductive disorders



Clinical signs

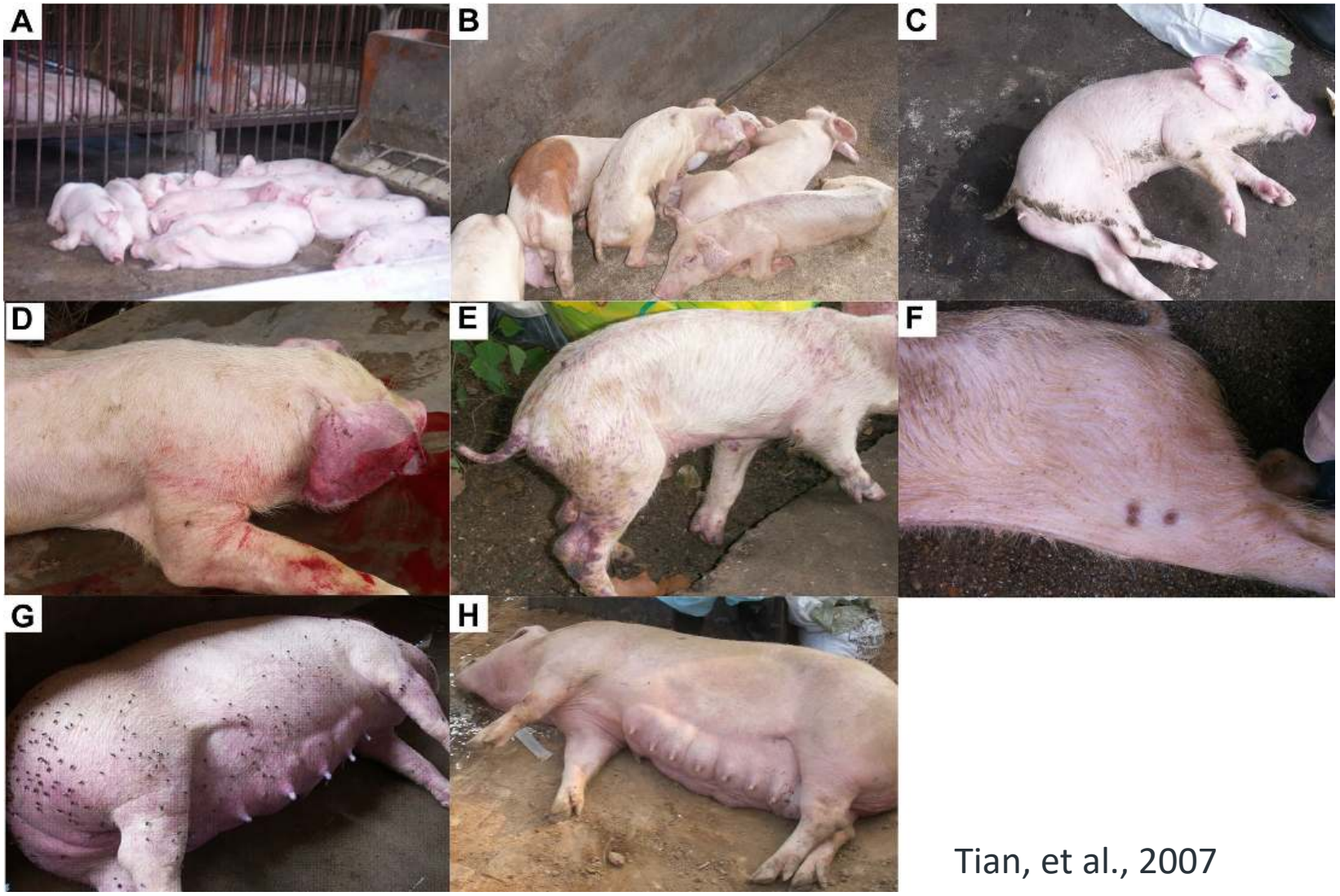


- Cutaneous signs may occur
  - Hyperemia or cyanosis of extremities
- Neurologic signs are rarely observed in sows.
- Reproductive, respiratory, and cutaneous manifestations resemble those of equine viral arteritis in horses.



Niederwerder, et al., 2015

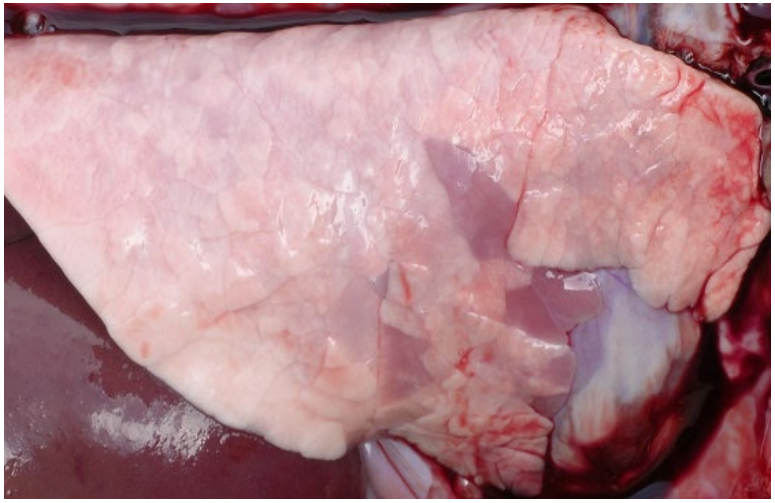
Clinical signs — HP-PRRS

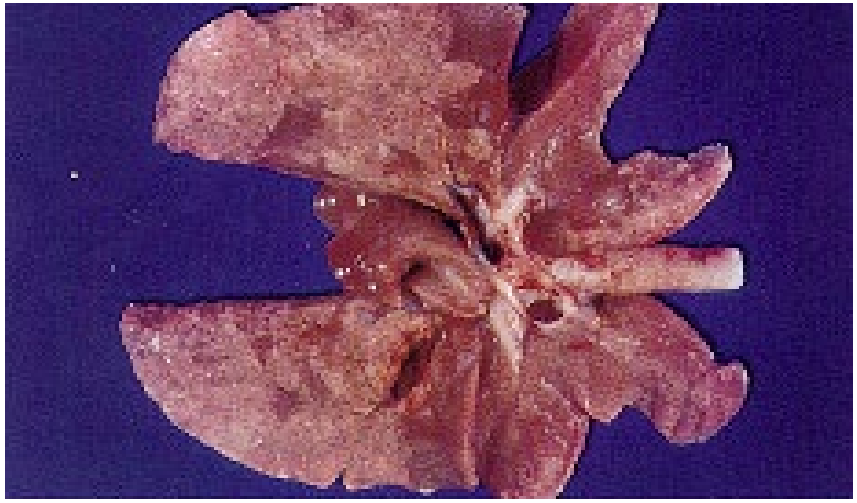


Tian, et al., 2007

Gross lesions

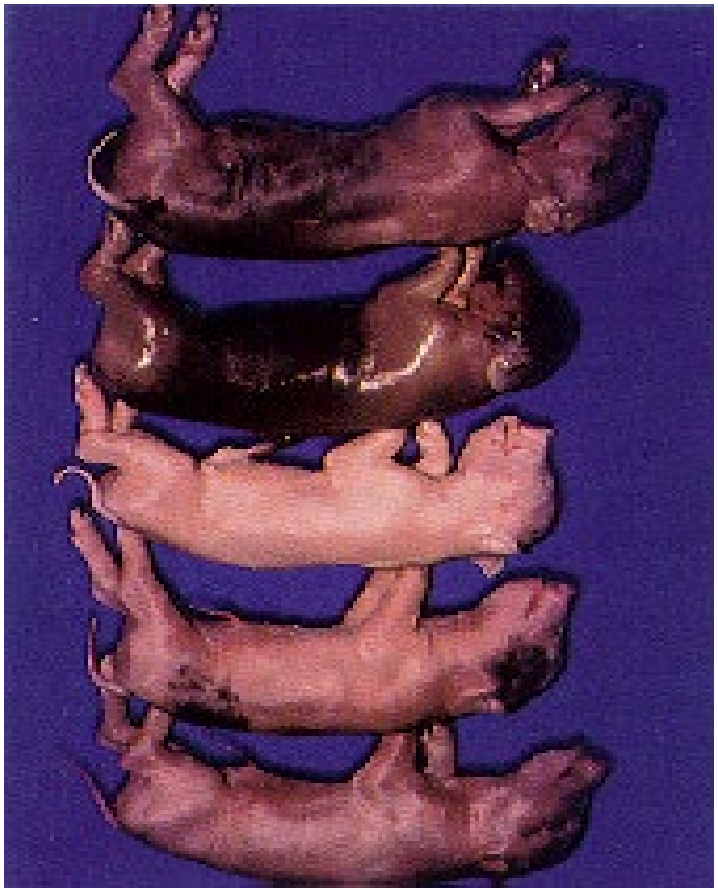
- In severe disease, lungs are mottled, tan and red, and fail to collapse.
- Lymph nodes are moderately to severely enlarged and tan in color and, for some strains of viruses, may be hemorrhagic.
- Under field conditions, most PRRSV-infected pigs are co-infected with one or more pathogens, which complicates the diagnosis of PRRS based on pathology.





Mottled, tan and red lungs

Reproductive disorders



Aborted fetuses

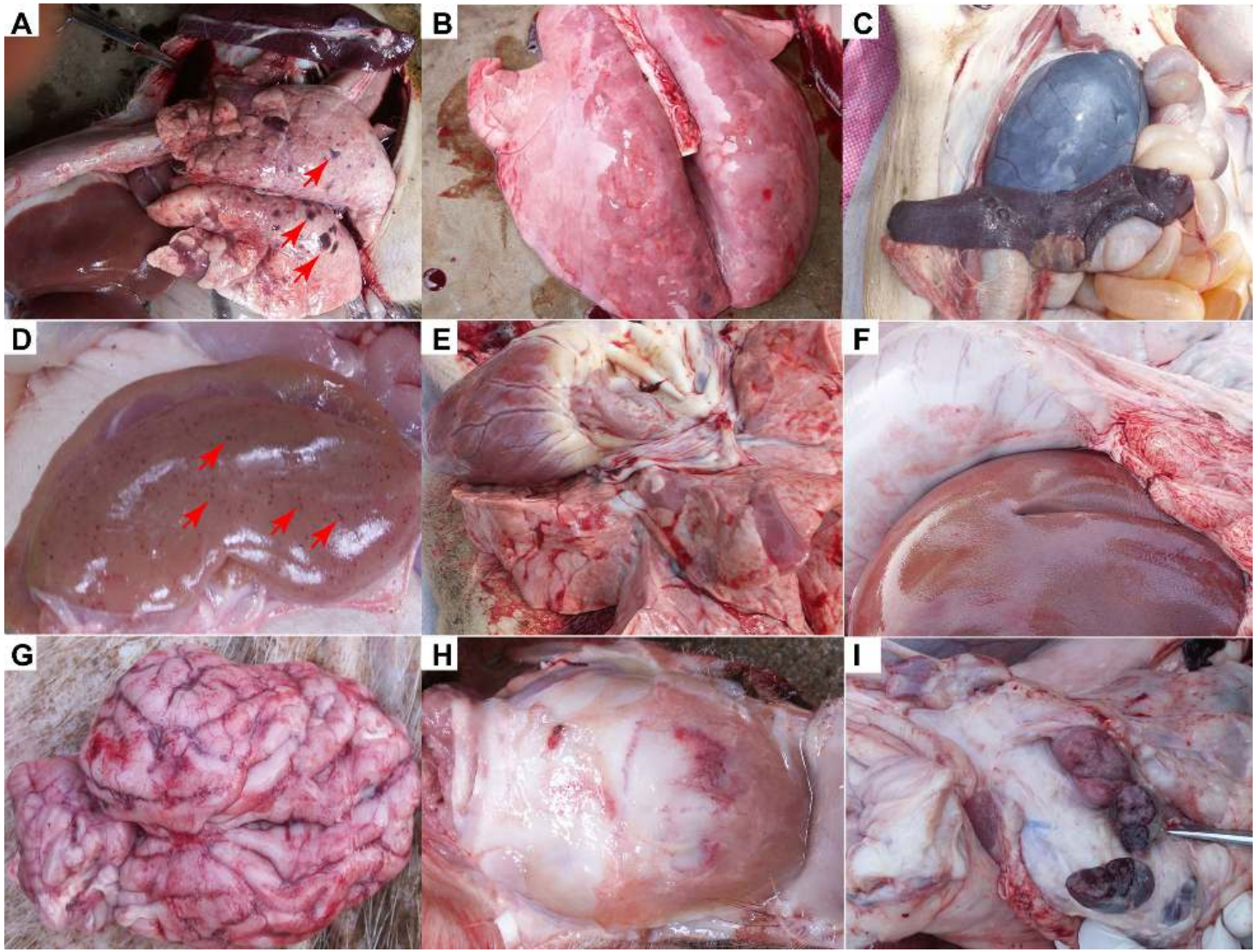
Stevenson, 1998



Umbilical lesions

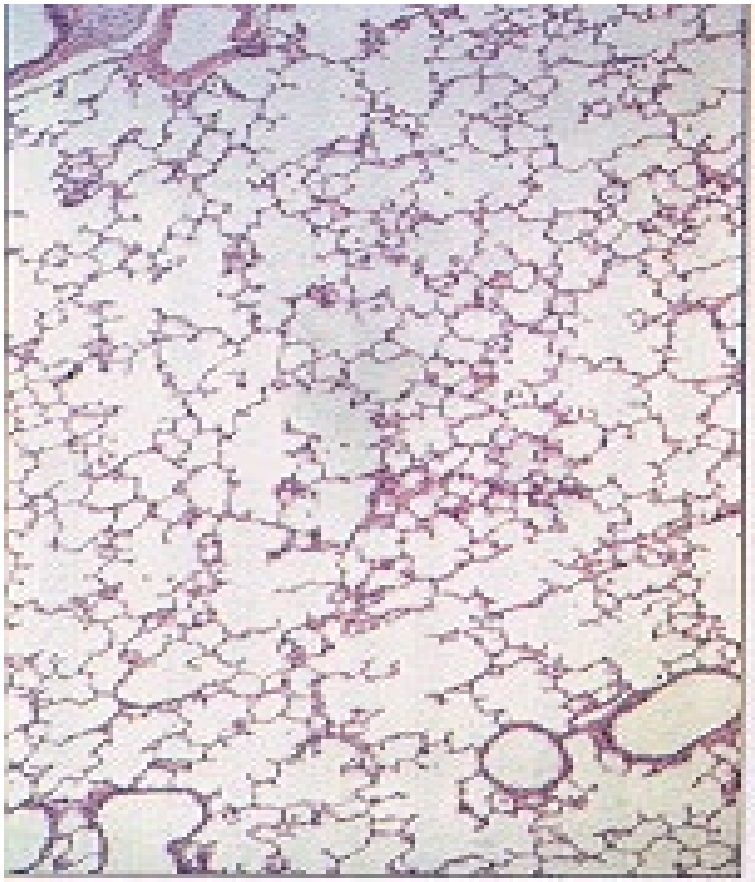
Lagar, 1998

Gross lesions — HP-PRRS

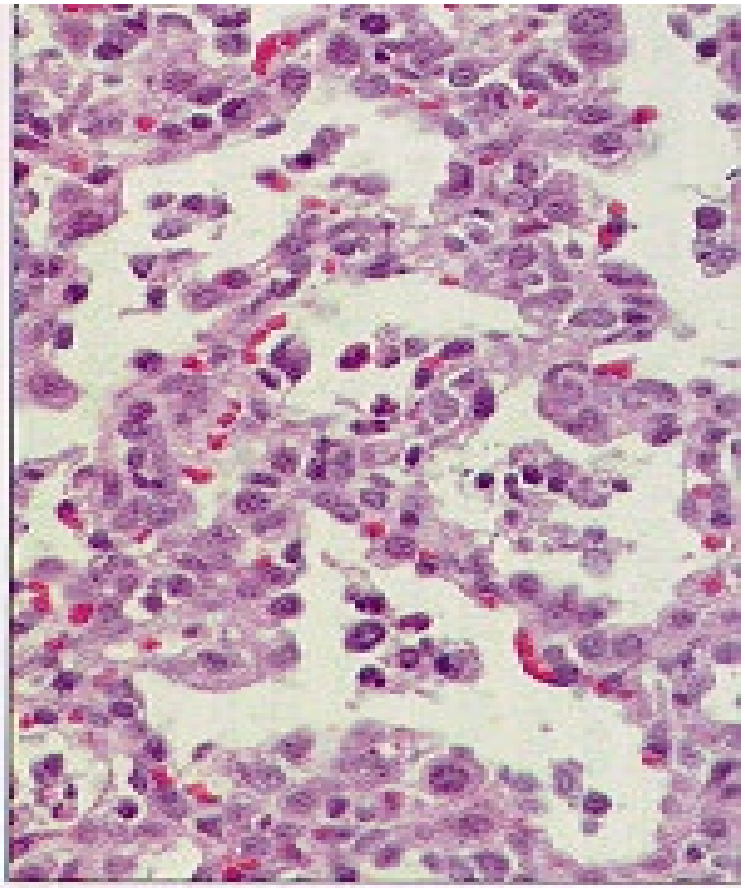


Tian, et al., 2007

Microscopic lesions



Normal



Interstitial pneumonia

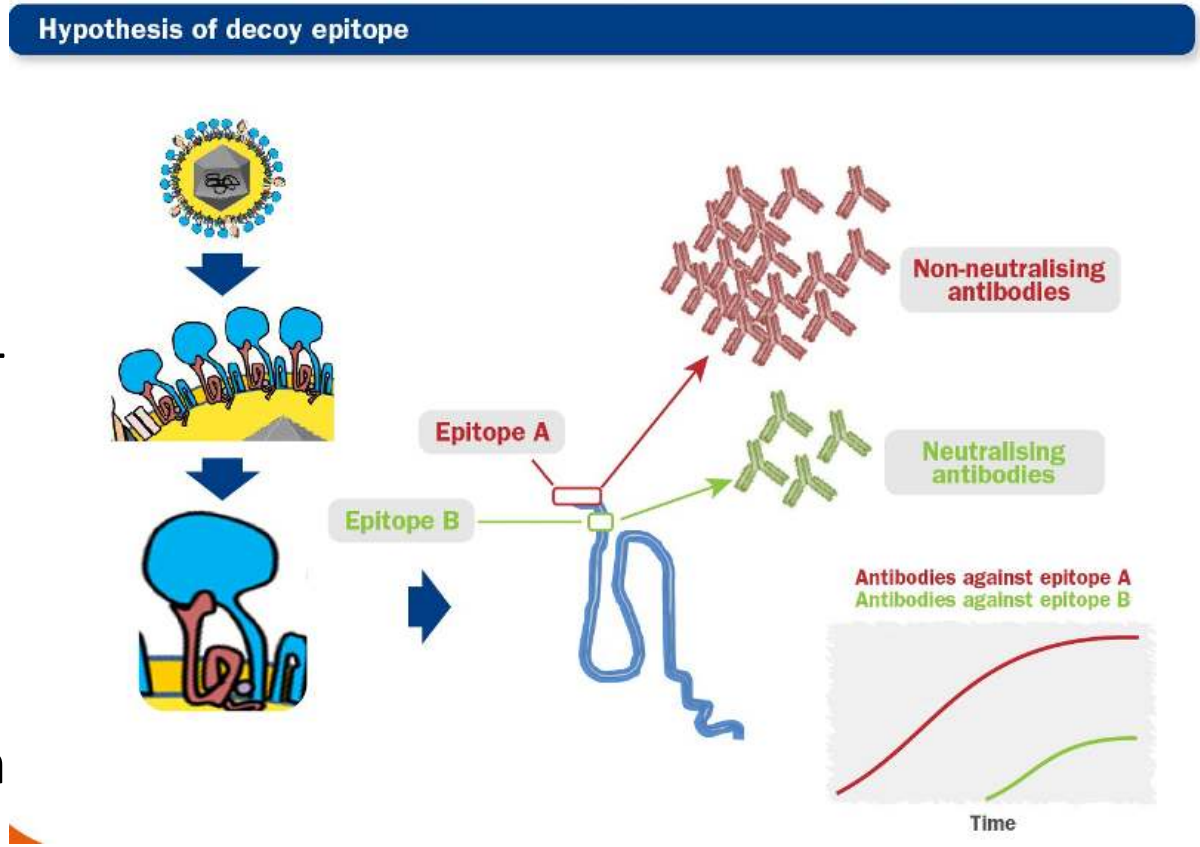
Halbur, 1998

Humoral immune response

- **Humoral**
  - Rapid onset of non-neutralizing antibody
    - 7-14 days: High titer of the specific antibody can be detected.
  - Neutralizing response is slow
    - 3-4 wks: Neutralizing antibodies begin to appear, usually at low titers.
  - The early antibody response is dominated by non-neutralizing antibodies, particularly against the **N** protein.
  - Neutralizing antibodies develop later and target mainly envelope glycoproteins, including **GP5**.

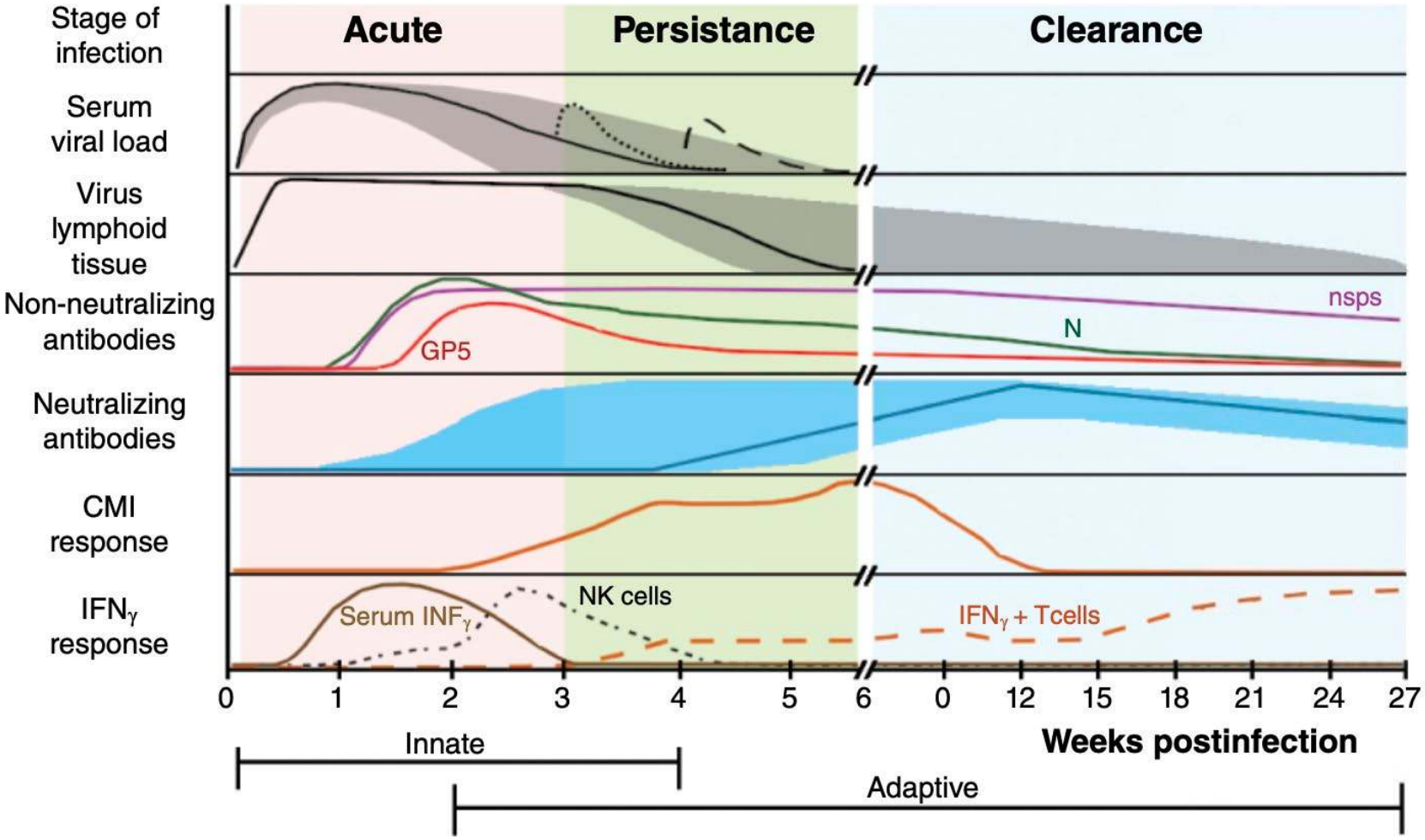
Production of neutralizing antibodies of PRRS infection

- **Epitope A**: induced a **high** titer immune response in the **early** stage after infection (14 dpi).
- **Epitope B**: induced a **low** titer immune response in the **late** stage (after 21 dpi) with neutralizing ability.



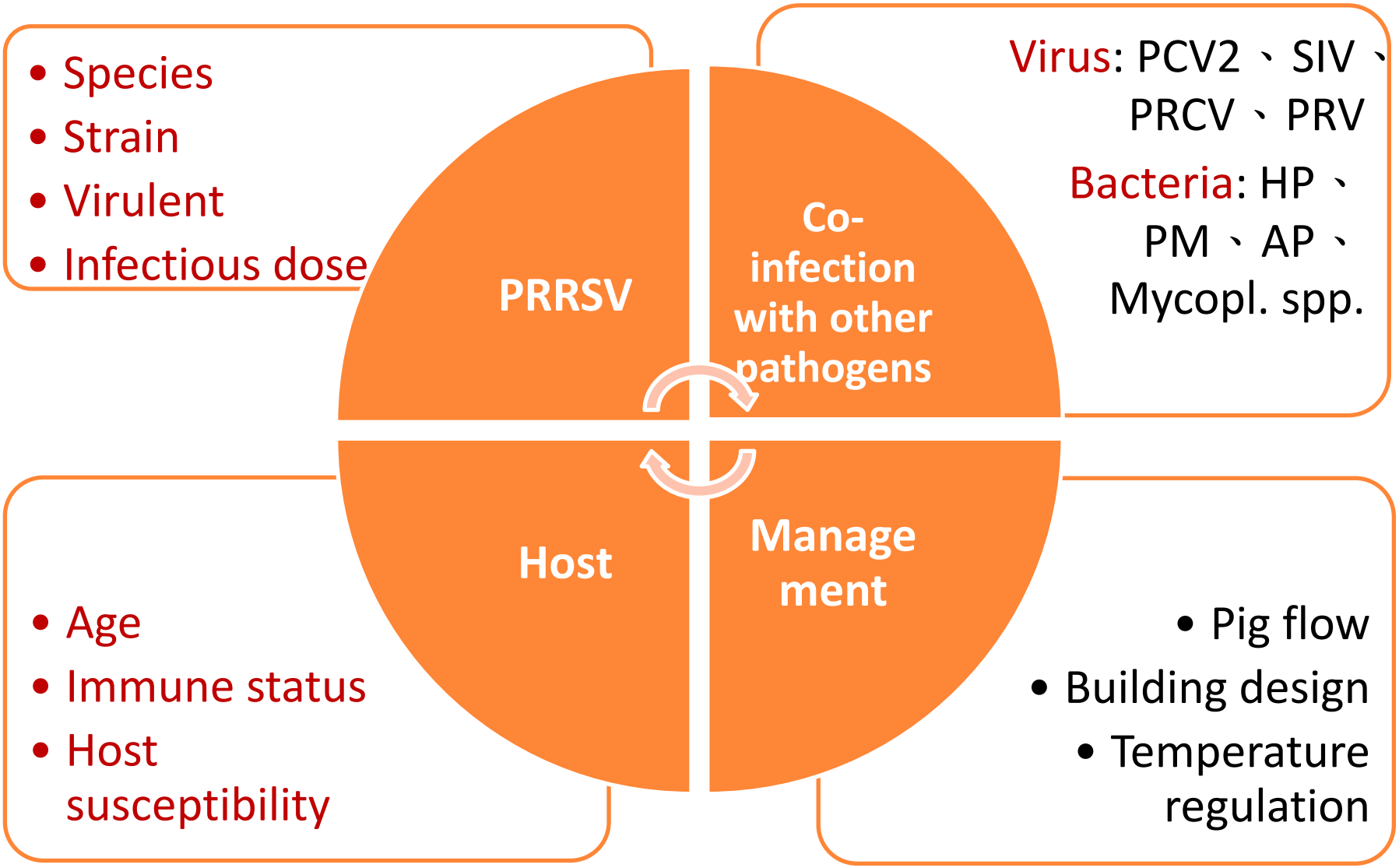
https://prrscontrol.com/immunology/why-nas-does-not-develop-as-strongly-and-early-as-non-neutralising-antibodies/

Viral dynamics and immunity at different stages of PRRSV infection



Diseases of Swine, 12th Edition

Influence factors of PRRS

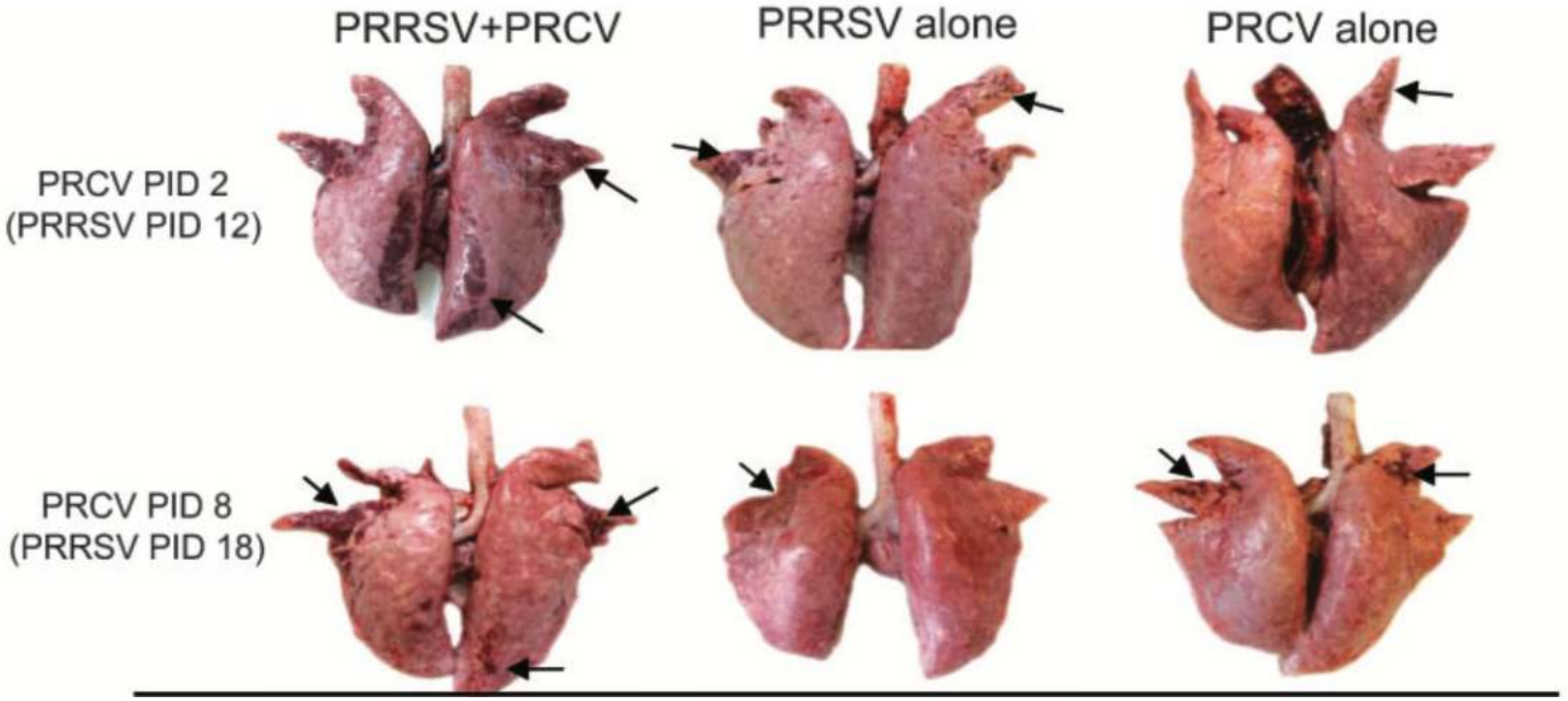


Interaction of PRRSV and other pathogens

- Classical swine fever virus
- Porcine circovirus 2
- Porcine respiratory coronavirus
- *Mycoplasma hyopneumoniae*
- Porcine parvovirus
- *Streptococcus suis*
- Other viral and bacterial pathogens

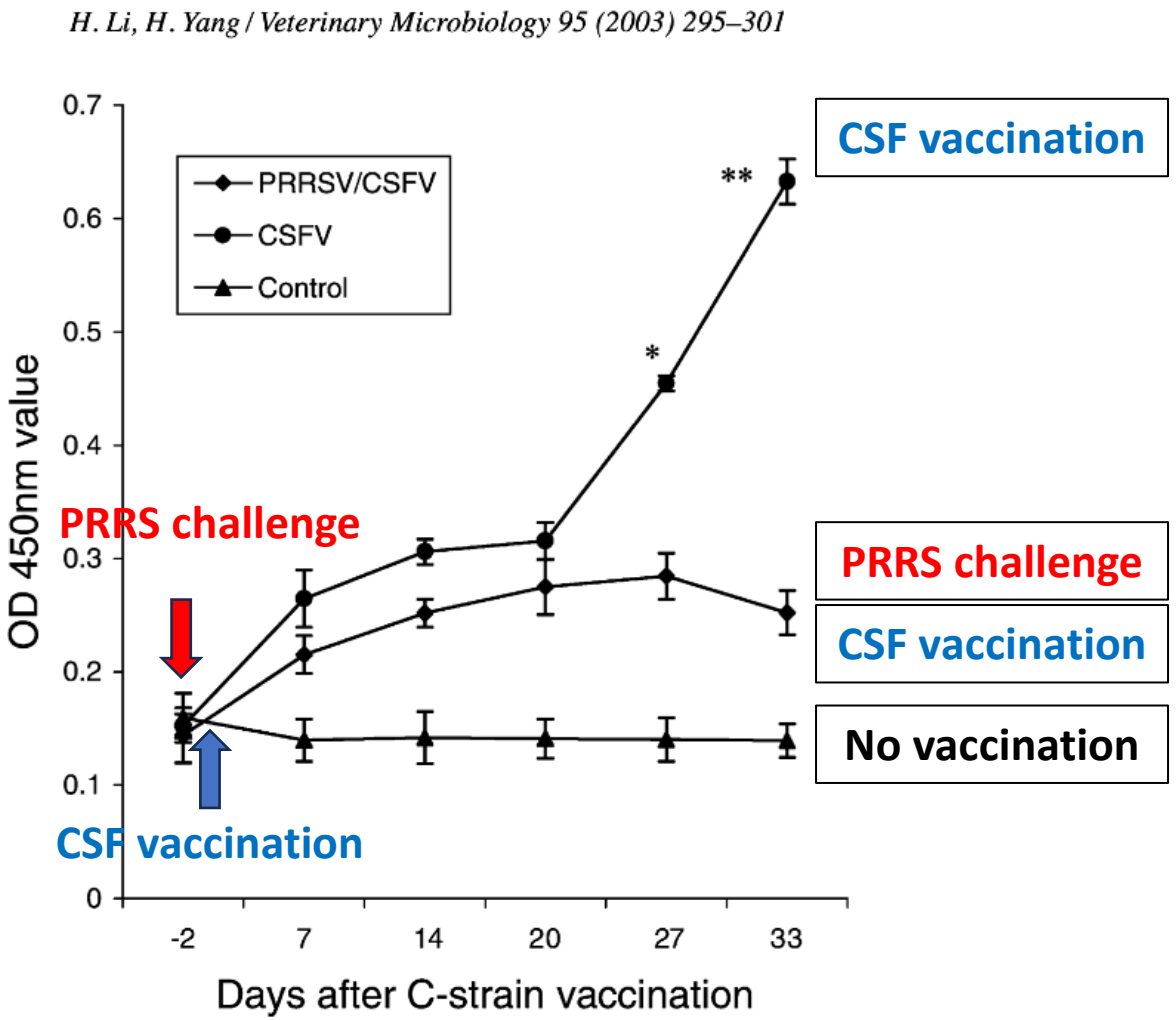
Co-infection with other pathogens

- Co-infection with Porcine respiratory coronavirus (PRCV)



Jung et al., 2009

PRRS infection decreases the efficacy of attenuated CSFV vaccine



Persistent infection of PRRSV

- Inapparent infection is common.
- Infection may be prolonged or persistent.
- PRRSV may persist for several months after infection.
- Virus shedding can be intermittent.
- Persistently infected pigs may contribute to virus circulation, reinfection within the herd, and environmental contamination.
- **PRRSV is difficult to eliminate from infected herds**

PRRSV evolution

- PRRSV evolves at an unusually rapid rate, even for an RNA virus.
- PRRSV evolves at approximately 10^{-2} substitutions/site/year, about 1–3 orders of magnitude faster than the rates reported for many other RNA viruses (10^{-3}–10^{-5} substitutions/site/year).
- This rapid evolution may result from more frequent viral replication rather than a higher replication error rate than that of other RNA viruses.

Diversity of PRRSV

- PRRSV exists as a quasispecies.
- Genetic variation occurs among farms, among pigs within a farm, and within individual pigs.
- Distribution of genetic variation:
  - Among farms: ~93%
  - Among pigs within farms: ~4%
  - Within individual pigs: ~3%
- Multiple viral variants can coexist within an individual pig.
- In one study, no two pigs from the same farm had identical PRRSV sequences.

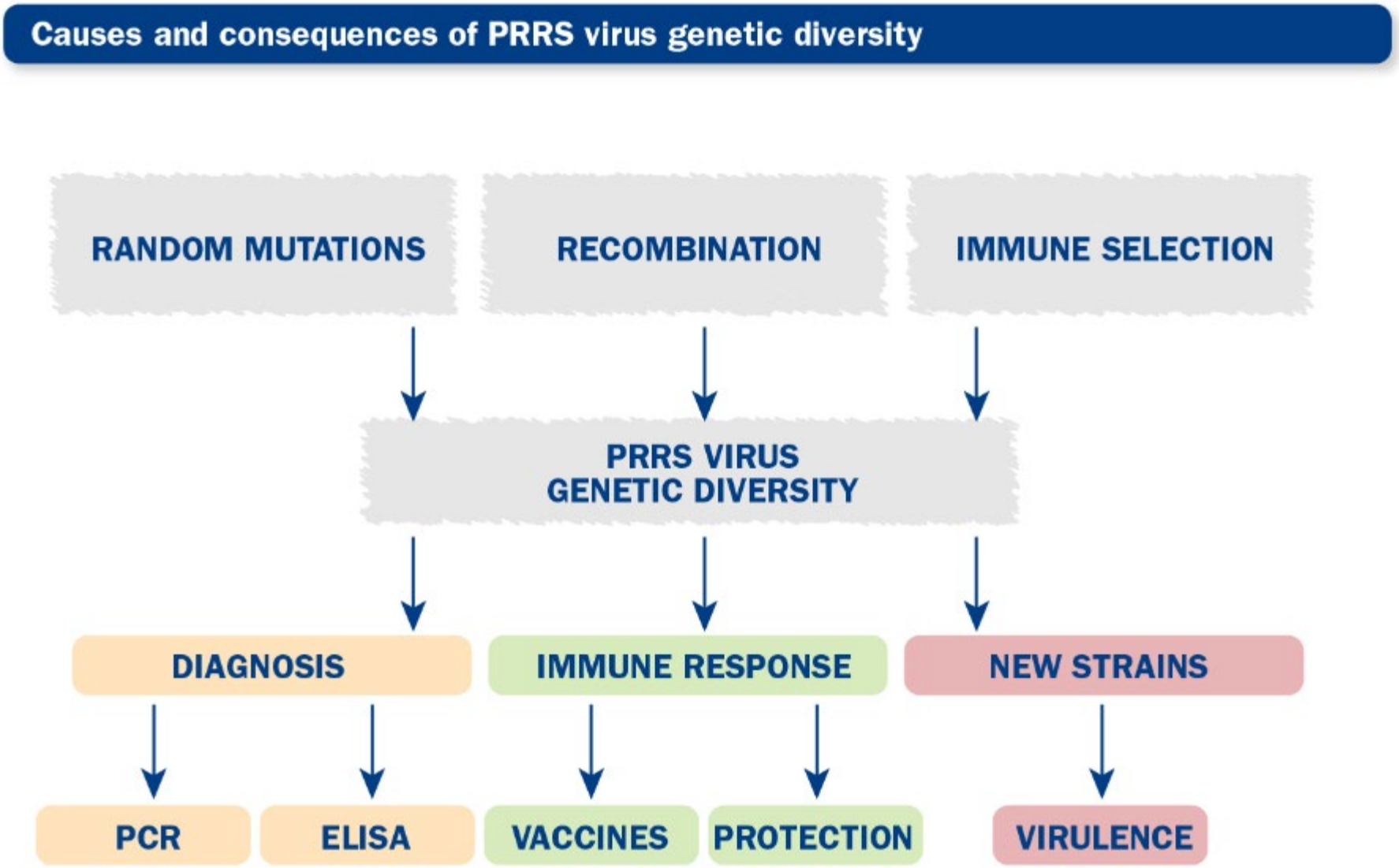
Goldberg et al. 2003

Influence by viral evolution

- Changes in disease patterns and virulence
- Changes in cell tropism
- Facilitation of persistent infection
- Challenges for disease control

How does the rapid generation of genetic variants contribute to the PRRS

- Frequent opportunities for the virus to evade host immune responses.
- New variants may cause disease in susceptible subpopulations within an otherwise immune herd.
- Persistent PRRSV may continue to evolve and potentially escape existing host immunity, leading to rebound viremia or recurrent disease.



https://prrscontrol.com/the-prrs-virus/causes-of-prrs-virus-genetic-diversity/

Differential diagnosis

Reproductive disease	Respiratory and postweaning disease
– classical swine fever	– swine influenza
– African swine fever	– enzootic pneumonia
– leptospirosis	– proliferative and necrotising pneumonia
– porcine parvovirus	– *Haemophilus parasuis* infection
– porcine enterovirus	– haemagglutinating encephalomyelitis virus
– haemagglutinating encephalomyelitis virus	– porcine respiratory coronavirus
– Aujeszky's disease	– syncitial pneumonia and myocarditis
	– porcine circovirus-associated disease
	– Nipah virus infection

Source: WOAH

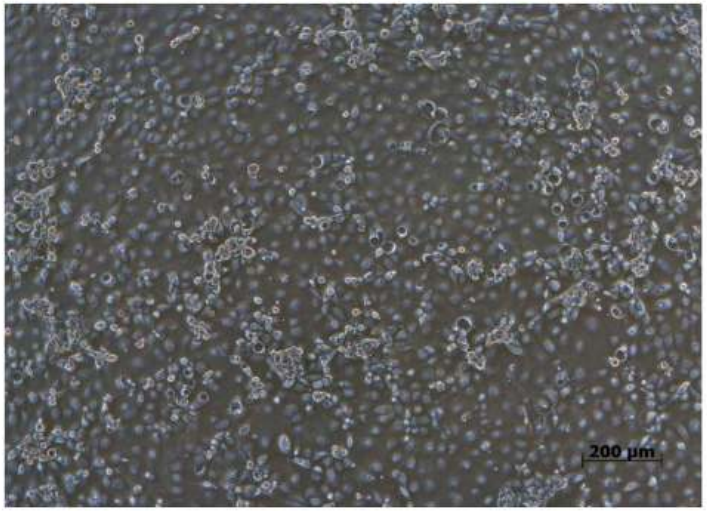
Diagnostic samples

- **Serum or whole blood** – acute infection, RT-PCR
- **Lung, tonsil, and lymph nodes** – necropsy and persistent infection
- **Bronchoalveolar lavage (BAL) fluid** – lower respiratory tract infection
- **Semen** – monitoring of breeding boars
- **Oral fluid** – convenient for herd-level surveillance

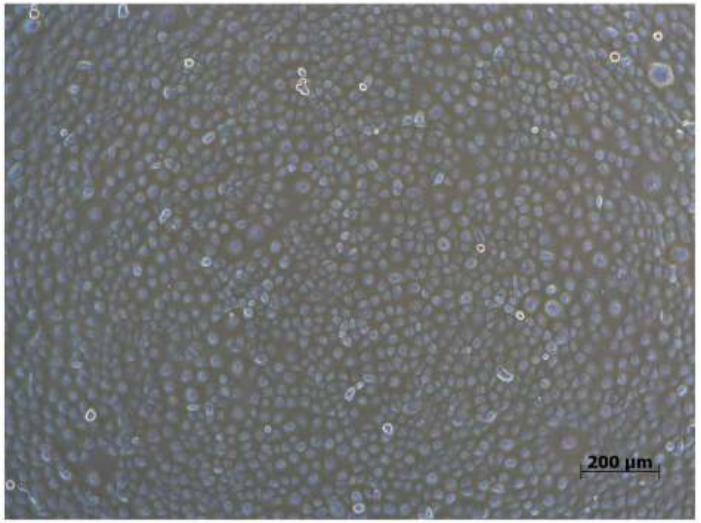
Diagnosis

- Detection of the agent
  - Virus isolation
  - Detection and characterization of viral nucleic acids





MARC-145 cells cultured with PRRSV



Uninfected MARC-145 cells

Lin et al., 2020

Diagnosis

- Detection of antibody
  - Commercial ELISA
  - Virus neutralizing test



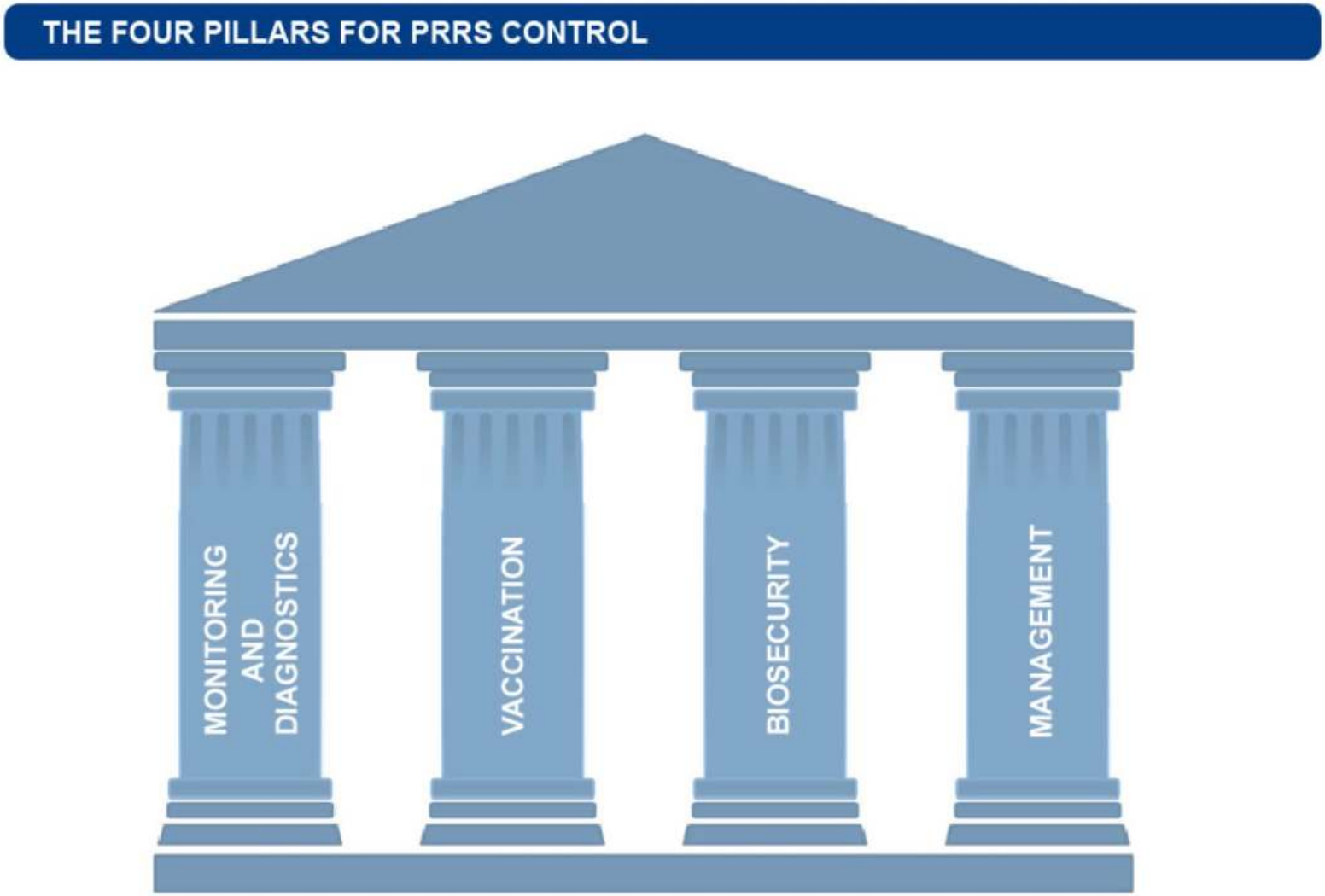
Diagnosis

Interpretation of the results

ASSAY	RESULT	PROBABLE SCENARIOS[1]
RT-PCR	Negative	a) Negative animal.
ELISA	Negative	b) A still-infected pig (non-viraemic) or a pig that has cleared the infection and becomes seronegative (≥6-12 months post-infection).
RT-PCR	Negative	a) Pig may still be infected, but it is not viraemic. Infection occurred some time ago (from days to months)[2].
ELISA	Positive	b) Piglet PRRSV negative that half antibodies from calostrum.
		c) Vaccinated pigs.
RT-PCR	Positive	a) Pig has recently infected / vaccinated (<14 days) and it has not yet seroconverted.
ELISA	Negative	
RT-PCR	Positive	a) Pig is still viraemic and has already seroconverted (>14 days).
ELISA	Positive	b) Re-infection of a pig with pre-formed antibodies from a previous infection/vaccination.

https://prrscontrol.com/diagnostics-and-monitoring/practical-approach-to-diagnosis/

Prevention and Control



https://prrscontrol.com/vaccinology/types-of-vaccines-against-prrs/

Prevention and Control

- In PRRS-endemic countries or regions
  - Farm-level biosecurity is essential to prevent the introduction of new PRRSV strains.
  - The PRRSV status of replacement gilts and boars should be known before introduction.
  - Incoming animals should undergo proper quarantine and acclimatization.
  - Animals should be tested before or upon entry into quarantine and retested before introduction into the breeding herd.

Prevention and Control

- When PRRS occurs on a farm
  - Depopulation and repopulation may be used to eliminate the virus.
  - Thorough cleaning and disinfection and proper disposal of carcasses are essential before restocking.
  - Multisite production, segregated early weaning, and test-and-removal strategies have also been used for PRRS control or elimination.

Vaccination

- Commercial vaccines generally provide good virological and clinical protection against genetically **homologous** strains.
- However, current vaccines do not provide complete or universal protection.
- Protection against **heterologous** strains is variable and often incomplete.





Compared with PRRSV vaccines

Types of Vaccine	Advantage	Disadvantage
MLV vaccine	Strong immunogenicity	Low safety; cannot provide complete protection
Inactivated vaccine	Convenient storage; high security; maternal antibody interference is relatively weak; the vaccine can be used in pigs with weakened immunity	Poor efficacy; multiple vaccinations are needed to strengthen immunity
Subunit vaccine	No active ingredients; high security; the vaccine can be used in pigs with weakened immunity	High production costs; long development time; low immunogenicity
Live vector vaccine	Good immune effect; high-concentration expression of antibodies can be achieved after vaccination; a single dose is sufficient to stimulate long-term protection	Vector viruses may also trigger immune responses, thereby reducing vaccine effectiveness; low production efficiency and high cost
DNA vaccine	The preparation process is simple; the production cycle is short; high security	Low immunogenicity; insufficient resolution of the heterogeneity of PRRSV
Gene-deletion vaccine	Good safety; long immune period, especially suitable for local vaccination	Difficulty in preparation; high cost
Synthetic peptide vaccine	Good safety and stability; low production cost	The protective ability of a single linear antigen epitope may not be good for pigs; low immunogenicity
VLP vaccine	Strong immunogenicity; low immune dose	Difficulty in preparation; high cost

Zhang et al., 2023

Effectiveness of PRRSV MLV vaccines

Type	Vaccine Strain Name	Approval Time	Source of Vaccine Strains	Company	Vaccine Efficacies	Immune Responses
PRRSV-1	Porcilis PRRS	2000	DV	Merck	Minor local reactions	Humoral and cellular immune responses
	UNISTRAIN PRRS	2013	VP- 046 BIS	Hipra	The mean duration of fever was shortened and the number of fever days was significantly lower	Humoral immune response
	Ingelvac PRRSFLEX EU	2015	94811	Boehringer Ingelheim	Increases in Average Daily Weight Gain; reduces the appearance of clinical signs	Humoral and cellular immune responses
	ReproCyc PRRS EU	2015	94881	Boehringer Ingelheim	Reduces viremia	Humoral immune response
	Suvaxyn PRRS MLV	2018	96V198	Zoetis	No local reactions	Humoral and cellular immune responses
PRRSV-2	Ingelvac PRRS MLV	1994	ML (VR2332)	Boehringer Ingelheim	Reduce viremia and lung lesions	Cellular immune responses
	Fostera PRRS	2012	/	Zoetis	Reduces viremia and lung lesions	Humoral and cellular immune responses
	PrimePac PRRS	2014	/	Merck	Reduces rectal temperatures, clinical respiratory scores, lung lesion scores, and levels of virus load in serum and lung tissue	Cellular immune responses
	Prevacent PRRS	2018	/	Eli Lilly	/	/

Zhang et al., 2023

Take-home messages

- PRRSV causes major reproductive failure in sows and respiratory disease in growing pigs.
- Persistent infection and intermittent shedding contribute to long-term virus circulation within herds.
- PRRSV evolves rapidly and exists as a quasispecies, resulting in substantial genetic and antigenic diversity.
- Diagnosis requires appropriate interpretation of virological and serological results.
- Vaccination reduces clinical disease and virus replication but does not provide complete cross-protection against all strains.
- Effective control requires an integrated approach combining monitoring, vaccination, biosecurity, and herd management.

Porcine Parvovirus infection



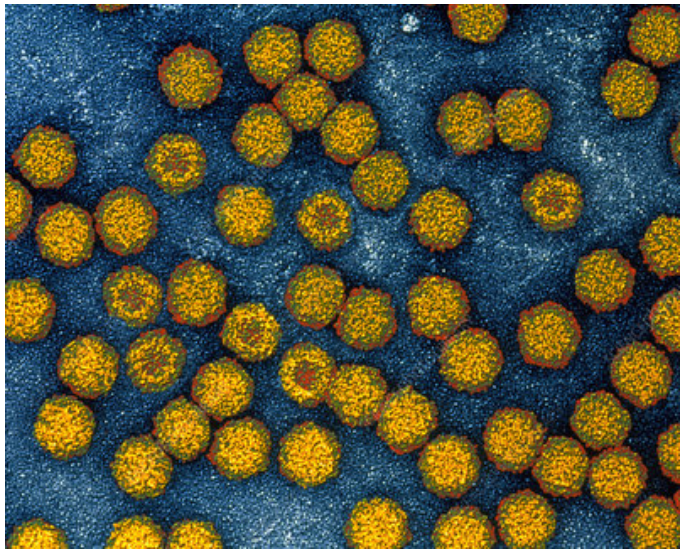


Porcine parvovirus (PPV)

- Porcine parvovirus type 1 (PPV1) is an important infectious cause of **reproductive failure** in pigs.
- PPV1 was first identified in 1965.
- PPV1 infection causes reproductive losses characterized by **stillbirth, mummification, embryonic death, and infertility (SMEDI)**.
- More recently, PPV2–PPV8 have been identified in pigs in different countries.
- The prevalence of these novel PPVs varies with virus type, age group, geographic region, and study population.

Porcine parvovirus

- PPVs 1–8 belong to the family *Parvoviridae*.
- PPV virions are small, approximately 20–30 nm in diameter, and non-enveloped.
- PPVs contain a linear single-stranded DNA genome, approximately 4–6 kb in length.



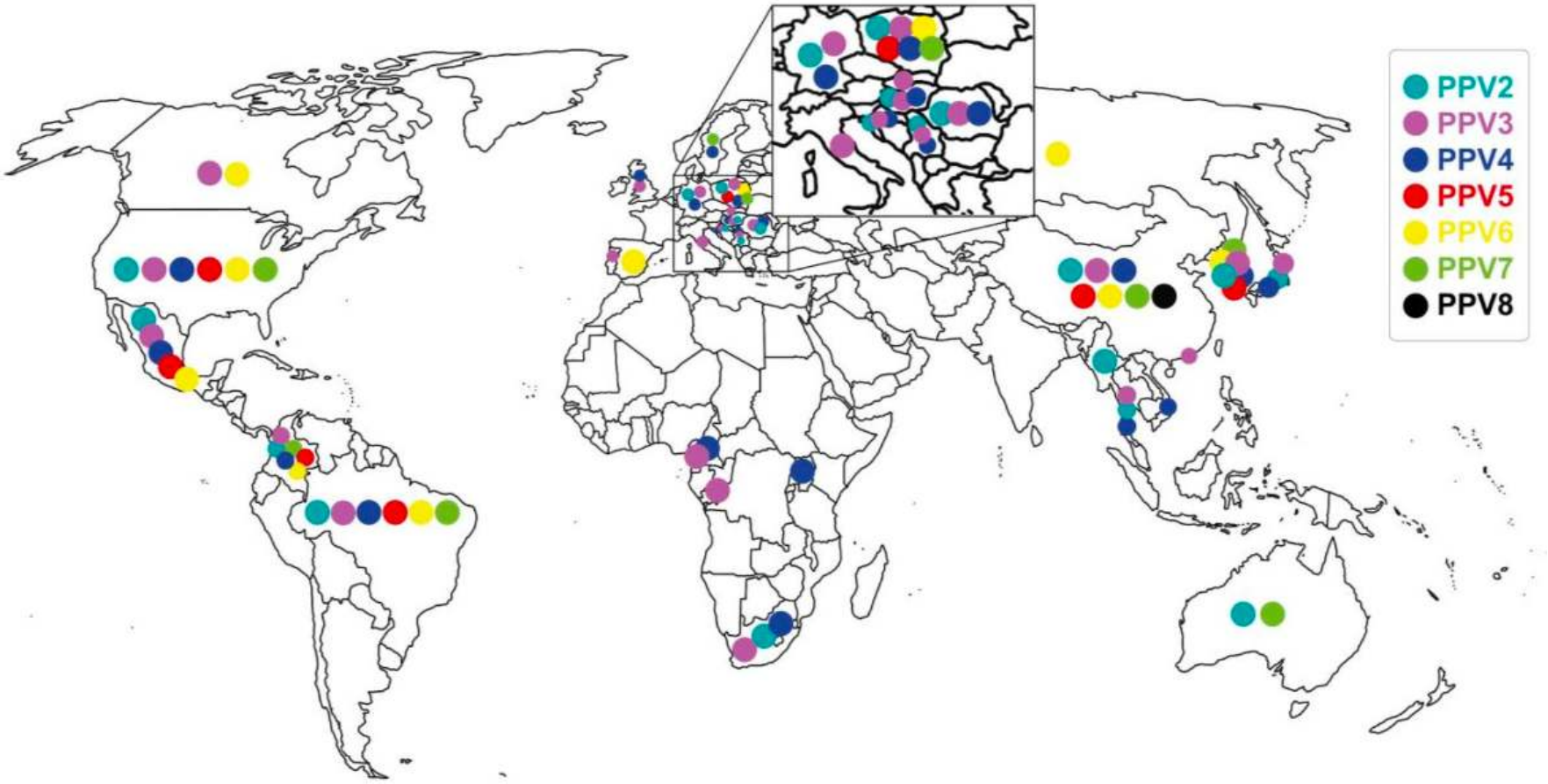
Virus inactivation

- PPV is highly resistant to environmental conditions and is relatively heat-stable.
- It may survive brief exposure to dry heat at 90°C; moist heat is generally more effective than dry heat.
- Effective disinfectants include
  - Peracetic acid 過氧乙酸
  - Aldehyde-based disinfectants 醛類
  - Sodium hypochlorite 次氯酸鈉

Epidemiology

- **PPV1** is endemic in swine populations worldwide.
- PPV2–PPV7 have been detected in multiple regions worldwide.
- PPV2 was first identified in Myanmar, PPV3 in Hong Kong, PPV4 and PPV5 in the United States, PPV6 in China, and PPV7 in the United States.
- PPV8 was first reported in China and has subsequently been detected in Europe.

Epidemiology



Vargas-Bermudez et al., 2023

Transmission

- PPV1 can replicate in susceptible pigs of all ages, but clinical disease is primarily observed in pregnant females as reproductive failure.
- Infected pigs may shed the virus in feces and other secretions.
- PPV1 can remain infectious for months in the environment and on contaminated equipment, providing a persistent source of infection.

Transmission

- PPV1 can spread between herds via **fomites**, including clothing, boots, and equipment.
- Rodents may contribute to the mechanical transmission of PPV1 between herds.
- Infected boars may shed PPV1 in **semen**, which can serve as a source of virus introduction.

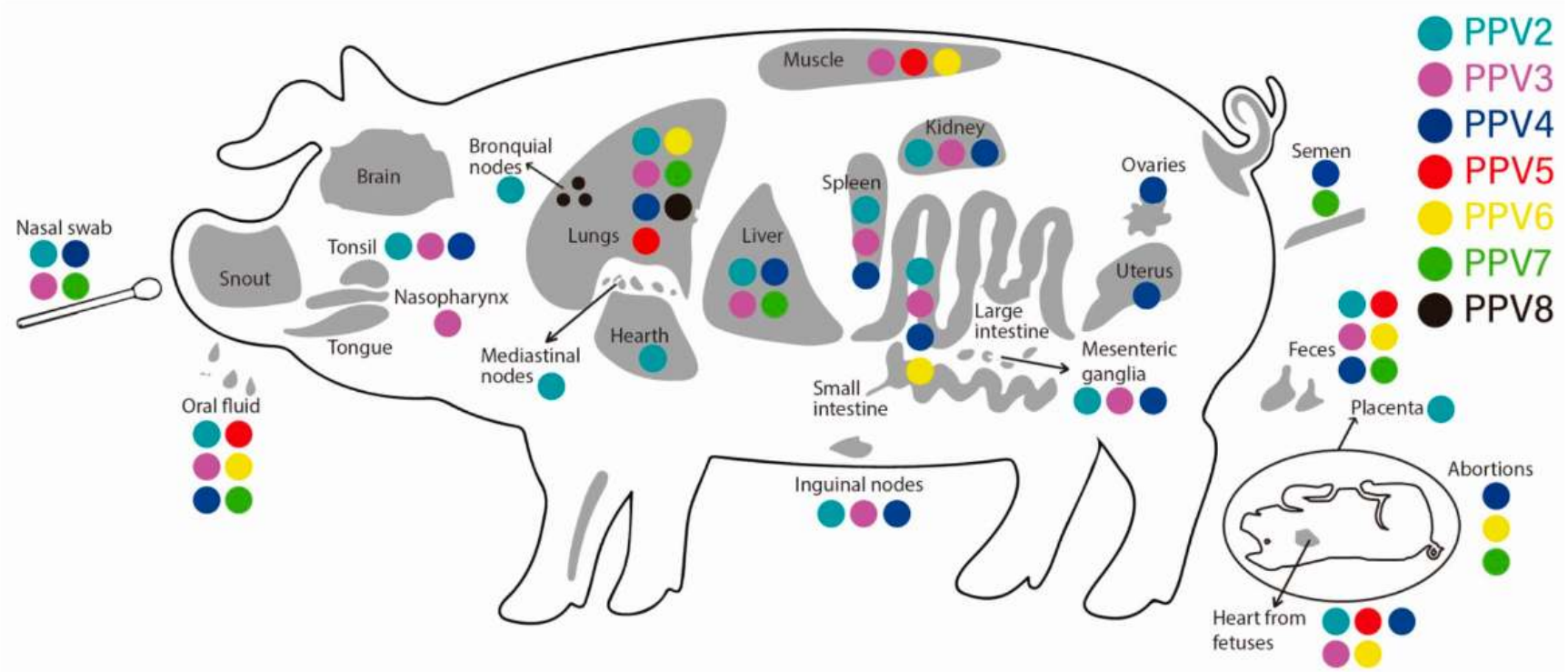
Pathogenesis

- Following primary replication in lymphoid tissues, PPV1 spreads systemically via cell-free viremia.
- PPV1 can be detected in multiple tissues and organs; however, replication is favored in rapidly dividing cells rather than a single specific tissue.
- Reproductive disease results from the ability of PPV1 to infect the embryo or fetus.
- PPV1 may reach the fetus through
  - Free virus in maternal blood or lymph
  - Progressive replication across placental cell layers
  - Virus-infected immune cells, such as macrophages or lymphocytes

Pathogenesis

- The pathogenicity of PPV1 may be influenced by coinfection with other viruses.
  - Coinfection with PCV2 and PPV1 can increase the severity of PCV2-associated disease and associated lesions.
  - However, PPV1 coinfection is **not** required for the development of PCV2-associated disease.
- The pathogenic significance of PPV2–PPV8 remains incompletely understood.

Pathogenesis



Vargas-Bermudez et al., 2023

Clinical signs

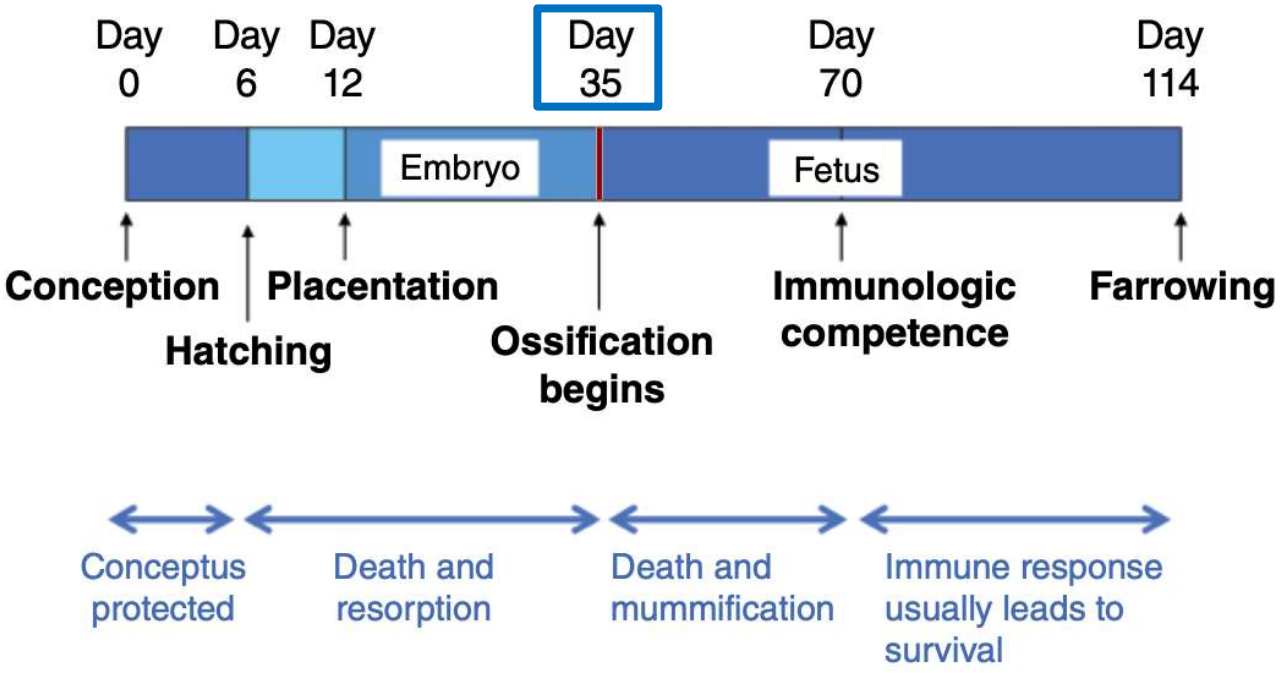
- **Reproductive failure** is the major and only well-established clinical consequence of PPV1 infection.
- Reproductive losses are usually low in vaccinated herds but can be severe in susceptible, unvaccinated herds.
- Typical reproductive losses include **stillbirth, mummification, embryonic death, and infertility (SMEDI)**.





Clinical signs

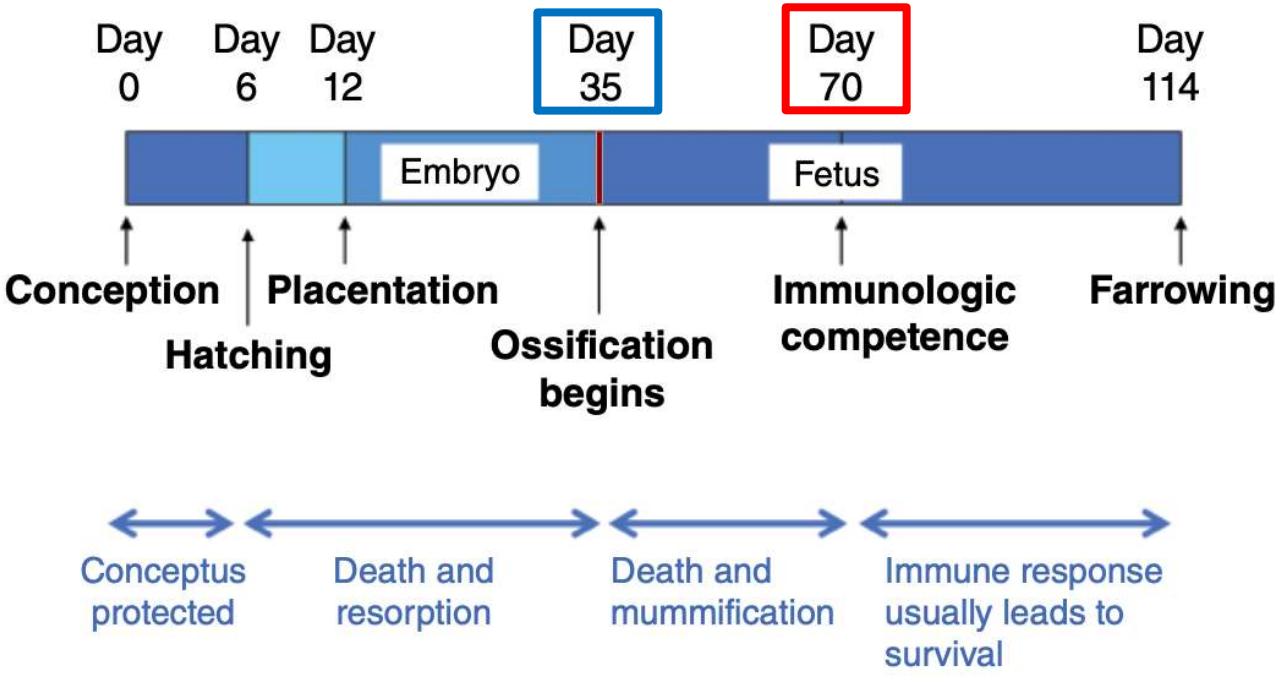
- Clinical outcome depends on the stage of gestation.
- Very early embryos are protected by the zona pellucida.
- Before day **35**: embryonic death and resorption



Diseases of Swine, 12th

Clinical signs

- Days **35**–**70**: fetal death and **mummification**
- Around day **70**: the fetus becomes immunocompetent
- After day **70**: infection is usually subclinical; piglets may be born with **anti-PPV1 antibodies**



Diseases of Swine, 12th Edition

Gross lesions

- Embryonic death and resorption are common outcomes of early PPV1 infection.
- Affected fetuses may show congestion, edema, hemorrhagic discoloration, and mummification.

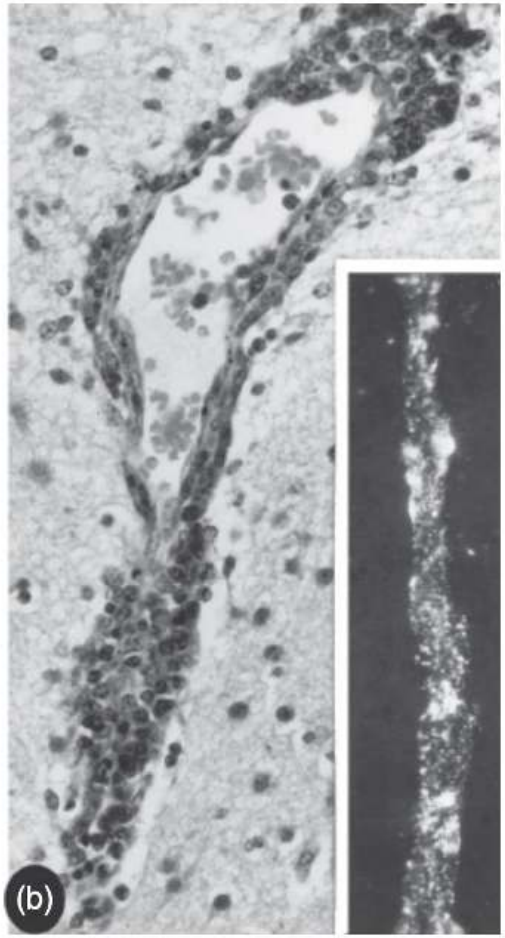


Diseases of Swine, 12th Edition

Microscopic lesions



Liver: focal necrosis



Cerebrum: perivascular mononuclear cell infiltration

Diseases of Swine, 12th

Diagnosis of PPV1 – Clinical symptoms

- PPV1 should be considered in cases of reproductive failure.
  - An increased **return-to-estrus** rate may be observed.
  - Affected litters may show reduced litter size and increased numbers of mummified fetuses, particularly in gilts or young sows.
  - A mixture of normal piglets and mummified fetuses of different sizes within the same litter is highly suggestive of PPV1 infection.
- PPV1 infection generally causes **no** clinical signs in adult pigs.

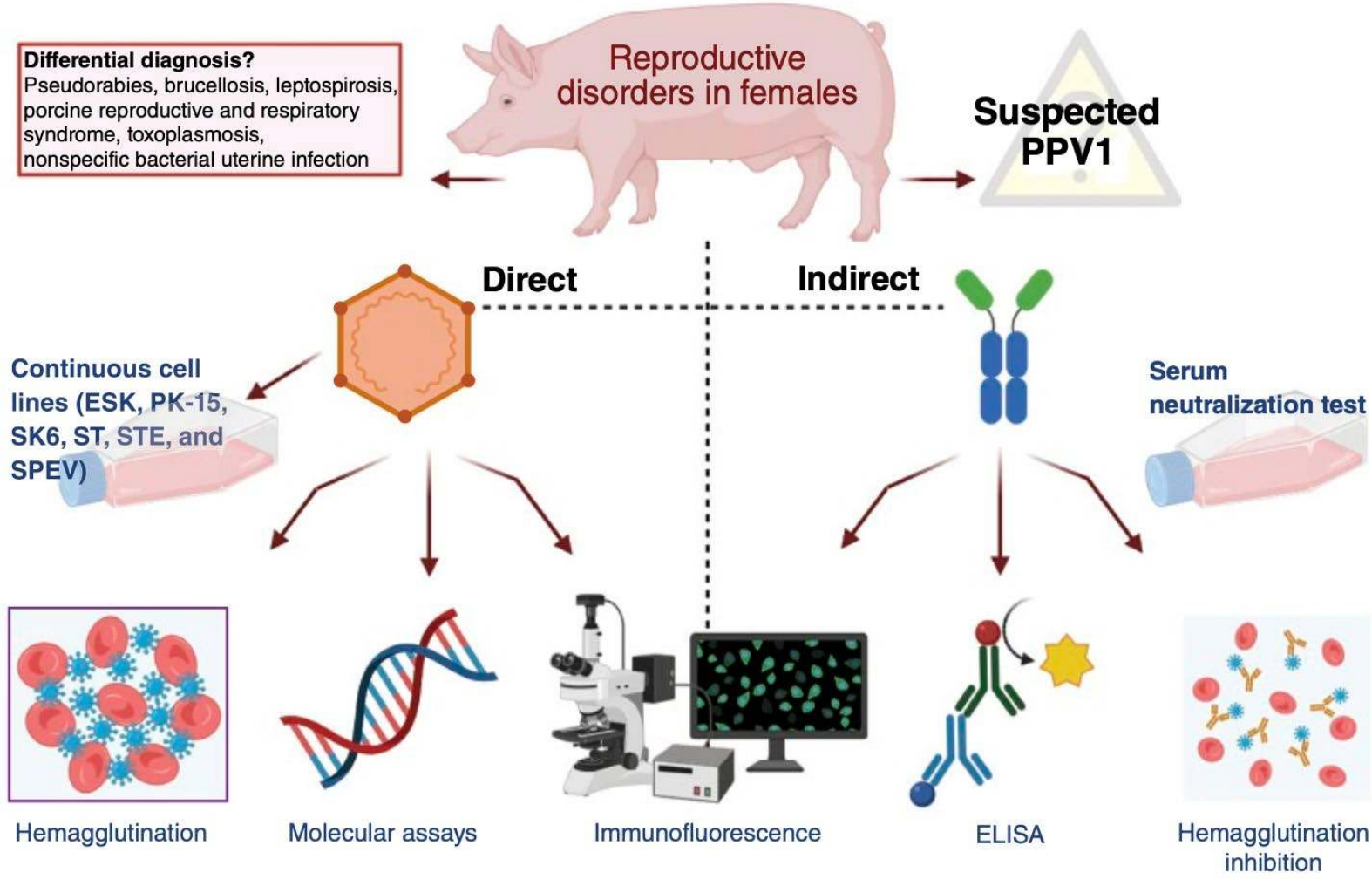
Differential diagnosis

- Porcine reproductive and respiratory syndrome
- Japanese encephalitis
- Classical swine fever
- Pseudorabies
- Brucellosis
- Leptospirosis
- Toxoplasmosis
- Other infectious and noninfectious causes of reproductive failure

Diagnostic samples

- **Mummified fetuses, stillborn piglets, and aborted fetuses** – primary diagnostic specimens
- **Fetal lung, liver, kidney, spleen, and lymphoid tissues** – PCR or antigen detection
- **Placenta** – virus detection in reproductive cases
- **Fetal fluids or serum** – antibody detection in immunocompetent fetuses
- **Serum from sows** – serology; paired samples may help demonstrate recent infection

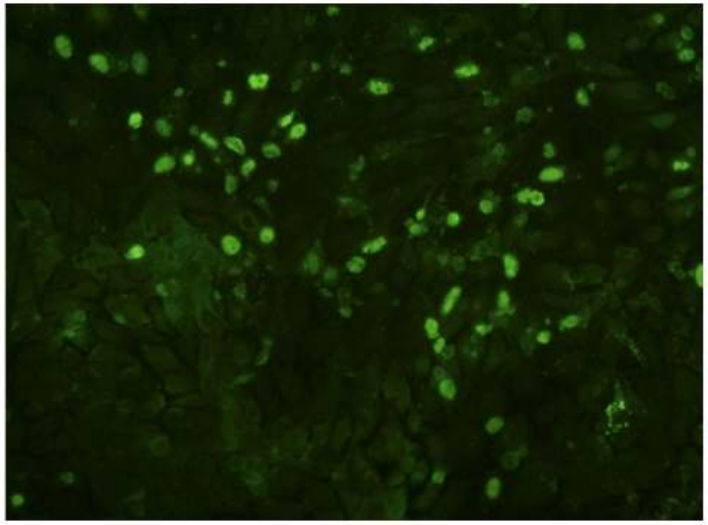
Diagnosis of PPV1 – Laboratory



Diseases of Swine, 12th

Diagnosis of PPV – Virus/antigen detection

- PCR is the most useful method for routine detection of PPV1 in fetal tissues, semen, and other samples.
- Immunofluorescence (IF) can be used to detect PPV1 antigen in fetal tissues.
- Detection of PPV2–PPV8 relies mainly on molecular assays, such as PCR.



Diagnosis of PPV – Antibody

- Hemagglutination inhibition (HI) can detect and quantify PPV1-specific antibodies.
- ELISA is suitable for standardized and high-throughput antibody testing.
- Antibodies in fetal fluids, stillborn piglets, or precolostral serum may indicate in utero infection.

PPV1 immunity

- **Passive immunity**
  - Piglets acquire maternal antibodies through colostrum.
  - Maternal antibodies gradually decline and may persist for approximately 14–26 weeks.
- **Active immunity**
  - PPV1-specific antibodies can be detected from approximately 6 days post-infection and peak around 14–21 days.
  - Immunity after infection or vaccination is generally long-lasting, and antibodies may persist for years.
  - Cell-mediated immunity also develops, although antibodies play a major role in protection.

Prevention and Control – Vaccination

- **Inactivated vaccines**
  - Most commercial PPV1 vaccines are inactivated.
  - Effectively prevent reproductive disease but may not completely prevent infection or virus shedding
- **Modified live vaccines**
  - Can induce strong and long-lasting immunity
  - Use is limited by safety concerns
- **Current challenges**
  - Emerging PPV1 genotypes and antigenic variants should be continuously monitored.
  - Improved vaccines with broader and longer-lasting protection are needed.



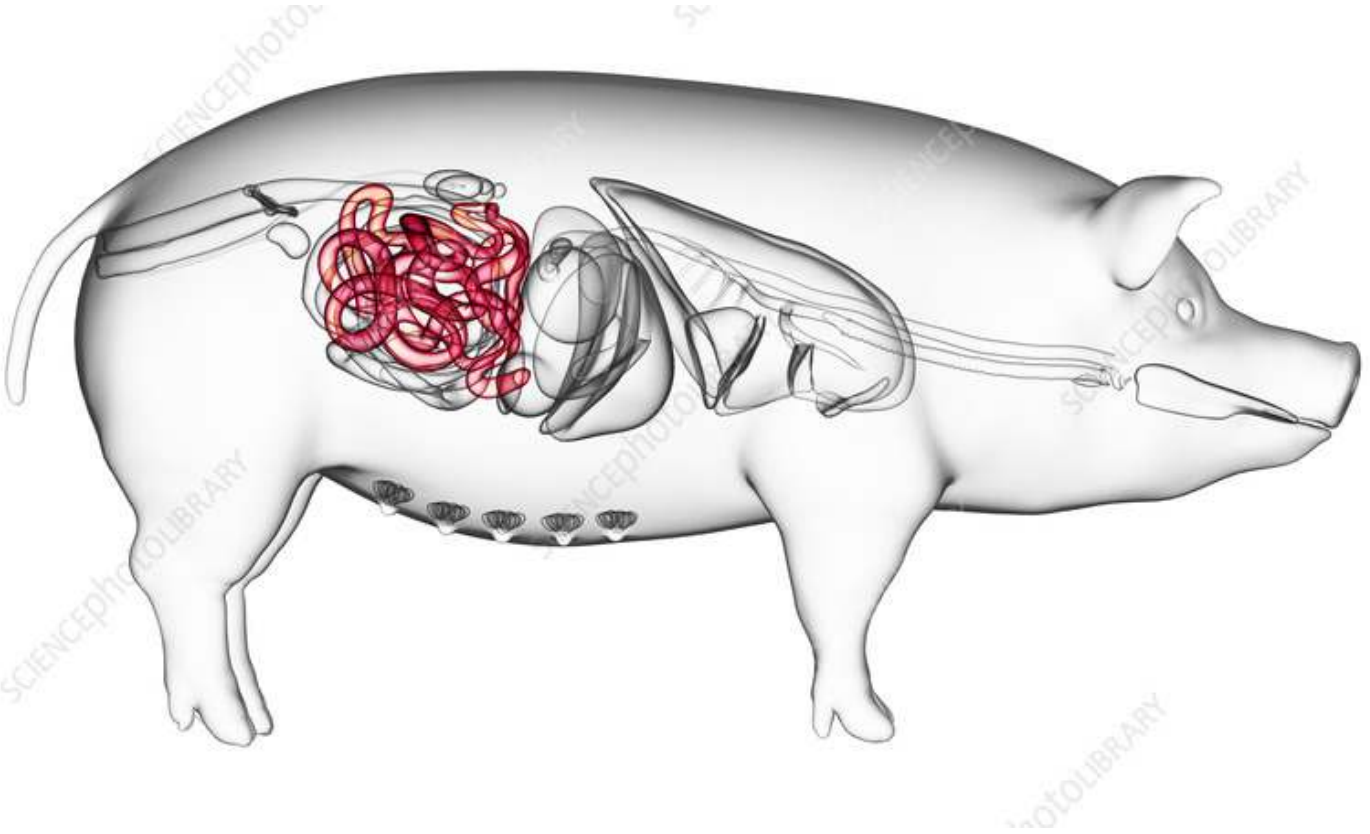
Prevention and Control

- Establish protective immunity before breeding
- Routine vaccination of replacement gilts and breeding females
- Introduction of pigs and semen from reliable sources
- Appropriate quarantine and health monitoring of replacement animals
- Effective cleaning and disinfection
- Good farm biosecurity to prevent virus introduction

Take-home messages

- PPV1 infection is common, but reproductive disease primarily occurs when susceptible females are infected during pregnancy.
- The outcome of fetal infection is strongly determined by the stage of gestation.
- Fetal mummification is a hallmark of PPV1-associated reproductive failure.
- Effective control depends on establishing protective immunity before breeding.
- Detection of PPV2–PPV8 does not necessarily indicate disease, and their pathogenic roles remain less well defined than that of PPV1.

Porcine Rotavirus infection



Porcine rotavirus infection

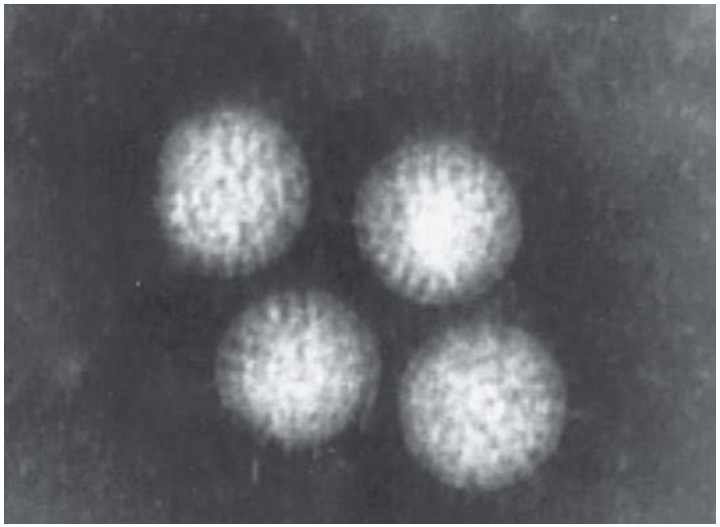
- Rotaviruses (RVs) are major causes of **diarrhea** in humans and animals, including pigs.
- RV infections are **endemic** in swine populations and may be clinical or subclinical.
- Coinfections with enteric bacteria or other pathogens can increase disease severity.
- **RVA, RVB, RVC, and RVH** have been well documented in pigs.
- RVE-like viruses have also been reported historically in pigs.

Clinical significance by RV type

- **RVA** – the most prevalent and best-characterized porcine rotavirus
- **RVB** – detected across age groups, including older pigs
- **RVC** – particularly important in neonatal piglets
- **RVH** – pathogenic significance remains less well defined

Porcine rotavirus

- Rotaviruses belong to the genus *Rotavirus* of the family *Sedoreoviridae*.
- They are non-enveloped, icosahedral viruses with a segmented double-stranded RNA (dsRNA) genome.
- Virions are approximately 70–100 nm in diameter.



Diseases of Swine, 12th Edition

Rotavirus classification and genotyping

- RV species are distinguished mainly based on **VP6** and genomic sequence characteristics.
- RVA genotypes are commonly classified using
  - G genotype = VP7
  - P genotype = VP4
  - G9P[13]
    - G9 → VP7 genotype 9
    - P[13] → VP4 genotype 13
- Whole-genome sequencing can further classify all 11 genome segments.

Virus inactivation

- RV is highly **stable** in the environment, and complete drying may not inactivate all virions.
- Effective disinfectants include chlorine-based compounds and glutaraldehyde when used at appropriate concentrations and contact times.

Public health

- **RVA** has the greatest zoonotic potential among porcine rotaviruses.
- Porcine-like and human–porcine reassortant **RVA** strains have been detected in **humans**.
- These findings suggest occasional interspecies transmission and reassortment between pigs and humans.
- **RVB, RVC, and RVH** appear to be more host-restricted, with limited evidence of pig-to-human transmission.
- Pigs may serve as a reservoir for novel RVA genes and genotypes with zoonotic potential.
- Zoonotic events appear to be sporadic, but reassortment may facilitate adaptation of animal-origin strains to humans.

Human RVs of porcine origin

Table 1. Human RV genotypes of suspect or confirmed porcine origin via direct transmission or multiple re-assortment events.

Porcine RV Species	G and/or P Genotype	Geographic Region	Year Samples Collected	Epidemiological Status and Medical Relevance
A	G1-G4, G9, G12 and P8	Worldwide	2000s	Commonly seen in humans *
	G3–G5, G9 and G11, as well as P[6]	Denmark, France, Hungary, Italy, Slovenia	2003–2007	G3-G5—common in humans, G5—regional in humans, P6—rare in humans
	G1 and G4	Brazil	2007	Common
	G1P[8]	China; MD, USA	2004–2009	Common
	G1, G1P[6]	Japan	2001	Common
	G1P[6]	Japan	1997	Rare
	G1P[6], G4P[6] and G12P[6]	Democratic Republic of the Congo	2007–2010	Common
	G1P[19]	India	1992	Rare
	G2	Europe	1992	Uncommon
	G3P[6], G4P[6] and G4P[8]	China, Italy, Slovenia	2003–2013	Common
	G3P[25]	Taiwan	2009	Rare
	G4P[6] strains, one G5P[6]	Taiwan	2006–2012	Common
	G4P[6]	Hungary, China, Argentina, Madagascar	2006–2007 2008–2009	Sporadic identification in humans worldwide
	G5P[6]	Japan, Bulgaria	2011 2006	Rare
	G5P[8]	Brazil, Argentina, Paraguay, Cameroon, China, Thailand, and Vietnam	1986–2005	Common in Asian, African and South American countries
	G9	NE, USA; India	1980s, 1190s, 1997–2000	Uncommon, emerging worldwide
	G9P[6]	India	2007	Unusual
	G9P[19]	Thailand, India	2012–2013 1989–1990	Rare
	G9P[19] and G9P[13]	Taiwan	2014–2015	Rare
	G9P[19] and G10P[14]	Vietnam	2007–2008	Rare
	G11P[4], G11P[6], G11P[8] G11P[25]	Nepal, Bangladesh	2001–2004	Rare
	G11P[25]	India	2005–2009	Uncommon
	G12P[6] and G12P[8]	Kenya, Myanmar	2010, 2011	Common
	G26P[19]	Vietnam	2009–2010	Atypical in humans
B	N/A	Brazil	2000s	Regional significance
C	N/A	Japan, Brazil	1982–1986, 2000–2007	Regional significance

Vlasova et al., 2017

Epidemiology

- Mixed infections with multiple RV species or strains may occur, particularly as pigs are repeatedly exposed with increasing age.
- **Age** is an important factor influencing both the prevalence of infection and disease severity.
- **Young piglets**, particularly those with inadequate maternal immunity, are highly susceptible to RV-associated diarrhea.
- Increased RV shedding by **sows** around farrowing may increase environmental contamination and exposure of neonatal piglets.

Porcine RVA genotypes

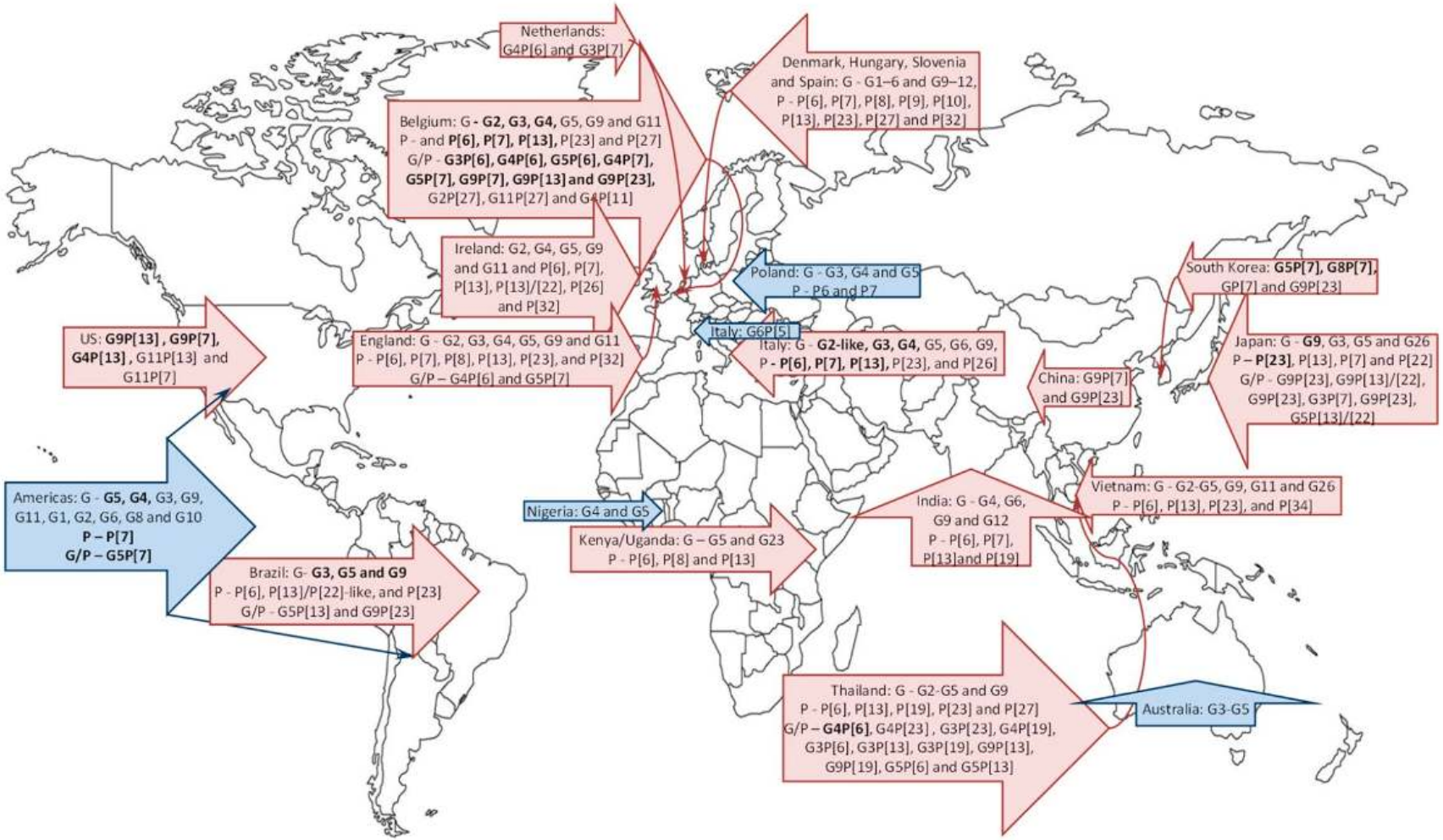


Figure 2. Global genotype distribution of porcine RVA strains reported in historic (1976–2011, blue figure arrows) and current (after 2000, pink figure arrows) studies. Porcine RVAs are also detected in Germany and Russia, but no genotyping data is available.

Vlasova et al., 2017

Porcine RVs genotypes

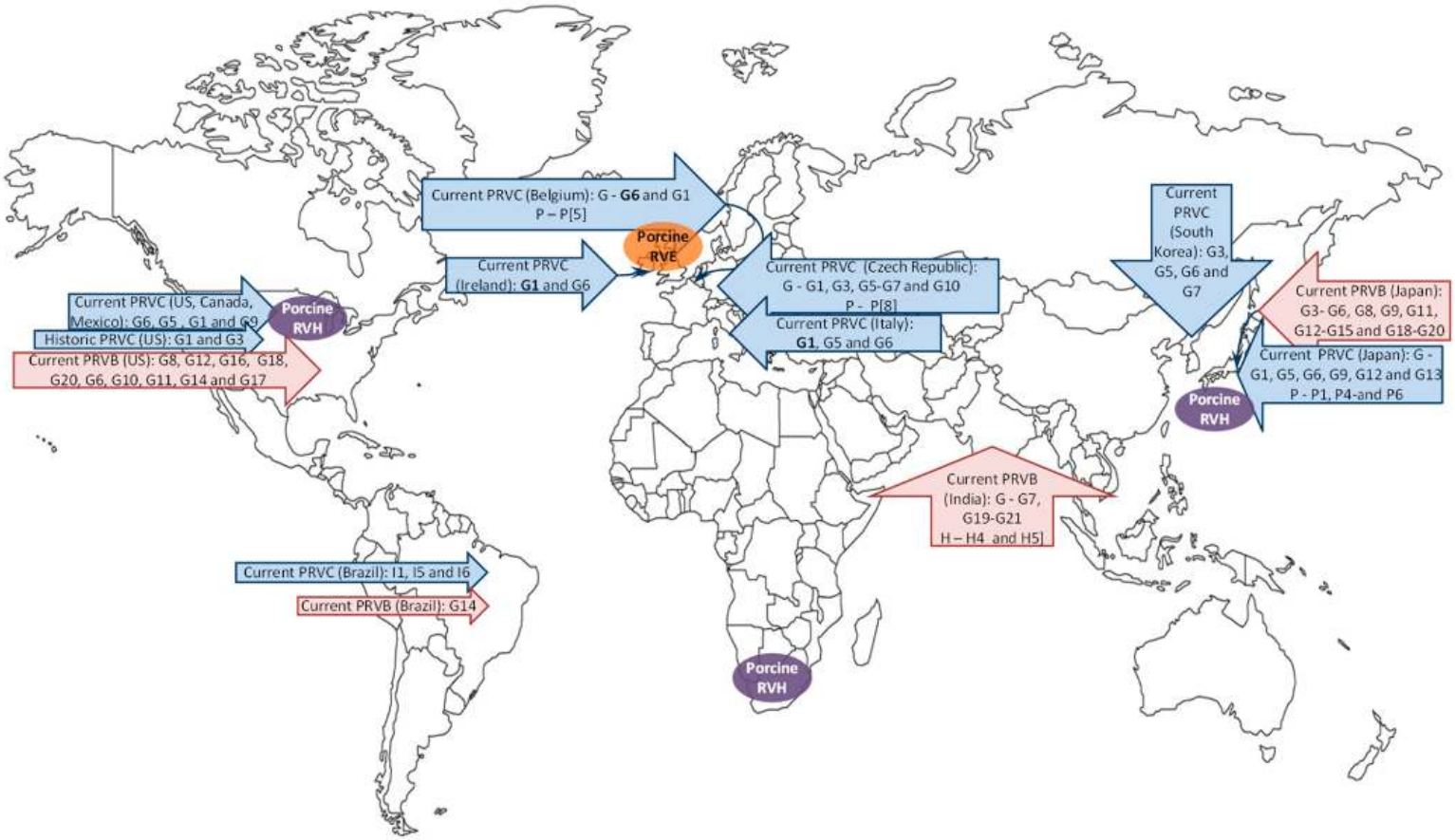


Figure 3. Global genotype distribution of porcine RVB (pink figure arrows) and RVC (blue figure arrows) strains and porcine RVE (bolded, orange circle)/RVH (bolded, purple circles) occurrence in different countries reported in historic (1976–2011) and current (after 2000) studies. Porcine RVCs are also detected in Germany and China, and porcine RVB is confirmed in Germany and Czech Republic, but no porcine RVC/RVB genotyping data is available for these countries.

Vlasova et al., 2017

Porcine RVA genotypes in Taiwan

Contents lists available at ScienceDirect

Infection, Genetics and Evolution

journal homepage: www.elsevier.com/locate/meegid

Prevalence and diversity of rotavirus A in pigs: Evidence for a possible reservoir in human infection☆

Fang-Tzy Wu[a,*], Luke Tzu-Chi Liu[a], Baoming Jiang[b], Ting-Yu Kuo[a], Ching-Yi Wu[a], Ming-Huei Liao[c,d]

- High genetic similarity between **porcine RVA** and unusual RVA strains detected in **children** suggests that pigs may serve as a reservoir for sporadic zoonotic infection.

Wu et al., 2022

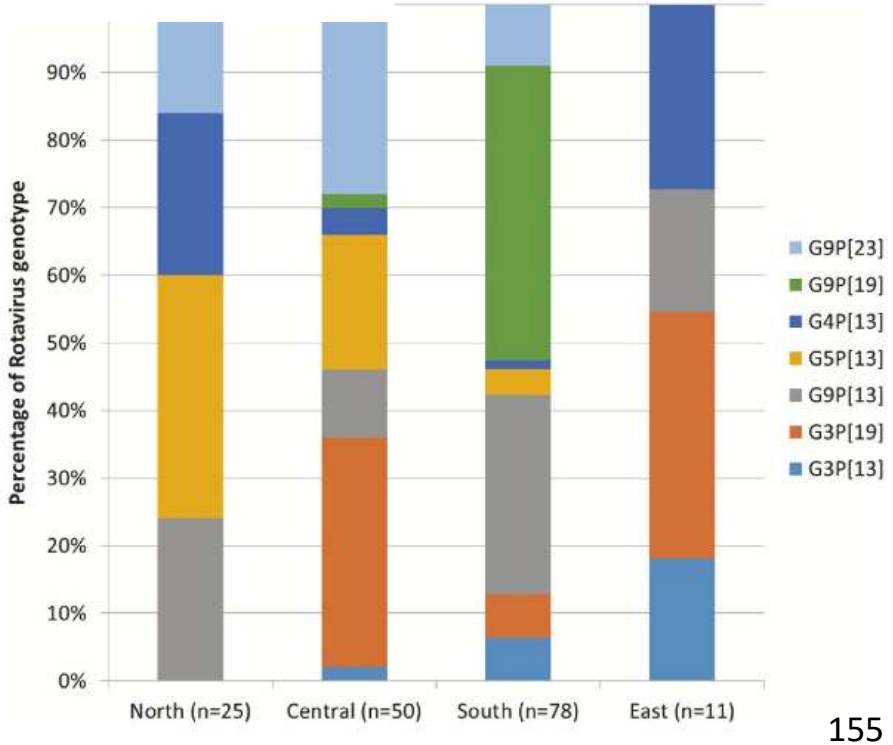


Fig. 2. Major porcine rotavirus G/P genotypes by region, Taiwan, 2014–2017.

Transmission

- RV particles are shed in large numbers in the **feces** of infected pigs.
- RVs are highly infectious; as few as 90 virus particles have induced diarrhea and virus shedding in colostrum-deprived piglets.
- Feces may contain up to approximately $1x10^{10}$ virus particles/g, facilitating rapid transmission.
- Transmission occurs mainly by the **fecal–oral route** through contaminated feces, fomites, feed, or water.
- RVs are environmentally stable and can persist in contaminated facilities and equipment.

Pathogenesis

- RV primarily infects mature enterocytes in the **jejunum** and **ileum**.
- Viral replication causes villus blunting and atrophy, particularly at the villus tips.
- The major lesions are confined mainly to the **small intestine**.
- Diarrhea results primarily from impaired absorption following enterocyte loss and villus atrophy, with additional secretory effects mediated by viral proteins.

Clinical signs



- RV infection ranges from subclinical infection to severe disease, depending on the virus strain, age, immune status, and concurrent infections.
- The incubation period is usually 1–4 days in piglets.
- Clinical signs include watery diarrhea, lethargy, anorexia, dehydration, and rapid weight loss; vomiting may also occur.
- Feces are often yellow to pale and may contain undigested milk.

Clinical signs

- Diarrhea in **neonatal piglets** may persist for several days and can result in marked dehydration and weight loss.
- Clinical disease is generally milder and shorter in older piglets, and mortality is lower.
- **Subclinical infection** is common in older pigs.
- Disease severity may be increased by dietary factors and coinfection with other enteric pathogens.

Clinical signs





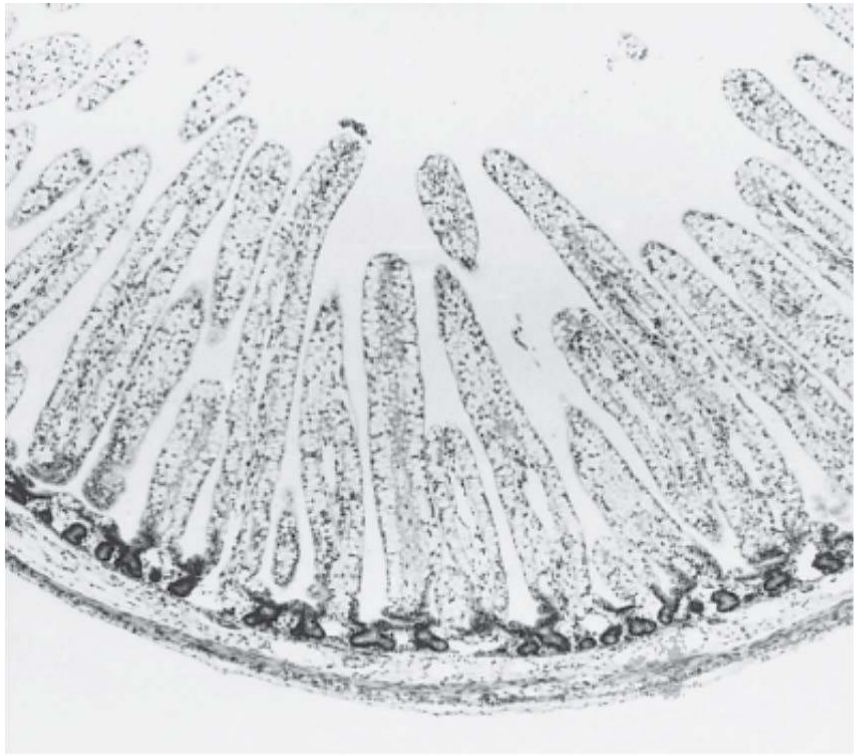
Clinical signs



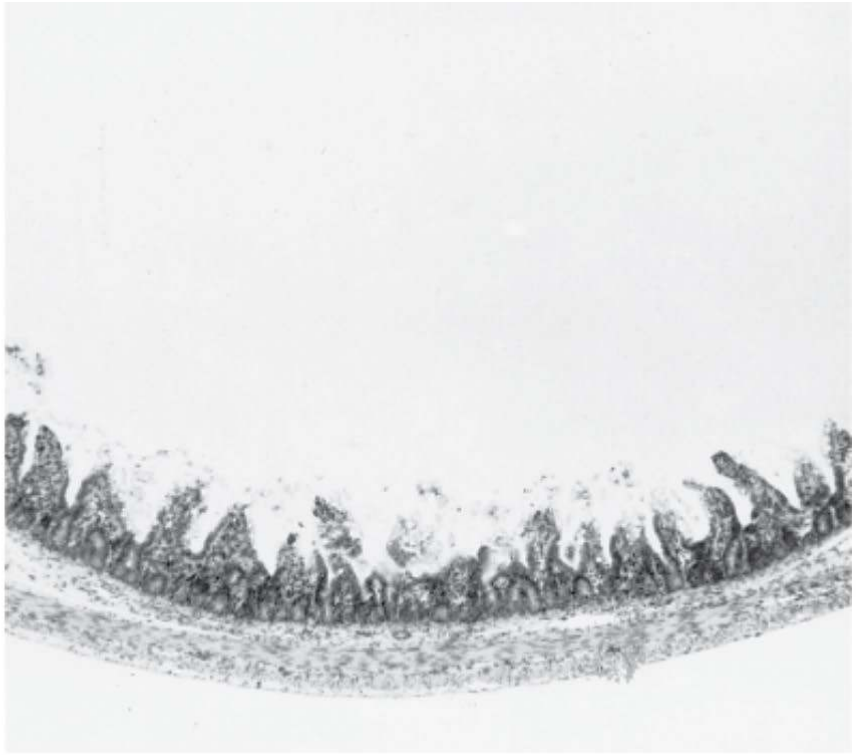
Gross lesions

- The **small intestine** is often dilated and thin-walled, with excessive watery, yellow to gray contents.
- The **large intestine** may also contain increased fluid.
- The **stomach** may be distended with undigested milk.

Microscopic lesions



Normal villi



Villous atrophy

Diseases of Swine, 12th Edition

Differential diagnosis

- RV primarily affects neonatal and suckling piglets and should be considered in the differential diagnosis of **enteric** disease.
- Clinical signs are nonspecific; laboratory confirmation is required.
- Because subclinical RV infections are common, detection of RV alone does not always prove that it is the cause of diarrhea.

Differential diagnosis

- PEDV
- TGEV
- PDCoV
- Enterotoxigenic E. coli
- *Clostridium perfringens*
- *Cystoisospora suis*
- Other enteric coinfections

Diagnostic samples

- **Feces or rectal swabs** – RT-PCR and antigen detection
- **Intestinal contents** – virus detection in acute enteric cases
- **Jejunum and ileum** – histopathology, antigen detection, and RT-PCR
- **Fresh and fixed intestinal tissues** should be collected from necropsied piglets.

Diagnosis – Laboratory

- **Direct detection**
  - RT-PCR – most commonly used for RV detection and typing
  - Antigen detection – feces or intestinal contents
  - Immunohistochemistry (IHC) – intestinal tissues
  - *In situ* hybridization (ISH) – viral RNA in tissues
  - Cell culture with immunofluorescence – mainly for virus isolation/research
- **Antibody detection**
  - ELISA
  - Hemagglutination inhibition (HI)
  - Plaque reduction neutralization (PRN)
  - Fluorescent focus neutralization (FFN)

Immunity

- **Active immunity**
  - RV infection induces systemic and mucosal immune responses.
  - **VP7** and **VP4** are the major targets of neutralizing antibodies.
  - Intestinal **IgA** is an important correlate of protection.
- **Passive immunity**
  - Neonatal piglets depend mainly on maternal immunity.
  - **IgG** is mainly acquired through colostrum, whereas **IgA** is continuously supplied through milk.
  - Inadequate colostrum intake or early weaning increases susceptibility to severe diarrhea.

Cross-protection

- Within the same species, protection is generally stronger when **VP7** and **VP4** of the vaccine and circulating strains are closely related.
- Cross-protection between different genotypes or lineages is variable.
- Vaccines should ideally include locally prevalent genotypes.

Prevention and Control – Vaccination

- Commercial vaccines are available mainly for **RVA**.
- Modified live or attenuated vaccines generally induce stronger mucosal immunity than inactivated vaccines.
- Maternal antibodies may interfere with the replication and immune response of live vaccines in piglets.
- Vaccination of **sows** before farrowing is commonly used to enhance lactogenic immunity in piglets.



Prevention and Control

- Enhance maternal immunity before farrowing
- Ensure adequate colostrum intake
- Maintain good farrowing-house hygiene
- Reduce environmental contamination between litters
- Appropriate temperature and supportive care for affected piglets
- Control coinfections

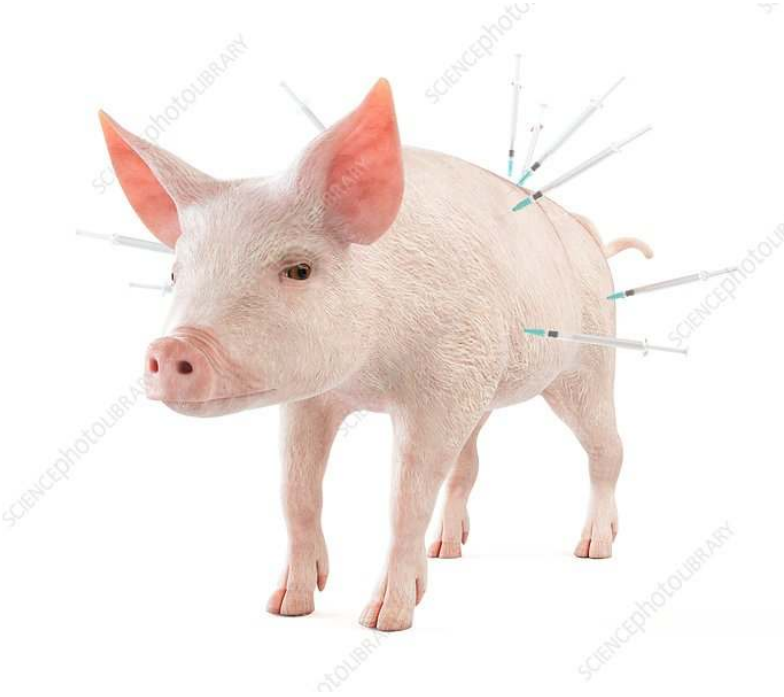
Treatment

- Supportive care includes maintaining an appropriate environmental temperature, particularly for neonatal piglets.
- Fluid and electrolyte replacement is important to prevent dehydration.
- Adequate nutritional support should be provided during recovery.
- Antibiotics may be used when secondary bacterial infection is suspected or confirmed.

Take-home messages

- Rotavirus is an important cause of diarrhea in neonatal and suckling piglets.
- Disease severity is strongly influenced by age, maternal immunity, and coinfection with other enteric pathogens.
- RVA is the most important and genetically diverse porcine rotavirus, with occasional zoonotic potential through interspecies transmission and reassortment.
- Protection of young piglets depends largely on maternal antibodies, especially lactogenic IgA.
- Effective control requires maternal vaccination, good hygiene, and supportive care.

Vaccination of pigs





Global animal vaccine industry

Animal Vaccines Market Size & Share 2026-2035

Report ID: GMI3117 | Published Date: March 2026 | Report Format: PDF/Excel/Dashboard/Platform

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Animal Vaccines Market Size

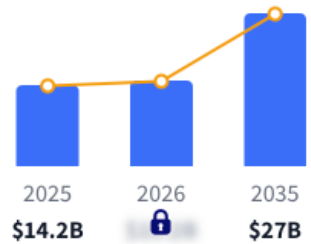
The global animal vaccines market was valued at USD 14.2 billion in 2025. The market is expected to grow from USD 15.2 billion in 2026 to USD 27 billion in 2035, growing at a CAGR of 6.6% during the forecast period, as per the latest report published by Global Market Insights Inc.

Animal Vaccines Market Key Takeaways

Share

2025 Market Size	2026 Market Size	2035 Forecast Market Size	CAGR (2026–2035)
$ 14.2 Billion	$ 15.2 Billion	$ 27 Billion	6.6%

Market Size Trend



Regional Dominance

Largest Market

North America

Key Players

- **Market Leader:** Zoetis led with over 22% market share in 2025.
- **Leading Players:** Top 5 players in this market include **Zoetis, Boehringer Ingelheim International, Elanco Animal Health Incorporated, Merck & Co., Inc., Virbac**, which collectively held a market share of **65%** in 2025.

Global animal vaccine industry – Swine vaccines

Swine Vaccines Market Size & Share 2026-2035

Report ID: GMI10445 | Published Date: March 2026 | Report Format: PDF/Excel/Dashboard/Platform

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Swine Vaccines Market Size

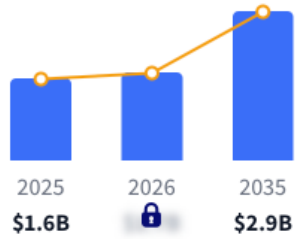
The global swine vaccines market was valued at USD 1.6 billion in 2025. The market is expected to grow from USD 1.7 billion in 2026 to USD 2.9 billion in 2035, at a CAGR of 6.3% during the forecast period, according to the latest report published by Global Market Insights Inc.

Swine Vaccines Market Key Takeaways

Share

2025 Market Size	2026 Market Size	2035 Forecast Market Size	CAGR (2026–2035)
$ 1.6 Billion	$ 1.7 Billion	$ 2.9 Billion	6.3%

Market Size Trend



Regional Dominance

Largest Market

North America

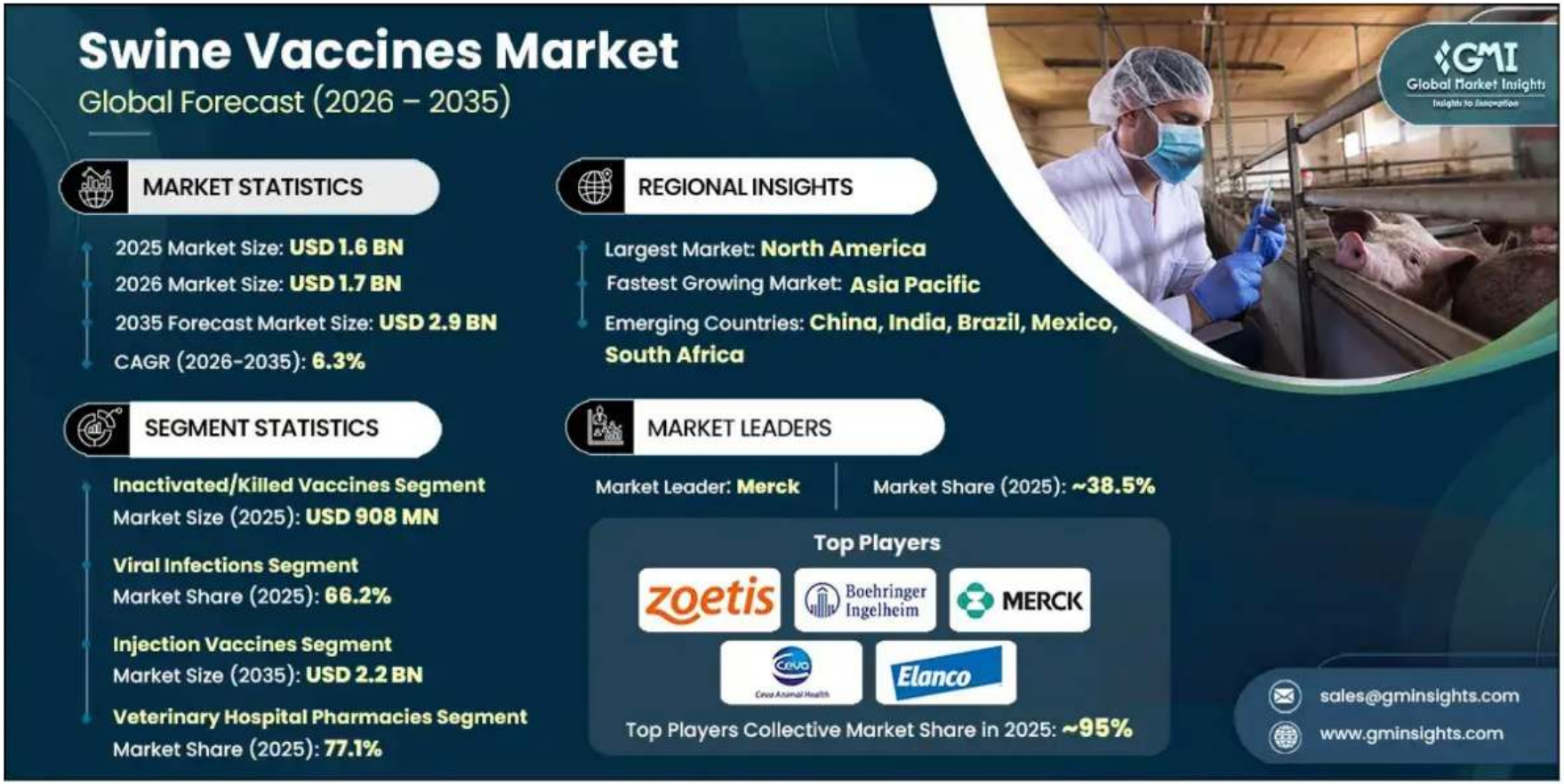
Fastest Growing Region

Asia Pacific

Key Players

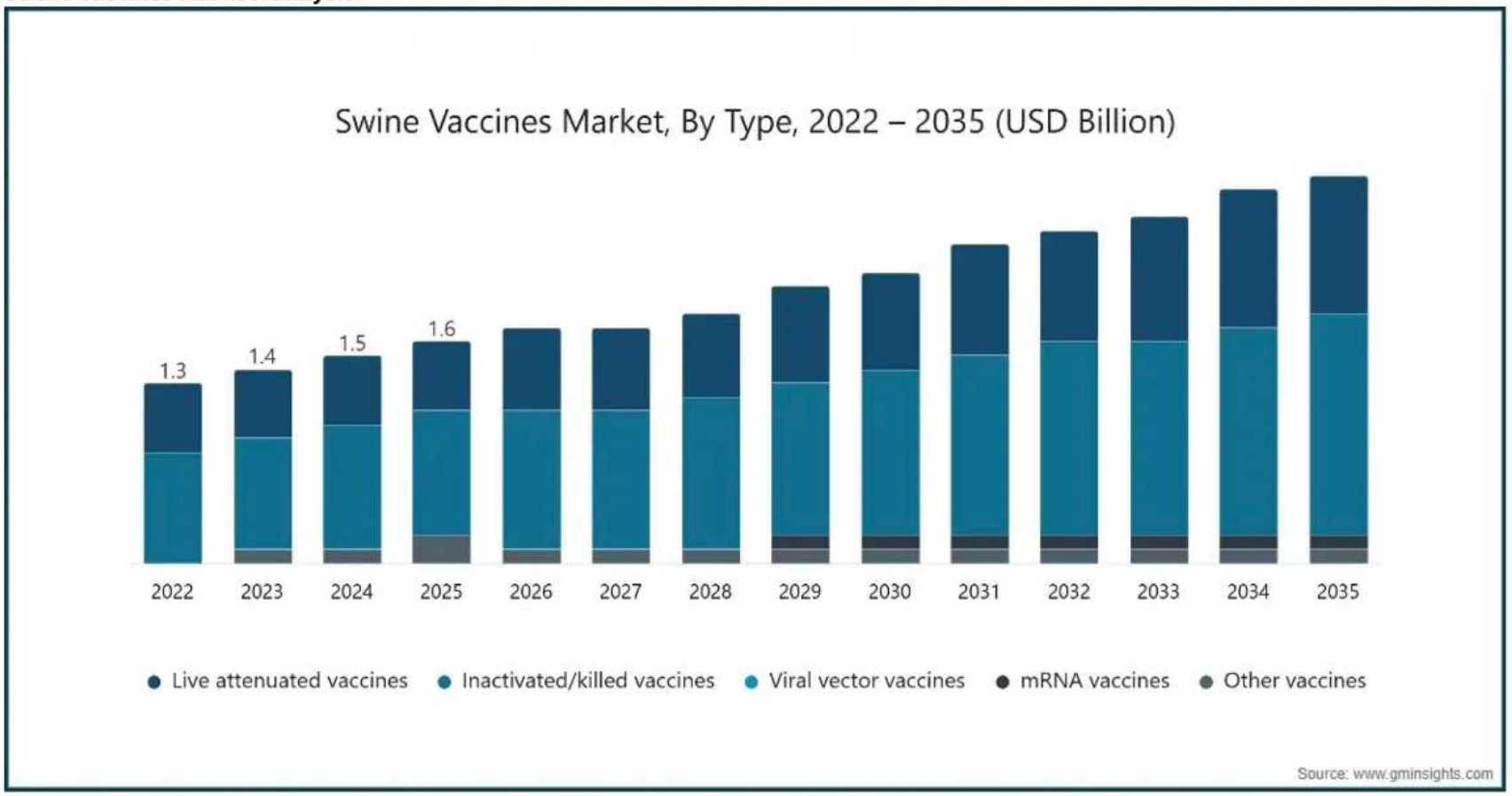
- **Market Leader:** Merck led with over **38.5%** market share in 2025.
- **Leading Players:** Top 5 players in this market include **Zoetis, Boehringer Ingelheim International, Merck, Ceva, Elanco Animal Health Incorporated**, which collectively held a market share of **95%** in 2025.

Global animal vaccine industry – Swine vaccines

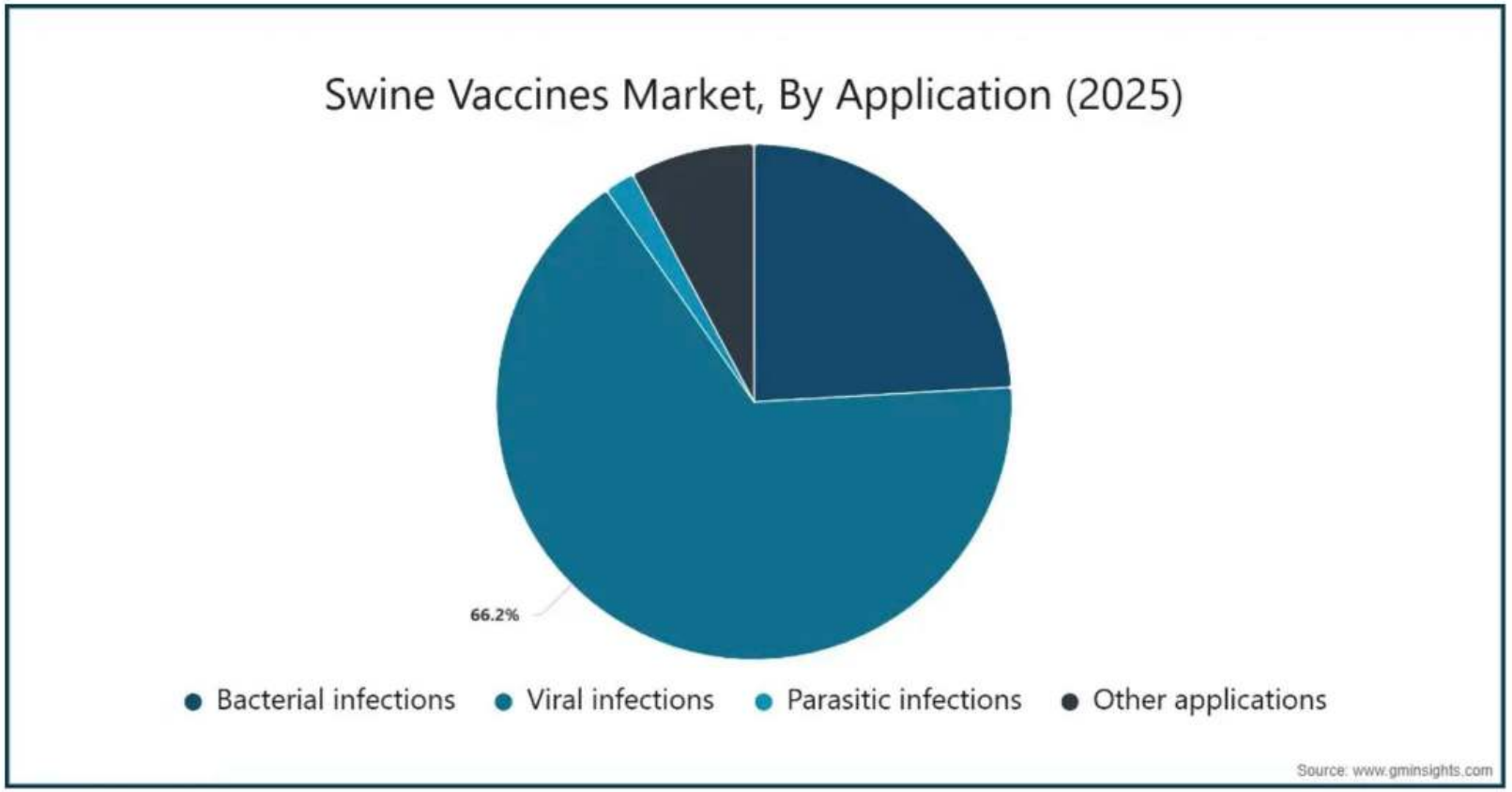


Global animal vaccine industry – Swine vaccines

Swine Vaccines Market Analysis

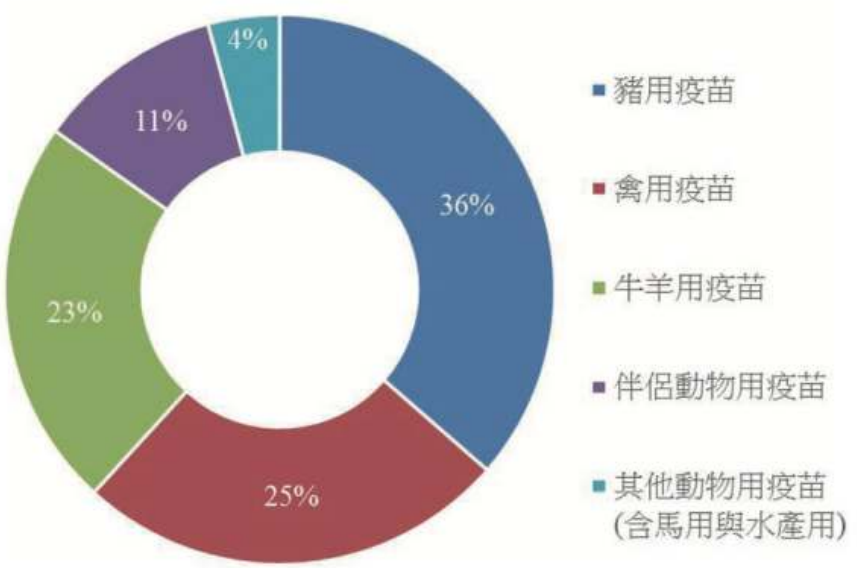


Global animal vaccine industry – Swine vaccines



Animal vaccine industry – Species

Global, 2015



資料來源：MarketsandMarkets(2015)；台灣經濟研究院生物科技產業研究中心整理繪製。

圖一　全球各動物用疫苗市場規模占比

Taiwan, 2020

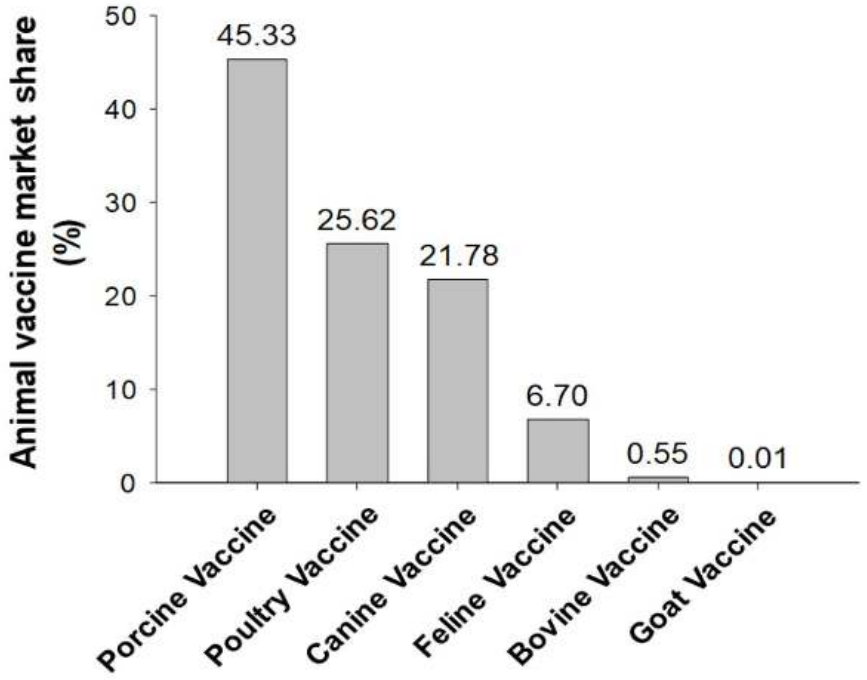


TABLE 18.1 ■ **Vaccination of Piglets Under 6 Months of Age**

Vaccine	Vaccine Timing	Comments
Porcine parvovirus	Vaccinate with 2 doses, 3–5 weeks apart with the second dose 2–4 weeks before breeding.	Revaccinate annually before breeding. Revaccinate boars semiannually. Withdrawal time 21 days.
Pseudorabies	Vaccinate piglets 1–7 days old intranasally, followed by intramuscular vaccination of all the other pigs on the farm.	Revaccinate breeding animals semiannually. Withdrawal time 21 days.
Porcine circovirus	Vaccinate at 3 weeks of age, revaccinate at 6 weeks. Some producers prefer 1 and 3 weeks of age or even at 2 and 4 weeks.	Withdrawal time 21 days.
Porcine respiratory and reproductive syndrome virus	Vaccinate piglets over 3 weeks of age with a single dose. Another such vaccine may be given at 1 day of age. The autogenous intranasal vaccine may be administered at 7–10 days and at weaning (3 weeks).	Boosted semiannually. Withdrawal time 21 days.
Porcine epidemic diarrhea	Vaccinate piglets over 3 weeks of age and revaccinate 3 weeks later.	Withdrawal time 21 days.
Influenza	Vaccinate piglets over 3–5 weeks intramuscularly, followed by a second dose 2–3 weeks later. Vaccinate piglets over 1 day of age. intranasally.	Revaccinate semiannually. Withdrawal time 21 days.
Rotavirus	Vaccinate with the oral or injectable vaccine at 7–10 days preweaning.	If using the oral vaccine do not return the piglet to the sow for at least 30 minutes. Withdrawal time 21 days.

TABLE 18.2 ■ **Vaccination of Adult Pigs**

Vaccine	Vaccine Timing	Comments
Antiviral Vaccines		
Porcine parvovirus	Vaccinate breeding sows and boars 3–4 weeks apart. Sows should be boosted at each weaning. Boars should receive 6 monthly boosting.	Withdrawal time 21 days.
Pseudorabies	Vaccinate sows and piglets intranasally when 1–7 days old, followed by intramuscular vaccination of all the other pigs on the farm.	Withdrawal time 21 days.
Porcine respiratory and reproductive syndrome virus	Vaccinate sows and gilts 3–4 weeks before breeding or during pregnancy to prevent the reproductive form of the disease.	Withdrawal time 21 days.
Porcine epidemic diarrhea	Vaccinate pregnant sows, 5 and 3 weeks before farrowing.	In subsequent pregnancies give a single dose 2 weeks before farrowing.
Influenza	Vaccinate healthy breeding pigs in 2 doses 3–5 weeks apart. Revaccinate 2–4 weeks before breeding.	Revaccinate boars semiannually and sows annually before breeding. Withdrawal time 21 days.
Transmissible gastroenteritis	Vaccinate sows and gilts orally with two doses 3–5 weeks before farrowing and boost with the intramuscular (IM) vaccine one week before farrowing. Otherwise give the IM vaccine 5 and 2 weeks before each farrowing.	Prime with the oral vaccine and boost with an intramuscular vaccine. Withdrawal time 21 days.
Rotavirus	Used with transmissible gastroenteritis. Vaccinate sows 2–5 weeks before farrowing.	Prime with the oral vaccine and boost with an intramuscular vaccine. Withdrawal time 21 days.

Principles of vaccination in pigs

- Vaccination aims to prevent clinical disease and reduce pathogen transmission, but may not completely prevent infection.
- Vaccination programs should be designed according to herd health status, disease risk, and production stage.
- Vaccine timing should consider **maternal antibodies** and the period of greatest disease risk.
- Breeding animals may be vaccinated to protect themselves or to provide maternal immunity to their offspring.

Factors affecting vaccine efficacy

- **Host factors**
  - Maternal antibody interference
  - Age and immune status of the pig
  - Concurrent infection, stress, or immunosuppression
- **Vaccine and pathogen factors**
  - Vaccine type, dose, route, and timing
  - Antigenic differences between vaccine and field strains
- **Management factors**
  - Cold chain and proper vaccine handling
  - Adherence to the recommended vaccination schedule

Questions?

